



Bachelor Thesis – Cosmic Backreaction

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Abstract

This thesis investigates inhomogeneous cosmology and cosmic backreaction, and how it can explain the observed acceleration of the expansion of the Universe. First, three local equations are considered; the continuity equation, the Raychaudhuri equation and the Hamiltonian constraint. From these equations, the Buchert equations[1] are derived to show how a backreaction term can arise from inhomogeneity. Then a two-region toy model is created to get an intuition of how this occurs, and how a simple, inhomogeneous model can describe a universe evolving similarly to the Λ CDM standard model of cosmology. Further, a standard Lemaître–Tolman–Bondi (LTB) model is derived analytically from Einstein’s field equations to examine the backreaction in a more realistic model. To evaluate this model, dynamical equations for both the areal radius, the density and the backreaction are derived, culminating in no backreaction in the standard LTB model. Another LTB model, the Tardis model[2], is therefore used to examine the backreaction in an LTB model. The backreaction in this model should be non-zero because of a non-monotonic areal radius function. The volume fraction and the theoretical backreaction term are considered to examine this. The volume fraction indicates significant backreaction, whereas the theoretical backreaction term leads to a small backreaction. To further examine this, the density parameters of the model are calculated, leading to an insignificant backreaction density parameter. Conclusively, the Tardis model has contradictory results concerning the backreaction.

Resumé

Dette bachelorprojekt undersøger inhomogen kosmologi og *cosmic backreaction*, og hvordan dette kan forklare den observerede acceleration af universets udvidelse. Først undersøges tre lokale ligninger: kontinuitetsligningen, Raychaudhuri-ligningen og den Hamiltonske begrænsning. Fra disse udledes Buchert-ligningerne[1] for at vise, hvordan backreaction kan opstå udfra inhomogenitet. Derefter udarbejdes en to-regions model for at få en intuition for, hvordan dette opstår, og hvordan en simpel, inhomogen model kan beskrive et univers, der udvikler sig sammenligneligt med den kosmologiske standardmodel, Λ CDM. For at undersøge backreaction i en mere realistisk model betragtes en standard Lemaître–Tolman–Bondi (LTB) model, og denne udledes analytisk fra Einsteins feltligninger. Modellen undersøges ved at udlede dynamiske ligninger for areal radius, densitet samt backreaction. Disse indikerer, at backreaction ikke opstår i en standard LTB model. Derfor betragtes endnu en LTB model, Tardis-modellen[2]. *Backreaction* forventes i denne model at være forskellig fra nul, da modellens areal radius er ikke-monoton. For at undersøge denne backreaction betragtes både udvidelsen af volumen-elementer samt det teoretiske udtryk for backreaction. Udvidelsen af volumen-elementer indikerer signifikant backreaction i Tardis modellen, hvorimod det teoretiske udtryk indikerer, at backreaction i modellen er neglegibelt. Dette undersøges yderligere ved at bestemme modellens densitetsparametre, hvilket resulterer i en insignifikant densitetsparameter for backreaction. Overordnet set stemmer resultaterne for backreaction i Tardis-modellen ikke overens.

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1 Introduction

In the beginning of the 20th century, Edwin Hubble observed that the Universe is expanding.[3] In the late 1990s this expansion was through observations found to be accelerating.[4][5] This accelerating expansion of the Universe has since puzzled cosmologists. In the standard cosmological model, referred to as Λ CDM, this is explained by introducing a cosmological constant Λ , also called dark energy. Cosmologists have however not yet explained what this dark energy is. One attempt tries to explain dark energy as the vacuum energy of empty space. However, calculations of this vacuum energy turn out to be 124 orders of magnitudes larger than the expected value for Λ .¹[6]

An alternative explanation for the accelerating expansion of the universe was introduced by Thomas Buchert in 2000.[1] The standard model of cosmology assumes the Universe to be homogeneous and isotropic. While this is true on large scales, the Universe is inhomogeneous at small- and intermediate scales, as it contains both dense- and underdense regions. Buchert took this inhomogeneity into account, suggesting that it could explain the accelerated expansion without dark energy. This property of inhomogeneity is called *cosmic backreaction*.

Buchert proposed a set of equations, known as the Buchert equations.[1] By deriving a set of local equations before taking the spatial average, these equations take inhomogeneity into account, contrary to what is done in the standard model of cosmology. The dynamical equations of the standard model describing the evolution of the Universe are called the Friedmann equations. In these equations, the average is taken before deriving the local equations.² The Buchert equations are similar to the Friedmann equations but an additional term called the *backreaction* appears. This term contains the variance of the *expansion rate* and the *shear*. A large variance of the expansion rate leads to an average accelerating expansion of the entire Universe, despite the decelerating expansion of local regions, and can therefore be used as an alternative or additional explanation of the observed accelerating expansion of the Universe.

This thesis examines cosmic backreaction by first diving into the theoretical background by explaining the principles of both Einstein’s theory of general relativity, and homogeneous- and inhomogeneous model universes. This is done in section 2. The Buchert equations are then derived from the local equations (the continuity equation, the Raychaudhuri equation and the Hamiltonian constraint) in section 3.

After this a simple two-region toy model is implemented, where inhomogeneity is examined by modelling an overdense and an underdense region, and seeing how they expand individually and on average. This can be seen in section 4.

A more complex and realistic model, a so-called Lemaître–Tolman–Bondi (LTB) model, is introduced by assuming an irrotational dust universe and solving Einstein’s field equations for the LTB metric. This is derived in section 5. In this section, a standard LTB model is simulated enabling a further examination of the cosmic backreaction.

Finally, another LTB model, the Tardis model[2], is examined. This is done similarly to the standard LTB model by deriving the dynamical equations of the Tardis model, which can be seen in section 6

The results are discussed in section 7 and a conclusion is made in section 8.

¹This has been dubbed the biggest error in all of science.[6]

²Averaging and solving for the local equations does not commute, since Einstein’s field equations are non-linear.[7]

2 Theoretical background

Before diving into inhomogeneous cosmology it is important to be familiar with general relativity and the standard cosmological model, and why alternative models are examined. These different models are all based on Einstein's theory of general relativity.[8]

2.1 Einstein's theory of general relativity

General relativity was first posed by Albert Einstein in 1915[9] and gave a new perspective on gravity and introduced spacetime. He applied this theory to model the Universe in 1917[10] and introduced the cosmological constant Λ in his field equations. Before understanding these, one has to be familiar with concepts such as tensors, the metric tensor, and comoving coordinates. These are defined together with other useful definitions and notations which are used throughout this thesis. Additionally, it should be noted that this thesis uses units in which $c = 1$.³

2.1.1 Tensors

Tensors are important mathematical objects which are used to describe different physical properties. The simplest form of a tensor is the rank zero tensor, which is a scalar, and then a rank one tensor, which is a vector. The number of indices will tell which rank the tensor has. When referring to tensors in general relativity one often means rank two tensors and higher. The reason tensors are an essential part of general relativity is that an equation containing the same type of tensors is basis invariant, thus remaining the same in all coordinate systems.[8]

In this thesis, tensors are notated using subscripts and superscripts. A tensor with lower indices is called *covariant*, one with upper indices is called *contravariant*, and one with both upper- and lower indices is simply called *mixed*. [11] For a rank two tensor, this could be $T_{\mu\nu}$ (covariant), $T^{\mu\nu}$ (contravariant) or T^μ_ν (mixed), which describe $n \times n$ matrices, and for each μ, ν one component of the tensor is described. As a common practice, the thesis will use Latin letters for spatial indices: $a, b, c, \dots$ which go from 1 to 3 and Greek letters for spacetime indices: $\alpha, \beta, \gamma, \dots$ which go from 0 to 3. Further, Einstein summation is used: [8]

$$\sum_{\alpha=0}^3 a_\alpha b^\alpha = a_0 b^0 + a_1 b^1 + a_2 b^2 + a_3 b^3 \equiv a_\alpha b^\alpha . \quad (2.1)$$

This rule is used when contracting indices, meaning that the same index will appear both as a superscript and a subscript. Some tensors are either symmetric or anti-symmetric in their indices or some indices. A symmetric tensor can be denoted with () around the indices in which the tensor is symmetric. It has the following trait:

$$T_{(\alpha\beta)} \equiv \frac{1}{2}(T_{\alpha\beta} + T_{\beta\alpha}) . \quad (2.2)$$

For tensors, which are symmetric in multiple components, all permutations of the indices will be added together and the constant in front will change accordingly. For anti-symmetric tensors, the principles are the same and can be denoted with [] and they have the property:

$$T_{[\alpha\beta]} \equiv \frac{1}{2}(T_{\alpha\beta} - T_{\beta\alpha}) . \quad (2.3)$$

³In the numerical calculations $c = 306.5926758$ Mpc/Gyr is added back in equations by dimensional analysis (see appendix A.2).

For a rank two tensor, the two definitions can be combined to describe any tensor which is neither symmetric nor anti-symmetric:

$$T_{(\alpha\beta)} + T_{[\alpha\beta]} = \frac{1}{2}(T_{\alpha\beta} + T_{\beta\alpha}) + \frac{1}{2}(T_{\alpha\beta} - T_{\beta\alpha}) = T_{\alpha\beta} . \quad (2.4)$$

Since tensors are physical properties it is essential that they can be differentiated. The derivative, ∇_μ maps a tensor to the tensor at the adjacent point on a manifold.⁴ For example, if the tensor depends on time, the tensor mapped to will be the tensor at the next time-step. The derivative has to satisfy different properties, leading to the covariant derivative which is defined as follows: [11]

$$\nabla_\beta u^\alpha \equiv \partial_\beta u^\alpha + \Gamma_{\beta\gamma}^\alpha u^\gamma , \quad (2.5)$$

where the partial derivative is the derivative in the direction of x^β and can also be written using “,”:

$$\partial_\beta u^\alpha = u^\alpha_{, \beta} = \frac{\partial}{\partial x^\beta} , \quad (2.6)$$

where $\Gamma_{\beta\gamma}^\alpha$ is called a Christoffel symbol and it is symmetric in its lower indices. These symbols are not tensors, since they do not transform like tensors.[11] The covariant derivative can be written in shorthand notation using “,”: [8]

$$\nabla_\beta u^\alpha = u^\alpha_{, \beta} = u^\alpha_{, \beta} + \Gamma_{\beta\gamma}^\alpha u^\gamma . \quad (2.7)$$

The Christoffel symbols are constructed from the metric tensor (see eq. 2.21). [12]

2.1.2 The metric tensor

The metric tensor is denoted with a g with two indices and is thus a rank two tensor. It is a tensor field which is defined everywhere on the spacetime manifold. The sign convention used in this thesis is $(-+++)$. In general relativity both the infinitesimal displacement of local coordinates dx^μ and the metric tensor $g_{(\mu\nu)}$ are used to define the line element: [12]

$$ds^2 = g_{(\mu\nu)} dx^\mu dx^\nu , \quad (2.8)$$

where ds^2 is called the proper distance. The metric is used to raise and lower indices by contraction:

$$x_\mu = g_{\mu\nu} x^\nu . \quad (2.9)$$

Further, the contravariant of the metric tensor is the inverse of the covariant metric tensor: [12]

$$g_{\mu\nu} g^{\mu\nu} = \delta^\mu_\mu = 4 . \quad (2.10)$$

2.1.3 Comoving coordinates

Another important concept is comoving coordinates, which are used to describe the spacetime manifold. The meaning of comoving is that the coordinates remain the same along a worldline.⁵ This is done by choosing a surface S which intersects all worldlines exactly once. These spatial coordinates are then called x^i . [8] The time component can be chosen as the proper time τ along the worldline, making the comoving local coordinates of the form of: [13]

$$x^\mu = (\tau, x^i) . \quad (2.11)$$

⁴Where a manifold is defined as a space which can be curved but in local areas look and act like an Euclidean space.[12]

⁵A worldline is the trajectory of a particle moving through spacetime. This trajectory will be curved in a curved manifold like the one in general relativity.[8]

The proper time is defined as the time in the centre of mass frame of the particle. This simplifies the equations by not changing the coordinates of each particle with time and the expansion of the universe, but instead remaining the same as time evolves. From the comoving coordinates, the four-velocity can be expressed as the proper time derivative of the comoving local coordinates: [13]

$$u^\mu = \frac{dx^\mu}{d\tau} , \quad (2.12)$$

where the difference between the covariant and contravariant of four-velocities is that they are each other's inverse, and there is a sign difference. This results in the normalisation:

$$u_\mu u^\mu = -1 . \quad (2.13)$$

The four-velocity is the tangent vector of the particle's worldline.[13]

2.1.4 Projection tensor, expansion tensor and shear tensor

To describe spacetime in three dimensions, holding time constant, a projection tensor is necessary. The definition of the projection tensor is: [8]

$$h_{\mu\nu} = g_{\mu\nu} + u_\mu u_\nu . \quad (2.14)$$

A specific projection tensor $h_{\mu\nu}$ describes a specific hypersurface which slices the four-dimensional spacetime into three dimensions. This is equivalent to describing all of space at a specific time. The covariant derivative projected into three dimensions is defined for a rank one tensor V_α as: [8]

$$\bar{\nabla}_\beta V_\alpha = h^\gamma_\alpha h^\delta_\beta \nabla_\delta V_\gamma , \quad (2.15)$$

which can be generalised for any tensor of any rank.

Using the projection tensor, the expansion tensor describing the expansion of spacetime is defined as:

$$\Theta_{\mu\nu} \equiv \sigma_{\mu\nu} + \frac{1}{3}\Theta h_{\mu\nu} , \quad (2.16)$$

where $\sigma_{\mu\nu}$ is the shear tensor, which is the projected symmetric tracefree part⁶, and the expansion rate Θ is the trace of $\Theta_{\mu\nu}$. The expansion tensor is simply the covariant derivative projected into three dimensions of the four-velocity:

$$\Theta_{\mu\nu} = \bar{\nabla}_{(\mu} u_{\nu)} , \quad (2.17)$$

making the expansion rate the trace of the covariant derivative of the four-velocity. It should be noted that the expansion tensor is symmetric in the lower indices, $\Theta_{\mu\nu} = \Theta_{(\mu\nu)}$. [8]

2.1.5 Einstein's field equations

Using the metric $g_{\mu\nu}$, the Einstein tensor $G_{\mu\nu}$ and the energy-momentum tensor $T_{\mu\nu}$ it is possible to write Einstein's field equations. These equations describe curvature on the left-hand side and how it is affected by the energy and momentum which is present on the right-hand side: [8]

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = 8\pi G T_{\mu\nu} , \quad (2.18)$$

where G is the Newtonian gravitational constant and Λ is the cosmological constant. The Einstein tensor is defined from the Ricci tensor $R_{\mu\nu}$ and the Ricci scalar $\mathcal{R}$:

$$G_{\mu\nu} \equiv R_{\mu\nu} - \frac{1}{2}\mathcal{R}g_{\mu\nu} . \quad (2.19)$$

⁶Shorthand notation is PSTF part.[8]

The Ricci tensor is a contracted version of the Riemann tensor and is defined as: [12]

$$R_{\mu\nu} = R^\alpha_{\mu\alpha\nu} = \partial_\alpha \Gamma^\alpha_{\nu\mu} - \partial_\nu \Gamma^\alpha_{\alpha\mu} + \Gamma^\alpha_{\alpha\lambda} \Gamma^\lambda_{\nu\mu} - \Gamma^\alpha_{\nu\lambda} \Gamma^\lambda_{\alpha\mu} . \quad (2.20)$$

The Riemann tensor is thus defined from the Christoffel symbols which are defined as:

$$\Gamma^\gamma_{\alpha\beta} \equiv \frac{1}{2} g^{\gamma\lambda} (\partial_\alpha g_{\lambda\beta} + \partial_\beta g_{\lambda\alpha} - \partial_\lambda g_{\alpha\beta}) . \quad (2.21)$$

The Riemann tensor is built from the Christoffel symbols and thus describes the curvature. Therefore, it is also sometimes called the curvature tensor.[12] To solve Einstein's field equations, the Christoffel symbols are first calculated by choosing a metric tensor. These are then used to compute the Riemann tensor. The Riemann tensor is then contracted to obtain the Ricci tensor which again can be contracted to find the Ricci scalar. This can be collected in the Einstein tensor which is then inserted into the field equations with the metric. The right-hand side contains the energy-momentum tensor which is determined by the considered components of the Universe.

In this gravitational theory matter and energy curve spacetime. This curvature tells matter how to move. This is in contrast to Newtonian physics where matter defines the strength of the gravitational field.

Einstein's field equations are only analytically solvable for simple cases and a solution was first found by Karl Schwarzschild in 1916.[14] This solution is known as the Schwarzschild metric and describes black holes. Another known solution is one assuming a homogeneous and isotropic universe. This solution is discussed in the following section.

2.2 Homogeneous universes

Alexander Friedmann derived a solution[15] to Einstein's field equations (eq. 2.18), describing a dynamical cosmological model. This was in contrast to what Einstein originally suggested, namely that the universe was static. Friedmann's model was further backed up by Edwin Hubble's 1929 discovery that far-away galaxies are receding from us[3], which had already been predicted by Georges Lemaître in 1927[16]. One explanation of Hubble's discovery was the Friedmann–Lemaître–Robertson–Walker model, which imposed a Big Bang as the start of the universe. This theory was further backed up by the discovery of the cosmic microwave background (CMB) in 1965.[17] The CMB is in thermal equilibrium and must thus once all have been connected for this equilibrium to exist, making the Big Bang a possible explanation of the start of the Universe.[6] This model, the Λ CDM model, is still considered the standard cosmological model, where Λ refers to dark energy, and CDM refers to cold dark matter. Cold dark matter is matter that does not interact electromagnetically nor moves close to the speed of light.[6] This model is based on the assumption of the Universe being both homogeneous and isotopic meaning no direction nor location is preferred. The standard model and observations⁷ results in a model of the Universe containing both dark energy and dark matter besides the visible matter. According to the Benchmark model, dark energy is 68% of the energy density of the Universe, dark matter is 27%, and visible matter is only 5%.[6]

2.2.1 Accelerated expansion of homogeneous universes

The expansion of the universe was first confirmed by Edwin Hubble's observations of the redshift of galaxies.[6] Through further observations of supernovae, the expansion of the universe was found to be

⁷The observations mentioned are observations supporting the acceleration of the Universe and the presence of dark matter; Hubble's law, the CMB, the Bullet Cluster, galaxy rotation curves, velocities of galaxies in galaxy clusters and simulations on structure development.[6][18]

accelerating.[3] In standard cosmology, this acceleration is explained by dark energy which is described by the cosmological constant Λ in Einstein's field equations. Considering the Λ CDM model the only component of the Universe, which leads to an accelerated, positive expansion, is the cosmological constant. Thus, in this model, dark energy is crucial to describe the accelerated expansion of the Universe that we observe.

2.2.2 Friedmann–Lemaître–Robertson–Walker universe

The Friedmann–Lemaître–Robertson–Walker (FLRW) universe is described by the line element of the FLRW-metric: [6]

$$ds^2 = -dt^2 + a(t)^2 \left[\frac{dr^2}{1 - \kappa r^2/R_0^2} + r^2 d\Omega^2 \right], \quad (2.22)$$

where the angular differential $d\Omega^2$ is defined as: [6]

$$d\Omega^2 = d\theta^2 + \sin^2 \theta d\phi^2. \quad (2.23)$$

The scale factor $a(t)$ describes the expansion of a line element in the FLRW universe. Thus considering a spatial geodesic, that is a spatial hypersurface at constant time (thus $dt^2 = 0$), only the radial component of ds^2 remains, as the angular components θ and ϕ are constant such that $d\Omega^2 = 0$. Equation 2.22 therefore reduces to: [6]

$$ds = a(t)dr. \quad (2.24)$$

Integrating both sides leads to the following relation between the proper distance d_p and the comoving distance r : [6]

$$d_p(t) = a(t)r, \quad (2.25)$$

where the comoving distance r describes the radial distance in comoving coordinates. This therefore remains constant as the comoving coordinates scale with the expansion of the Universe. The proper distance $d_p(t)$ describes the physical distance between two points and depends on time as this scales with the expansion of the Universe. This is done by taking the scale factor $a(t)$ at time t into account. At present time t_0 , the scale factor is set to equal one, meaning that proper distance equals the comoving distance $d_p(t_0) = r$.

Further, in the FLRW-metric (eq. 2.22), R_0 denotes the radius of curvature and κ denotes the curvature constant of the universe. This can take the values $\kappa = -1$ for a negatively curved (open) universe, $\kappa = 0$ for a flat universe and $\kappa = 1$ when considering a positively curved (closed) universe.[8]

The FLRW model assumes the universe to be statistically isotropic and homogeneous. Solving Einstein's field equations (eq. 2.18), when assuming the FLRW universe to contain perfect fluid⁸, leads to a set of equations called the Friedmann equations. These describe the dynamics of the FLRW model by the Friedmann equation: [6]

$$H(t)^2 = \left(\frac{\dot{a}}{a} \right)^2 = \frac{8\pi G}{3} \rho - \frac{\kappa}{a^2} + \frac{\Lambda}{3}, \quad (2.26)$$

and the acceleration equation: [6]

$$\frac{\ddot{a}}{a} = -\frac{4\pi G}{3}(\rho + 3p) + \frac{\Lambda}{3}, \quad (2.27)$$

⁸A perfect fluid is characterized by having no viscosity, a constant density and pressure, thus ensuring homogeneity and isotropy.[8]

where $H(t) = \frac{\dot{a}}{a}$ is the Hubble parameter, which describes the expansion rate of the universe at time t . In equation 2.26 and 2.27, the energy density is denoted ρ and depends on the components of the universe considered e.g. matter and radiation. Further, p is the pressure. When considering a dust universe, it is assumed to contain perfect fluid, that is assumed to be pressureless $p = 0$. The energy-momentum tensor for dust cosmology therefore reduces to: [13]

$$T_{\mu\nu} = \rho_0 u_\mu u_\nu , \quad (2.28)$$

where ρ_0 is the density of the dust, and u_μ, u_ν describe the four-velocities of the dust. In comoving coordinates this is: $u^\lambda = (1, 0, 0, 0)$.

The acceleration equation 2.27 shows that only the cosmological constant can lead to a positively, accelerated expansion rate in standard cosmology when considering dust cosmology or discarding the possibility of components resulting in negative pressure.

Further, to describe the different components of the Universe and when they dominate, the *density parameter* is introduced: [12]

$$\Omega \equiv \frac{8\pi G}{3H^2} \rho = \frac{\rho}{\rho_{\text{crit}}} , \quad (2.29)$$

with the critical density defined as $\rho_{\text{crit}} \equiv \frac{3H^2}{8\pi G}$. The density parameter can also be used to rewrite the Friedmann equation 2.26: [12]

$$\Omega - 1 = \frac{\kappa}{H^2 a^2} , \quad (2.30)$$

where Ω is the sum of all the components' density parameters. From this, the definition of the curvature constant κ can be used to determine whether the Universe is open ($\Omega < 1 \Leftrightarrow \kappa < 0$), flat ($\Omega = 1 \Leftrightarrow \kappa = 0$) or closed ($\Omega > 1 \Leftrightarrow \kappa > 0$). [12] Additionally, the Friedmann equation can alternatively be expressed by dividing with H_0^2 and the dependency on the scale factor of the different components' energy density: [6]

$$\frac{H^2}{H_0^2} = \frac{\dot{a}^2}{H_0^2 a^2} = \frac{\Omega_{r,0}}{a^4} + \frac{\Omega_{m,0}}{a^3} + \Omega_{\Lambda,0} + \frac{1 - \Omega_0}{a^2} , \quad (2.31)$$

where $\Omega_{r,0}$ is the density of radiation at present time t_0 , correspondingly this applies for matter $\Omega_{m,0}$, and the cosmological constant $\Omega_{\Lambda,0}$. The Benchmark model of the Universe has the total density parameter $\Omega_0 = 1$, indicating a flat universe. The Benchmark model further has the following density parameters of the Universe $\Omega_{m,0} = 0.30$, $\Omega_{r,0} = 8.4 \cdot 10^{-5}$ and $\Omega_{\Lambda,0} = 0.70$. [6]

2.2.3 Einstein–de Sitter universe

A simple case of an FLWR universe is the Einstein–de Sitter universe. This is a flat universe, with the curvature constant $\kappa = 0$, only containing non-relativistic matter, and thus no radiation nor a cosmological constant ($\Lambda = 0$). The scale factor of the Einstein–de Sitter universe depends on time as: [8]

$$a(t) = \left(\frac{t}{t_0} \right)^{\frac{2}{3}} , \quad (2.32)$$

where t_0 is the age of the universe at present time in this specific model: [6]

$$t_0 = \frac{2}{3} \frac{1}{H_0} , \quad (2.33)$$

with H_0 being the Hubble parameter at present time. The scale factor relates the proper distance $d_p(t, r)$ and the comoving distance r (eq. 2.25):

$$d_p(t, r) = a(t)r . \quad (2.34)$$

In the case of an Einstein–de Sitter universe, the Friedmann equation (eq. 2.26) reduces to:

$$\left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G}{3}\rho, \quad (2.35)$$

where the scale factor $a(t)$ and its derivative $\dot{a}(t)$ by equation 2.32 depend on time as:

$$a(t) = \left(\frac{t}{t_0}\right)^{2/3} \Rightarrow \dot{a}(t) = \frac{2}{3} \frac{t^{-1/3}}{t_0^{2/3}} \quad (2.36)$$

The Hubble parameter of the Einstein–de Sitter universe is therefore:

$$H(t) = \left(\frac{\dot{a}}{a}\right) = \frac{2}{3} \frac{t^{-1/3}}{t_0^{2/3}} = \frac{2}{3} \frac{1}{t} \Rightarrow H(t)^2 = \left(\frac{2}{3} \frac{1}{t}\right)^2 = \frac{4}{9} \frac{1}{t^2}. \quad (2.37)$$

Inserting the Hubble parameter in the Friedmann equation (eq. 2.35) then leads to:

$$\frac{4}{9} \frac{1}{t^2} = \frac{8\pi G}{3}\rho. \quad (2.38)$$

Thus isolating the density ρ leads to the density of the Einstein–de Sitter universe. Denoting $\rho = \rho_{EdS}$, this is:

$$\rho_{EdS}(t) = \frac{1}{6\pi G} \frac{1}{t^2}. \quad (2.39)$$

2.3 Inhomogeneous universes

The accelerated expansion of the Universe is explained by the introduction of a cosmological constant in section 2.2.1. This explanation is under the assumption that the Universe can be approximated as both statistically isotropic and homogeneous. Thus, assuming that the inhomogeneity at small- and intermediate scales does not affect the dynamical evolution of the Universe. The density distribution of the Universe is however not homogeneous on small- and intermediate scales.

2.3.1 Acceleration due to inhomogeneity

Another explanation of the accelerated expansion of the Universe than the cosmological constant is therefore considered. One possible explanation is cosmic backreaction, introduced by Thomas Buchert, suggesting that backreaction can occur due to small and intermediate-scale inhomogeneity of the Universe.[1] This takes inhomogeneity into account by first deriving the large-scale dynamical equations for the Universe before taking the spatial average of these. This is in contrast to the Friedmann equations (eq. 2.26 and eq. 2.27). In these equations the metric tensor $g_{\mu\nu}$ and the energy-momentum tensor $T_{\mu\nu}$ are averaged before deriving the dynamical equations.

2.3.2 The local equations

To derive the averaged, large-scale dynamical equations, the Buchert equations[1], for an inhomogeneous dust universe, the non-averaged local equations are to be considered. Assuming an irrotational dust universe the three non-averaged local equations are obtained. The first is the continuity equation: [19][7]

$$\dot{\rho} = -\Theta\rho, \quad (2.40)$$

where ρ is the density and Θ is the local volume expansion rate, which is defined as the trace of the expansion tensor Θ^μ_ν :

$$\Theta = \Theta^\mu_\mu. \quad (2.41)$$

The continuity equation describes the conservation of mass and energy.

The second equation is the Raychaudhuri equation: [7]

$$0 = \dot{\Theta} + \frac{1}{3}\Theta^2 + 2\sigma^2 + 4\pi G\rho - \Lambda , \quad (2.42)$$

where $\dot{\Theta}$ denotes the time-derivative of the local expansion rate. The Raychaudhuri equation depends on the shear squared σ^2 , which is defined as: [1]

$$\sigma^2 = \frac{1}{2}\sigma^\mu_\nu\sigma^\nu_\mu , \quad (2.43)$$

where the shear tensor σ^μ_ν is defined as the trace-free part of the symmetric expansion tensor (eq. 2.16).

The third non-averaged equation is the Hamiltonian constraint: [7]

$$^{(3)}\mathcal{R} = 2\left(\sigma^2 - \frac{1}{3}\Theta^2 + \Lambda + 8\pi\rho G\right) , \quad (2.44)$$

where $^{(3)}\mathcal{R}$ describes the spatial curvature of spacetime.

2.3.3 Spatial average of scalar fields

Before deriving the Buchert equations, it is of interest to determine the general spatial average of an arbitrary scalar field. Therefore a volume element of a given hyperspatial surface is considered. For the given metric $g_{\mu\nu}$, this is: [1]

$$dV = \sqrt{\det(^{(3)}g_{\mu\nu})}d^3X = Jd^3X , \quad (2.45)$$

where d^3X denotes the spatial differential, and the integrand is denoted $\sqrt{\det(^{(3)}g_{\mu\nu})} = J$. The total volume of the domain $\mathcal{D}$ is then found by integration:

$$V_{\mathcal{D}} = \int_{\mathcal{D}} \sqrt{\det(^{(3)}g_{\mu\nu})}d^3X = \int_{\mathcal{D}} Jd^3X . \quad (2.46)$$

Thus, the normalised spatial average of any scalar field Ψ over the domain $\mathcal{D}$ is: [1]

$$\langle\Psi\rangle_{\mathcal{D}} = \frac{1}{V_{\mathcal{D}}} \int_{\mathcal{D}} \sqrt{\det(^{(3)}g_{\mu\nu})}\Psi d^3X . \quad (2.47)$$

It is of interest to determine the spatial average of the expansion rate $\langle\Theta\rangle_{\mathcal{D}}$. First the time derivative of the volume $V_{\mathcal{D}}$ is determined:

$$\dot{V}_{\mathcal{D}} = \int_{\mathcal{D}} \frac{dJ}{dt}d^3\mathbf{x} = \int_{\mathcal{D}} \dot{J}d^3\mathbf{x} , \quad (2.48)$$

where $\dot{J} = \Theta J$ [1] which leads to:

$$\frac{\dot{V}_{\mathcal{D}}}{V_{\mathcal{D}}} = \frac{1}{V_{\mathcal{D}}} \int_{\mathcal{D}} \Theta Jd^3\mathbf{x} = \langle\Theta\rangle_{\mathcal{D}} , \quad (2.49)$$

where both sides are divided by $V_{\mathcal{D}}$ and the spatial average of Θ is recognized from equation 2.47.

The scale factor of the domain $\mathcal{D}$ depends on time and can be expressed in terms of the volume $V_{\mathcal{D}}$:

$$a_{\mathcal{D}} = \left(\frac{V_{\mathcal{D}}}{V_{\mathcal{D},0}}\right)^{1/3} , \quad (2.50)$$

where $V_{\mathcal{D},0}$ denotes the volume of the domain $\mathcal{D}$ at present time. The time derivative is computed by applying both the product rule and the chain rule:

$$\dot{a}_{\mathcal{D}} = \frac{1}{3} \frac{V_{\mathcal{D}}^{-2/3}}{V_{\mathcal{D},0}^{1/3}} \dot{V}_{\mathcal{D}} . \quad (2.51)$$

Solving for the term $\frac{\dot{V}_{\mathcal{D}}}{V_{\mathcal{D}}}$:

$$\frac{\dot{V}_{\mathcal{D}}}{V_{\mathcal{D}}} = 3\dot{a}_{\mathcal{D}} V_{\mathcal{D},0}^{1/3} \frac{V_{\mathcal{D}}^{2/3}}{V_{\mathcal{D}}} = 3\dot{a}_{\mathcal{D}} \left(\frac{V_{\mathcal{D},0}}{V_{\mathcal{D}}} \right)^{1/3} = 3 \frac{\dot{a}_{\mathcal{D}}}{a_{\mathcal{D}}} . \quad (2.52)$$

Thus, the spatial average of the expansion rate $\langle \Theta \rangle_{\mathcal{D}}$ in terms of both $V_{\mathcal{D}}$ and $a_{\mathcal{D}}$ is:

$$\langle \Theta \rangle_{\mathcal{D}} = \frac{\dot{V}_{\mathcal{D}}}{V_{\mathcal{D}}} = 3 \frac{\dot{a}_{\mathcal{D}}}{a_{\mathcal{D}}} . \quad (2.53)$$

The time derivative of any averaged scalar field (eq. 2.47), denoted $\langle \Psi \rangle_{\mathcal{D}}^{\bullet}$ is found using the product rule:

$$\begin{aligned} \langle \Psi \rangle_{\mathcal{D}}^{\bullet} &= \frac{d}{dt} \left(\frac{1}{V_{\mathcal{D}}} \right) \int_{\mathcal{D}} J \Psi d^3 X + \left(\frac{1}{V_{\mathcal{D}}} \right) \int_{\mathcal{D}} \frac{d}{dt} (J) \Psi d^3 X + \left(\frac{1}{V_{\mathcal{D}}} \right) \int_{\mathcal{D}} J \frac{d}{dt} (\Psi) d^3 X \\ &= -\frac{\dot{V}_{\mathcal{D}}}{V_{\mathcal{D}}^2} \int_{\mathcal{D}} J \Psi d^3 X + \frac{1}{V_{\mathcal{D}}} \int_{\mathcal{D}} \Theta J \Psi d^3 X + \frac{1}{V_{\mathcal{D}}} \int_{\mathcal{D}} J \dot{\Psi} d^3 X \\ &= -\langle \Psi \rangle_{\mathcal{D}} \langle \Theta \rangle_{\mathcal{D}} + \langle \Theta \Psi \rangle_{\mathcal{D}} + \langle \dot{\Psi} \rangle_{\mathcal{D}} , \end{aligned} \quad (2.54)$$

where $\dot{J} = \Theta J$ and the definition of the average expansion rate is used. Further, rearranging leads to the commutator rule for scalar fields:

$$\langle \Psi \rangle_{\mathcal{D}}^{\bullet} - \langle \dot{\Psi} \rangle_{\mathcal{D}} = \langle \Theta \Psi \rangle_{\mathcal{D}} - \langle \Psi \rangle_{\mathcal{D}} \langle \Theta \rangle_{\mathcal{D}} . \quad (2.55)$$

3 The Buchert equations

It is then possible to derive the Buchert equations[1], that is the averaged large-scale dynamical equations found without assuming homogeneity and isotropy.[20] Thus when assuming an inhomogeneous dust universe, where the dust is assumed to be irrotational⁹, the Buchert equations can be derived by taking the spatial average of the continuity equation (eq. 2.40), the Raychaudhuri equation (eq. 2.42) and the Hamiltonian constraint (eq. 2.44) respectively.

3.1 Averaging the continuity equation

Considering the non-averaged continuity equation (eq. 2.40):

$$\dot{\rho} = -\Theta \rho . \quad (3.1)$$

Averaging both sides of the equation leads to:¹⁰

$$\langle \dot{\rho} \rangle = -\langle \Theta \rho \rangle . \quad (3.2)$$

The commutator relation derived above (eq. 2.55) is inserted on the left-hand side:

$$\langle \rho \rangle^{\bullet} - \langle \Theta \rho \rangle + \langle \rho \rangle \langle \Theta \rangle = -\langle \Theta \rho \rangle . \quad (3.3)$$

Thus the averaged continuity equation is:

$$\langle \rho \rangle^{\bullet} = -\langle \rho \rangle \langle \Theta \rangle . \quad (3.4)$$

The averaged continuity equation relates the average density to the average expansion rate of the domain $\mathcal{D}$. The time derivative of the density is seen to be negative. Thus for the average density to increase, the average expansion has to be negative. The average density on the other hand decreases when the average expansion is positive.

⁹This assumption refers to a pressureless, vorticity-free fluid

¹⁰From here on, the $\mathcal{D}$ is dropped from the averages since all averages are over this domain. It is, however, kept when denoting the scale factor $a_{\mathcal{D}}$ to avoid confusing it with the scale factor a in the Einstein–de Sitter universe.

3.2 Averaging the Raychaudhuri equation

Considering the non-averaged Raychaudhuri equation (eq. 2.42):

$$\dot{\Theta} + \frac{1}{3}\Theta^2 + 2\sigma^2 + 4\pi G\rho - \Lambda = 0. \quad (3.5)$$

Both sides of the equation are averaged using the relation $\langle a + b \rangle = \langle a \rangle + \langle b \rangle$:

$$\langle \dot{\Theta} \rangle + \frac{1}{3}\langle \Theta^2 \rangle + 2\langle \sigma^2 \rangle + 4\pi G\langle \rho \rangle - \Lambda = 0. \quad (3.6)$$

The expansion rate Θ is a scalar field and therefore the averaged time-derivative $\langle \dot{\Theta} \rangle$ is found using the commutator relation (eq. 2.55):

$$\langle \dot{\Theta} \rangle = \langle \Theta \rangle^\bullet - \langle \Theta \Theta \rangle + \langle \Theta \rangle \langle \Theta \rangle \quad (3.7)$$

$$= \langle \Theta \rangle^\bullet - \langle \Theta^2 \rangle + \langle \Theta \rangle^2. \quad (3.8)$$

Substituting $\langle \dot{\Theta} \rangle$ back leads to:

$$0 = \langle \Theta \rangle^\bullet - \langle \Theta^2 \rangle + \langle \Theta \rangle^2 + \frac{1}{3}\langle \Theta^2 \rangle + 2\langle \sigma^2 \rangle + 4\pi G\langle \rho \rangle - \Lambda \quad (3.9)$$

$$= \langle \Theta \rangle^\bullet - \frac{2}{3}\langle \Theta^2 \rangle + \langle \Theta \rangle^2 + 2\langle \sigma^2 \rangle + 4\pi G\langle \rho \rangle - \Lambda. \quad (3.10)$$

The average of the expansion rate is given by equation 2.53. The time derivative of this is therefore found by applying the product rule:

$$\langle \Theta \rangle^\bullet = 3 \frac{\partial}{\partial t} \left(\frac{\dot{a}_{\mathcal{D}}}{a_{\mathcal{D}}} \right) \quad (3.11)$$

$$\begin{aligned} &= 3 \frac{\partial}{\partial t} \left(\frac{\partial a_{\mathcal{D}}}{\partial t} \frac{1}{a_{\mathcal{D}}} \right) \\ &= 3 \left[\frac{\partial^2 a_{\mathcal{D}}}{\partial t^2} \frac{1}{a_{\mathcal{D}}} + \frac{\partial a_{\mathcal{D}}}{\partial t} \frac{\partial}{\partial t} \left(\frac{1}{a_{\mathcal{D}}} \right) \right] \\ &= 3 \left[\frac{\ddot{a}_{\mathcal{D}}}{a_{\mathcal{D}}} + \dot{a}_{\mathcal{D}} \left(-\frac{1}{a_{\mathcal{D}}^2} \dot{a}_{\mathcal{D}} \right) \right] \\ &= 3 \left[\frac{\ddot{a}_{\mathcal{D}}}{a_{\mathcal{D}}} - \left(\frac{\dot{a}_{\mathcal{D}}}{a_{\mathcal{D}}} \right)^2 \right]. \end{aligned} \quad (3.12)$$

Recognizing the term $\frac{\dot{a}_{\mathcal{D}}}{a_{\mathcal{D}}} = \frac{1}{3}\langle \Theta \rangle$ leads to:

$$\langle \Theta \rangle^\bullet = 3 \left(\frac{\ddot{a}_{\mathcal{D}}}{a_{\mathcal{D}}} - \frac{1}{9}\langle \Theta \rangle^2 \right) = 3 \frac{\ddot{a}_{\mathcal{D}}}{a_{\mathcal{D}}} - \frac{1}{3}\langle \Theta \rangle^2. \quad (3.13)$$

The term $\langle \Theta \rangle^\bullet$ is then substituted back:

$$0 = 3 \frac{\ddot{a}_{\mathcal{D}}}{a_{\mathcal{D}}} - \frac{1}{3}\langle \Theta \rangle^2 - \frac{2}{3}\langle \Theta^2 \rangle + \langle \Theta \rangle^2 + 2\langle \sigma^2 \rangle + 4\pi G\langle \rho \rangle - \Lambda \quad (3.14)$$

$$= 3 \frac{\ddot{a}_{\mathcal{D}}}{a_{\mathcal{D}}} + \frac{2}{3}(\langle \Theta \rangle^2 - \langle \Theta^2 \rangle) + 2\langle \sigma^2 \rangle + 4\pi G\langle \rho \rangle - \Lambda. \quad (3.15)$$

The variance of the expansion rate Θ is defined as $\text{Var}(\Theta) \equiv \langle \Theta^2 \rangle - \langle \Theta \rangle^2$:

$$3 \frac{\ddot{a}_{\mathcal{D}}}{a_{\mathcal{D}}} = \frac{2}{3}\text{Var}(\Theta) - 2\langle \sigma^2 \rangle - 4\pi G\langle \rho \rangle + \Lambda. \quad (3.16)$$

Thus the averaged Raychaudhuri equation is:

$$3 \frac{\ddot{a}_{\mathcal{D}}}{a_{\mathcal{D}}} = -4\pi G\langle \rho \rangle + \Lambda + Q, \quad (3.17)$$

where the backreaction term Q is introduced and defined as:

$$Q \equiv \frac{2}{3}\text{Var}(\Theta) - 2\langle \sigma^2 \rangle. \quad (3.18)$$

The averaged Raychaudhuri equation is analogous to the acceleration equation (eq. 2.27).[7] This equation introduces a second possibility to get accelerated expansion other than dark energy, namely *backreaction*.

3.3 Averaging the Hamiltonian constraint

Considering the non-averaged Hamiltonian constraint (eq. 2.44):

$${}^{(3)}\mathcal{R} = 2 \left(\sigma^2 - \frac{1}{3} \Theta^2 + \Lambda + 8\pi\rho G \right). \quad (3.19)$$

Both sides of the equation are averaged using the relation $\langle a + b \rangle = \langle a \rangle + \langle b \rangle$ to evaluate each term of the sum:

$$\langle {}^{(3)}\mathcal{R} \rangle = 2 \left(\langle \sigma^2 \rangle - \frac{1}{3} \langle \Theta^2 \rangle + \Lambda + 8\pi \langle \rho \rangle G \right). \quad (3.20)$$

This can be rewritten using the definition of cosmic backreaction Q (eq. 3.18)

$$2\langle \sigma^2 \rangle = -Q + \frac{2}{3} (\langle \Theta^2 \rangle - \langle \Theta \rangle^2). \quad (3.21)$$

Inserting this leads to:

$$\langle {}^{(3)}\mathcal{R} \rangle = -Q + \frac{2}{3} (\langle \Theta^2 \rangle - \langle \Theta \rangle^2) - \frac{2}{3} \langle \Theta^2 \rangle + 2\Lambda + 16\pi \langle \rho \rangle G. \quad (3.22)$$

This is simplified by cancelling out $\frac{2}{3} \langle \Theta^2 \rangle$ and dividing by 2 on both sides:

$$\frac{1}{2} \langle {}^{(3)}\mathcal{R} \rangle = -\frac{1}{2} Q - \frac{1}{3} \langle \Theta \rangle^2 + \Lambda + 8\pi \langle \rho \rangle G. \quad (3.23)$$

Thus obtaining the averaged Hamiltonian constraint by recognizing $\langle \Theta \rangle^2 = \left(\frac{\dot{a}_{\mathcal{D}}}{a_{\mathcal{D}}} \right)^2 = H_{\mathcal{D}}^2$:

$$3 \left(\frac{\dot{a}_{\mathcal{D}}}{a_{\mathcal{D}}} \right)^2 = 3H_{\mathcal{D}}^2 = -\frac{1}{2} \langle {}^{(3)}\mathcal{R} \rangle - \frac{Q}{2} + \Lambda + 8\pi G \langle \rho \rangle, \quad (3.24)$$

which is analogous to the Friedmann equation (eq. 2.26).

3.4 Acceleration due to inhomogeneity revisited

In the case of no cosmological constant ($\Lambda = 0$), the Buchert equations; the continuity equation (eq. 2.40), the Raychaudhuri equation (eq. 2.42) and the Hamiltonian constraint (eq. 2.44), reduce to:

$$\langle \rho \rangle^\bullet = -\langle \rho \rangle \langle \Theta \rangle \quad (3.25)$$

$$3 \frac{\ddot{a}_{\mathcal{D}}}{a_{\mathcal{D}}} = -4\pi G \langle \rho \rangle + Q \quad (3.26)$$

$$3 \left(\frac{\dot{a}_{\mathcal{D}}}{a_{\mathcal{D}}} \right)^2 = -\frac{1}{2} \langle {}^{(3)}\mathcal{R} \rangle - \frac{Q}{2} + 8\pi G \langle \rho \rangle, \quad (3.27)$$

where the backreaction term Q (eq. 3.18) arising due to inhomogeneity is found to depend on both the variance of the expansion rate Θ and the averaged shear squared σ^2 :

$$Q \equiv \frac{2}{3} \text{Var}(\Theta) - 2\langle \sigma^2 \rangle. \quad (3.28)$$

The averaged Raychaudhuri equation (eq. 3.26) shows that the only contribution to the positively accelerated expansion is the variance of the expansion rate. Thus the backreaction term, occurring in the Buchert equations due to inhomogeneity, allows the average expansion to be accelerating (eq. 3.26) despite the expansion of local volume elements being decelerating due to the shear. This requires the variance of the expansion rate in the backreaction term Q to be large compared to negative terms in both equation 3.26 and equation 3.18, that is the mass density and the shear respectively. This requires a spatial inhomogeneous expansion.[7][13]

Additionally, the dynamics of the model can be expressed similarly to the Friedmann model by density parameters (eq. 2.31). To obtain the parameters equation 3.27 is divided by $3H_{\mathcal{D}}^2$: [1]

$$1 = \Omega_m + \Omega_{\mathcal{R}} + \Omega_Q , \quad (3.29)$$

with the definitions:

$$\Omega_m \equiv \frac{8\pi G}{3H_{\mathcal{D}}^2} \langle \rho \rangle_{\mathcal{D}} \quad (3.30)$$

$$\Omega_{\mathcal{R}} \equiv -\frac{\langle {}^{(3)}\mathcal{R} \rangle_{\mathcal{D}}}{6H_{\mathcal{D}}^2} \quad (3.31)$$

$$\Omega_Q \equiv -\frac{Q_{\mathcal{D}}}{6H_{\mathcal{D}}^2} . \quad (3.32)$$

4 Two-region model

4.1 Introduction to the two-region model

To further examine the backreaction occurring in the Buchert equations due to inhomogeneity a two-region toy model representing a simple, inhomogeneous dust universe with two distinct regions is constructed. The regions are assumed to be spherically symmetric with one representing a void and one representing a collapsing structure. Compared to a critical density, the first region is underdense (representing a void) and the second region is overdense (representing a collapsing structure), each region can be described by an independent FLRW universe.[7]

To describe the model a *deceleration parameter* is introduced for each of the two regions q_1 and q_2 . The deceleration parameter generally denotes the dynamics of the system and whether the expansion is accelerating, decelerating or neither. This is defined as: [7]

$$q_i \equiv -\frac{\ddot{a}_i a_i}{\dot{a}_i^2} , \quad (4.1)$$

where a_i denotes the scale factor of the i 'th region, thus a_1 and a_2 for the underdense- and overdense region respectively.

The underdense region can be described by an empty, negatively curved FLRW universe.[7] Since $\kappa = -1$ and $\rho = 0$, the Friedmann equation (eq. 2.26) for this region reduces to:

$$\left(\frac{\dot{a}_1}{a_1} \right)^2 = \frac{1}{a_1^2} . \quad (4.2)$$

Taking the square root and multiplying with a_1 on both sides, then integrating both sides with respect to time t :

$$\int \dot{a}_1 dt = \int 1 dt . \quad (4.3)$$

Thus, assuming the initial scale factor to equal zero, the scale factor a_1 of the underdense region is proportional to time: [6]

$$a_1 \propto t . \quad (4.4)$$

From this it follows that the deceleration parameter of the region must be zero: $q_1 = 0$ since $\ddot{a}_1 = 0$.

The overdense region can be described by a positively curved, matter-dominated FLRW universe.[7] The scale parameter of this region is:¹¹ [6]

$$a_2 \propto 1 - \cos(\varphi) , \quad (4.5)$$

¹¹The proportionality constant of this equation is also given in [6], and is given by $\Omega_0/(2\Omega_0 - 2)$ when $\Omega_0 > 1$, thus depending on the density parameter at $t = t_0$

where φ is the *development angle* related to time: $t \propto \varphi - \sin(\varphi)$. The development angle is introduced to make the calculations easier to do, as using the scale parameter a would require solving differential equations, since the time derivative $\dot{a}$ and double time derivative $\ddot{a}$ of a are needed at all times.

The deceleration parameter of the overdense region is (see A.1.2): [7]

$$q_2 = \frac{1 - \cos(\varphi)}{\sin^2(\varphi)} . \quad (4.6)$$

This means that locally, only deceleration can happen in the overdense region. Physically, the expansion rate of this region slows down until the development angle reaches $\varphi = \pi$, after which the expansion stops and the region starts collapsing. The collapse continues until the development angle reaches $\varphi = 2\pi$, where the density of the region diverges towards infinity¹². The two-region model considers the development angle in the range $\varphi = (0, 2\pi)$ as done in [7].

A free parameter in this model is the volume fraction f_2 , which determines the volume fraction of region 2 (the overdense region) at the turnover point $\varphi = \pi$. Varying this will change how quickly the void will begin to dominate the volume, and is thus a free parameter that can be changed to obtain the best fit to observed data.

4.1.1 Hubble parameters in the two-region model

The Hubble parameters for each of the two regions are denoted H_1 and H_2 , and depend on the scale factors of the corresponding regions respectively:

$$H_1 = \frac{\dot{a}_1}{a_1} \quad (4.7)$$

$$H_2 = \frac{\dot{a}_2}{a_2} . \quad (4.8)$$

The Hubble parameter describing the entire two-region model is defined as: [7]

$$H \equiv \frac{\dot{a}}{a} , \quad (4.9)$$

where a is the total scale parameter, which correspondingly describes the entire two-region model. This can be expressed as the sum of the volumes: [7]

$$a^3 = a_1^3 + a_2^3 \Rightarrow a = (a_1^3 + a_2^3)^{1/3} . \quad (4.10)$$

The above expression for the Hubble parameter (eq. 4.9) can be rewritten by introducing the following parameters using the definitions of t , a and $\dot{a}$: [7]

$$v = \frac{a_2^3}{a_1^3 + a_2^3} = \frac{a_2^3}{a^3} = \frac{(1 - \cos \varphi)^3}{(\varphi - \sin \varphi)^3 + (1 - \cos \varphi)^3} \quad (4.11)$$

$$h = \frac{H_2}{H_1} = \frac{\sin \varphi (\varphi - \sin \varphi)}{1 - \cos \varphi} . \quad (4.12)$$

Therefore H for the two-region model can be expressed as:

$$H = H_1(1 - v + vh) . \quad (4.13)$$

The derivation of this expression can be seen in appendix A.1.1.

¹²Physically, the region will stop collapsing earlier, ending with a finite density. This stop is often set to be at $3\pi/2$. [7] Thus, the calculated values between $3\pi/2$ and 2π are mathematical, not physical properties of the two-region model.

It is further possible to rewrite the above expression for the parameter v , by normalising it with respect to the turnover point $\varphi = \pi$, since only proportionalities for the scale factors a_1 and a_2 are considered. Therefore, the value of each of the scale factors cubed at $\varphi = \pi$ are computed:

$$a_1(\pi)^3 \propto (1 - \cos \pi)^3 = 8 \quad (4.14)$$

$$a_2(\pi)^3 \propto (\pi - \sin \pi)^3 = \pi^3 \quad (4.15)$$

Thus normalising v with respect to the values found at the turnover point leads to:

$$v = \frac{\frac{1}{8}(1 - \cos \varphi)^3}{\frac{1}{\pi^3}(\varphi - \sin \varphi)^3 + \frac{1}{8}(1 - \cos \varphi)^3} \quad (4.16)$$

Further, v is scaled with the volume fractions of each region at the turnover point, that is f_2 and $f_1 = 1 - f_2$ respectively:

$$v = \frac{f_2 \pi^3 (1 - \cos \varphi)^3}{f_1 8 (\varphi - \sin \varphi)^3 + f_2 \pi^3 (1 - \cos \varphi)^3} \quad (4.17)$$

Thus the parameter v can be expressed as:

$$v = \frac{\pi^3 f_2 (1 - \cos \varphi)^3}{8(1 - f_2)(\varphi - \sin \varphi)^3 + \pi^3 f_2 (1 - \cos \varphi)^3} \quad (4.18)$$

4.1.2 Deceleration parameters of the two-region model

The deceleration parameter q , which describes the rate of change of the expansion of the entire toy-model universe, is defined as: [12]

$$q \equiv -\frac{1}{H^2} \frac{\ddot{a}}{a} = \frac{-1}{H^2 a} \frac{d}{dt} \left(\frac{\dot{a}_1 a_1^2 + \dot{a}_2 a_2^2}{a^2} \right). \quad (4.19)$$

The time derivative is calculated using the product rule:

$$q = -\frac{1}{H^2 a} \left(\frac{da^{-2}}{dt} + \frac{d\dot{a}_1}{dt} \frac{a_1^2}{a^2} + \frac{da_1^2}{dt} \frac{\dot{a}_1}{a^2} + \frac{d\dot{a}_2}{dt} \frac{a_2^2}{a^2} + \frac{da_2^2}{dt} \frac{\dot{a}_2}{a^2} \right). \quad (4.20)$$

Calculating the time derivatives, the equation can be expressed as:

$$q = -\frac{2a_1^3 a_2^3}{H^2 a^6} (H_1 - H_2)^2 - \frac{\ddot{a}_1 a_1^2}{H^2 a^3} - \frac{\ddot{a}_2 a_2^2}{H^2 a^3}. \quad (4.21)$$

The deceleration parameters of each of the two regions are defined as:

$$q_1 \equiv -\frac{1}{H_1^2} \frac{\ddot{a}_1}{a_1}, \quad q_2 \equiv -\frac{1}{H_2^2} \frac{\ddot{a}_2}{a_2}. \quad (4.22)$$

In terms of the parameters v and h , the expression for the total deceleration parameter q can be further reduced to:

$$q = -\frac{2v(1-v)(1-h)^2}{(1-v+vh)^2} + q_1 \frac{1-v}{(1-v+vh)^2} + q_2 \frac{vh^2}{(1-v+vh)^2}. \quad (4.23)$$

For the void region, which is empty, $q_1 = 0$:

$$q = -\frac{2v(1-v)(1-h)^2}{(1-v+vh)^2} + q_2 \frac{vh^2}{(1-v+vh)^2}. \quad (4.24)$$

The complete derivation of the deceleration parameter q for the entire two-region model can be seen in appendix A.1.3.

4.1.3 Numerical implementation of the two-region model

The two-region model is then implemented numerically (see appendix A.5). First, the volume fraction v is plotted as a function of time for each region. Then, the deceleration parameter q is calculated from the volume fraction v and the fractional Hubble parameter h , both as functions of the development angle φ . The value for f_2 used in v is $f_2 = 0.3$ as it gives the best fit to standard cosmology.[7]

To compare the two-region model to a more realistic model of the Universe, the Λ CDM model is obtained by solving the differential equation 2.31 for $t(a)$.

$$t(a) = \frac{1}{H_0} \int_0^a \frac{da'}{a' \sqrt{\Omega_{m,0}/a'^3 + \Omega_{\Lambda,0}}} = \frac{2}{3H_0} \frac{1}{\Omega_{\Lambda,0}} \operatorname{arcsinh} \left(\sqrt{\frac{\Omega_{\Lambda,0}}{\Omega_{m,0}}} a^{3/2} \right), \quad (4.25)$$

Assuming a flat universe and the values $\Omega_{r,0} = 0$, $\Omega_{m,0} = 0.3$, $\Omega_{\Lambda,0} = 0.7$, and $H_0 = 0.07 \text{ Gyr}^{-1}$. The integral is solved using the method of substitution. Now, $a(t)$ is obtained from this:

$$a(t) = \left(\sinh \left(t \frac{3H_0}{2} \sqrt{\Omega_{\Lambda,0}} \right) \sqrt{\frac{\Omega_{m,0}}{\Omega_{\Lambda,0}}} \right)^{2/3}. \quad (4.26)$$

This is used to calculate the deceleration parameter $q = -\ddot{a}a/\dot{a}^2$, which can be compared to the two-region model solved for above.

Last, the development angle φ is converted to time in Gyr to compare it to the observed evolution of the Universe. This is done by finding the proportionality constant of $t \propto \varphi - \sin \varphi$, which is obtained using $\phi = 3\pi/2$ at time $t_0 = 13.77 \text{ Gyr}$, the current age of the Universe.[6] From this, both the Λ CDM model and the two-region model can be plotted as a function of both time and the development angle φ .

4.2 Results of the two-region model

To examine the expansion of the two regions, the volume fraction of each of the regions is first considered. This is plotted as a function of the development angle φ in figure 1. As seen, the volume is initially dominated by the overdense region but after some time (at $\varphi = 2.69$), the void starts to dominate. From this, it is clear that the average expansion will rise due to the gradual volume domination by void, which is expanding faster than the overdense region.

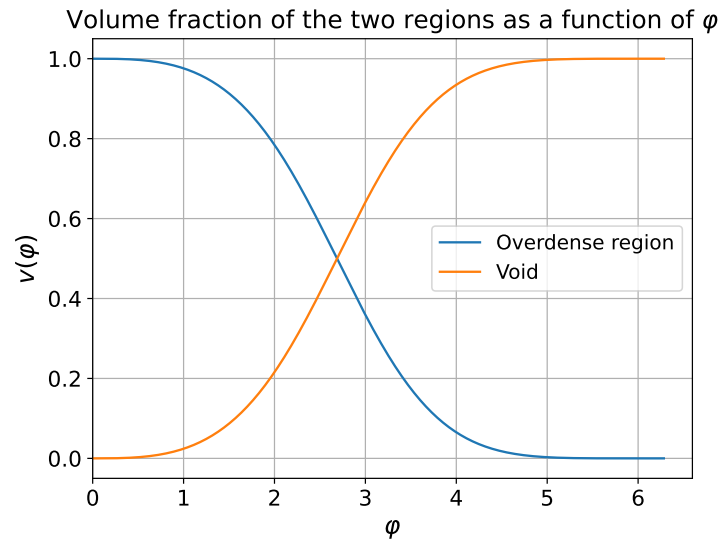


Figure 1: The volume fraction of the void and the overdense region as a function of φ , where the volume fraction at the turnover point is $f_2 = 0.3$.

The deceleration parameter q shown in figure 2 shows how the average deceleration changes with φ and time. This is compared to the Λ CDM model which has been constructed.[6] The two plots have the same overall shape, showing that the two-region toy model can be used to explain an accelerating universe similar to the standard model of cosmology and the observations of our universe. Thus, it shows that density differences can lead to an overall accelerating expansion despite local deceleration. This overall accelerating expansion is further examined in the following sections, where the models considered are complex and more realistic.

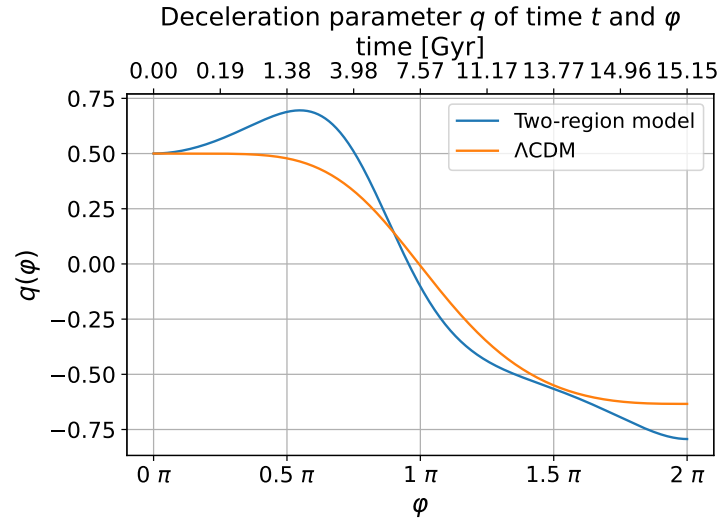


Figure 2: Deceleration parameter of the two-region toy model compared to Λ CDM as a function of time and φ .

Varying the volume fraction f_2 at the turnover point changes the deceleration parameter as shown in figure 3. If the overdense region covers a large fraction of the volume at $\varphi = \pi$, the collapse to zero volume at $\varphi = 2\pi$ will be steep, hence resulting in a steeper transition from a decelerating to an accelerating model universe. The opposite is true if f_2 is small. The value best resembling a Λ CDM universe is $f_2 = 0.3$, which is why this is the value chosen for the calculations in this section.

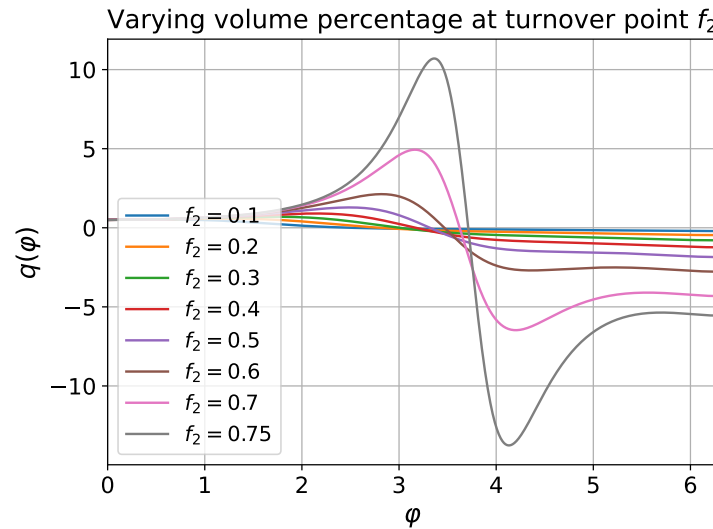


Figure 3: The deceleration parameter at different values of the volume fraction turnover point f_2 .

5 Lemaître–Tolman–Bondi model

The above-considered two-region model is one example of a toy model describing an inhomogeneous universe. The two-region model represents a universe with two distinct regions and thus represents a simple case of an inhomogeneous universe. This simplified description is not accurate for the large-scale structures observed in the real universe, and it is therefore of interest to consider more complex and representative toy models for inhomogeneous universes. Furthermore, shear should be taken into account by defining continuous models in contrast to the two-region model, where shear only occurs at the boundary between the distinct regions.^[13]

5.1 Introduction to LTB models

One example of models describing inhomogeneous universes are Lemaître–Tolman–Bondi (LTB) models. In this thesis, a standard LTB model is considered. A model is constructed such that it represents an inhomogeneous universe with two regions; a spherically symmetric void described by the metric of the considered LTB model, which reduces to an Einstein–de Sitter universe outside the void (thus described by the equations derived in section 2.2.3).

A line element in the LTB model can in spherical coordinates be expressed as: ^[20]

$$ds^2 = -dt^2 + \frac{R'(t, r)^2}{1 - k(r)} dr^2 + R(t, r)^2 d\theta^2 + R(t, r)^2 \sin^2(\theta) d\varphi^2, \quad (5.1)$$

where $k(r)$ is a curvature function describing the curvature of the universe at a given radius r . $R(t, r)$ denotes the areal radius, and its radial derivative is denoted $R'(t, r) = \frac{\partial R(t, r)}{\partial r}$.¹³ Considering a line element in the LTB model along a spatial hypersurface, thus where $dt^2 = d\theta^2 = d\varphi^2 = 0$ reduces equation 5.1 such that:

$$ds = \frac{R'}{\sqrt{1 - k}} dr \quad (5.2)$$

Assuming the curvature $k(r)$ to be negligible, that is $k(r) \ll 1$, leads to:

$$ds \approx R' dr \quad (5.3)$$

Thus integrating both sides, the areal radius $R(t, r)$ is seen to approximately describe the proper distance, that is the physical distance between two points (eq. 2.25): ^[19]

$$s \approx \int_0^r R'(t, r') dr' = R(t, r) - R(t, 0) = R(t, r) \quad (5.4)$$

The considered LTB line element leads to the following metric describing a spherically symmetric dust universe:

$$g_{\mu\nu} = \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & \frac{(R')^2}{1-k} & 0 & 0 \\ 0 & 0 & R^2 & 0 \\ 0 & 0 & 0 & R^2 \sin^2 \theta \end{pmatrix}. \quad (5.5)$$

The metric is diagonal in spherical coordinates which simplifies the analytical solutions of Einstein's field equations.

¹³This shorthand notation is used throughout the thesis for all functions which depend on r .

5.2 Solving Einstein's field equations for the LTB metric

First, it is of interest to derive the dynamical equations for the considered LTB model. This is done as described in section 2 by first computing the Christoffel symbols and using these to solve Einstein's field equations.

5.2.1 Calculating Christoffel symbols

The Christoffel symbols of the LTB metric (eq. 5.5) are calculated using the definition of the Christoffel symbols (eq. 2.21):

$$\Gamma_{\alpha\beta}^{\gamma} = \frac{1}{2}g^{\gamma\lambda}(\partial_{\alpha}g_{\lambda\beta} + \partial_{\beta}g_{\lambda\alpha} - \partial_{\lambda}g_{\alpha\beta}) . \quad (5.6)$$

First, the Christoffel symbols with time as the upper index ($\gamma = 0$) are calculated explicitly:

$$\Gamma_{\alpha\beta}^0 = \frac{1}{2}g^{0\lambda}(\partial_{\alpha}g_{\lambda\beta} + \partial_{\beta}g_{\lambda\alpha} - \partial_{\lambda}g_{\alpha\beta}) . \quad (5.7)$$

For this to be non-zero, λ must also be 0, as $g^{\mu\nu}$ is diagonal in the chosen coordinates:

$$\Gamma_{\alpha\beta}^0 = \frac{1}{2}g^{00}(\partial_{\alpha}g_{0\beta} + \partial_{\beta}g_{0\alpha} - \partial_0g_{\alpha\beta}) . \quad (5.8)$$

Considering the terms in the parentheses, a non-zero, diagonal element of $g_{\mu\nu}$ requires either $\alpha = 0$ or $\beta = 0$, or $\alpha = \beta = 0$. For the first two terms in the parentheses this leads to:

$$\partial_{\alpha}g_{0\beta} = \partial_{\beta}g_{0\alpha} \Rightarrow \partial_{\alpha}g_{00} = \partial_{\alpha}(-1) = 0 . \quad (5.9)$$

Thus, only the last term in the parentheses remains and is non-zero when $\alpha = \beta$:

$$\Gamma_{00}^0 = -\frac{1}{2}g^{00}\partial_0g_{00} = \frac{1}{2}\partial_t(-1) = 0 \quad (5.10)$$

$$\Gamma_{11}^0 = -\frac{1}{2}g^{00}\partial_0g_{11} = \frac{1}{2}\partial_t\left(\frac{R't^2}{1-k}\right) = \frac{R'\dot{R}'}{(1-k)} \quad (5.11)$$

$$\Gamma_{22}^0 = -\frac{1}{2}g^{00}\partial_0g_{22} = \frac{1}{2}\partial_tRt^2 = R\dot{R} \quad (5.12)$$

$$\Gamma_{33}^0 = -\frac{1}{2}g^{00}\partial_0g_{33} = \frac{1}{2}\partial_tRt^2\sin^2\theta = R\dot{R}\sin^2\theta , \quad (5.13)$$

where the time derivative of the areal radius $R(t, r)$ is denoted $\frac{\partial R(t, r)}{\partial t} = \dot{R}$ ¹⁴ and the derivative with respect to both time and space is denoted $\dot{R}' = \frac{\partial^2 R(t, r)}{\partial r \partial t}$. Using the same procedure, the Christoffel symbols with spatial components in their upper index are calculated. The non-zero Christoffel symbols

¹⁴This shorthand notation is used throughout the thesis for all functions which depend on t .

for the LTB metric are:

$$\Gamma^0_{11} = \frac{R' \dot{R}'}{1-k} \quad (5.13a)$$

$$\Gamma^0_{22} = R \dot{R} \quad (5.13b)$$

$$\Gamma^0_{33} = R \dot{R} \sin^2 \theta \quad (5.13c)$$

$$\Gamma^1_{01} = \frac{\dot{R}'}{R'} \quad (5.13d)$$

$$\Gamma^1_{11} = \frac{R''}{R'} + \frac{k'}{2(1-k)} \quad (5.13e)$$

$$\Gamma^1_{22} = -\frac{(1-k)R}{R'} \quad (5.13f)$$

$$\Gamma^1_{33} = -\frac{(1-k)R \sin^2 \theta}{R'} \quad (5.13g)$$

$$\Gamma^2_{02} = \frac{\dot{R}}{R} \quad (5.13h)$$

$$\Gamma^2_{12} = \frac{R'}{R} \quad (5.13i)$$

$$\Gamma^2_{33} = -\cos \theta \sin \theta \quad (5.13j)$$

$$\Gamma^3_{03} = \frac{\dot{R}}{R} \quad (5.13k)$$

$$\Gamma^3_{13} = \frac{R'}{R} \quad (5.13l)$$

$$\Gamma^3_{23} = \frac{1}{\tan \theta} . \quad (5.13m)$$

The Christoffel symbols are symmetric in their lower indices, meaning that $\Gamma^1_{10} = \Gamma^1_{01}$. Thus, a total of 19 non-zero Christoffel symbols are obtained.

5.2.2 Calculating components of the Ricci tensor

The Christoffel symbols can now be used to calculate the Ricci tensor (eq. 2.20), which is a contraction of the Riemann tensor:

$$R_{\mu\nu} = R^\alpha_{\mu\alpha\nu} = \partial_\alpha \Gamma^\alpha_{\nu\mu} - \partial_\nu \Gamma^\alpha_{\alpha\mu} + \Gamma^\alpha_{\alpha\lambda} \Gamma^\lambda_{\nu\mu} - \Gamma^\alpha_{\nu\lambda} \Gamma^\lambda_{\alpha\mu} . \quad (5.14)$$

Calculating the time component first, which is when $\mu = \nu = 0$:

$$R_{00} = R^\alpha_{0\alpha 0} = \partial_\alpha \Gamma^\alpha_{00} - \partial_0 \Gamma^\alpha_{\alpha 0} + \Gamma^\alpha_{\alpha\lambda} \Gamma^\lambda_{00} - \Gamma^\alpha_{0\lambda} \Gamma^\lambda_{\alpha 0} . \quad (5.15)$$

The Christoffel symbols with 0 in both their lower indices are zero. Therefore the time component reduces to:

$$R_{00} = -\partial_0 \Gamma^\alpha_{\alpha 0} - \Gamma^\alpha_{0\lambda} \Gamma^\lambda_{\alpha 0} . \quad (5.16)$$

Writing out the sum over α explicitly leads to:

$$R_{00} = -\partial_0 \Gamma^0_{00} - \partial_0 \Gamma^1_{10} - \partial_0 \Gamma^2_{20} - \partial_0 \Gamma^3_{30} - \Gamma^0_{0\lambda} \Gamma^\lambda_{00} - \Gamma^1_{0\lambda} \Gamma^\lambda_{10} - \Gamma^2_{0\lambda} \Gamma^\lambda_{20} - \Gamma^3_{0\lambda} \Gamma^\lambda_{30} . \quad (5.17)$$

Using the derived Christoffel symbols and keeping only the non-zero terms simplifies the expression to:

$$R_{00} = -\partial_0 \Gamma^1_{10} - \partial_0 \Gamma^2_{20} - \partial_0 \Gamma^3_{30} - \Gamma^1_{0\lambda} \Gamma^\lambda_{10} - \Gamma^2_{0\lambda} \Gamma^\lambda_{20} - \Gamma^3_{0\lambda} \Gamma^\lambda_{30} . \quad (5.18)$$

Then summing over λ , using the fact that Γ^λ_{i0} is only non-zero when $\lambda = i$:

$$R_{00} = -\partial_0 \Gamma^1_{10} - \partial_0 \Gamma^2_{20} - \partial_0 \Gamma^3_{30} - \Gamma^1_{01} \Gamma^1_{10} - \Gamma^2_{02} \Gamma^2_{20} - \Gamma^3_{03} \Gamma^3_{30} . \quad (5.19)$$

Inserting the Christoffel symbols (eq. 5.13d, 5.13h, and 5.13k):

$$R_{00} = -\partial_t \left(\frac{\dot{R}'}{R'} \right) - \partial_t \left(\frac{\dot{R}}{R} \right) - \partial_t \left(\frac{\dot{R}}{R} \right) - \left(\frac{\dot{R}'}{R'} \right)^2 - \left(\frac{\dot{R}}{R} \right)^2 - \left(\frac{\dot{R}}{R} \right)^2. \quad (5.20)$$

Next, the time derivatives are computed:

$$R_{00} = -\frac{\ddot{R}'}{R'} + \left(\frac{\dot{R}'}{R'} \right)^2 - 2\frac{\ddot{R}}{R} + 2\left(\frac{\dot{R}}{R} \right)^2 - \left(\frac{\dot{R}'}{R'} \right)^2 - 2\left(\frac{\dot{R}}{R} \right)^2. \quad (5.21)$$

Thus, the time component of the Ricci tensor is found for the LTB model:

$$R_{00} = -\frac{\ddot{R}'}{R'} - 2\frac{\ddot{R}}{R}. \quad (5.22)$$

The spatial components of the Ricci tensor R_{ii} can be calculated with a similar approach. Calculating R_{11} explicitly by summing over α :

$$R_{11} = R_{1\alpha 1}^\alpha = \partial_\alpha \Gamma_{11}^\alpha - \partial_1 \Gamma_{\alpha 1}^\alpha + \Gamma_{\alpha\lambda}^\alpha \Gamma_{11}^\lambda - \Gamma_{1\lambda}^\alpha \Gamma_{\alpha 1}^\lambda \quad (5.23)$$

$$\begin{aligned} &= \partial_0 \Gamma_{11}^0 - \partial_1 \Gamma_{01}^0 + \Gamma_{0\lambda}^0 \Gamma_{11}^\lambda - \Gamma_{1\lambda}^0 \Gamma_{01}^\lambda \\ &+ \partial_1 \Gamma_{11}^1 - \partial_1 \Gamma_{11}^1 + \Gamma_{1\lambda}^1 \Gamma_{11}^\lambda - \Gamma_{1\lambda}^1 \Gamma_{11}^\lambda \\ &+ \partial_2 \Gamma_{11}^2 - \partial_1 \Gamma_{21}^2 + \Gamma_{2\lambda}^2 \Gamma_{11}^\lambda - \Gamma_{1\lambda}^2 \Gamma_{21}^\lambda \\ &+ \partial_3 \Gamma_{11}^3 - \partial_1 \Gamma_{31}^3 + \Gamma_{3\lambda}^3 \Gamma_{11}^\lambda - \Gamma_{1\lambda}^3 \Gamma_{31}^\lambda. \end{aligned} \quad (5.24)$$

Summing over λ , keeping only non-vanishing terms:

$$\begin{aligned} R_{11} &= \partial_0 \Gamma_{11}^0 - \Gamma_{11}^0 \Gamma_{01}^1 \\ &+ \partial_1 \Gamma_{11}^1 - \partial_1 \Gamma_{11}^1 + \Gamma_{01}^1 \Gamma_{11}^0 - \Gamma_{01}^1 \Gamma_{11}^0 + \Gamma_{11}^1 \Gamma_{11}^1 - \Gamma_{11}^1 \Gamma_{11}^1 \\ &- \partial_1 \Gamma_{12}^2 + \Gamma_{02}^2 \Gamma_{11}^0 + \Gamma_{12}^2 \Gamma_{11}^1 - \Gamma_{12}^2 \Gamma_{12}^2 \\ &- \partial_1 \Gamma_{13}^3 + \Gamma_{03}^3 \Gamma_{11}^0 + \Gamma_{13}^3 \Gamma_{11}^1 - \Gamma_{13}^3 \Gamma_{13}^3 \\ &= \partial_0 \Gamma_{11}^0 - \partial_1 (\Gamma_{12}^2 + \Gamma_{13}^3) - \Gamma_{11}^0 (\Gamma_{01}^1 - \Gamma_{02}^2 - \Gamma_{03}^3) + \Gamma_{11}^1 (\Gamma_{12}^2 + \Gamma_{13}^3) - (\Gamma_{12}^2)^2 - (\Gamma_{13}^3)^2. \end{aligned} \quad (5.25)$$

Inserting the computed Christoffel symbols (eq.'s 5.13a, 5.13i, 5.13l, 5.13d, 5.13h, 5.13k, 5.13e):

$$R_{11} = \partial_t \left(\frac{R' \dot{R}'}{1-k} \right) - \partial_r \left(2 \frac{R'}{R} \right) - \left(\frac{R' \dot{R}'}{1-k} \right) \left(\frac{\dot{R}'}{R'} - 2 \frac{\dot{R}}{R} \right) + \left(\frac{R''}{R'} + \frac{k'}{2(1-k)} \right) 2 \frac{R'}{R} - 2 \left(\frac{R'}{R} \right)^2. \quad (5.26)$$

Calculating the time and radial derivatives and the parentheses are expanded:

$$R_{11} = \frac{(\dot{R}')^2}{1-k} + \frac{R' \ddot{R}'}{1-k} - 2 \frac{R''}{R} + 2 \frac{(R')^2}{R^2} - \frac{(\dot{R}')^2}{1-k} + 2 \frac{\dot{R} R' \dot{R}'}{R(1-k)} \quad (5.27)$$

$$+ 2 \frac{R''}{R} + \frac{R' k'}{R(1-k)} - 2 \frac{(R')^2}{R^2}. \quad (5.28)$$

Multiple terms cancel, thus simplifying the radial component of the Ricci tensor:

$$R_{11} = \frac{R' \ddot{R}'}{1-k} + 2 \frac{\dot{R} R' \dot{R}'}{R(1-k)} + \frac{R' k'}{R(1-k)}. \quad (5.29)$$

The two other spatial components of the symmetrical Ricci tensor are calculated similarly, resulting in the following:

$$R_{22} = k + \frac{R k'}{2R'} + R \ddot{R} + \frac{R \dot{R} \dot{R}'}{R'} + \dot{R}^2 \quad (5.30)$$

$$R_{33} = \frac{(R' (k + R \ddot{R} + \dot{R}^2) + R (\frac{1}{2} k' + \dot{R} \dot{R}')) \sin^2 \theta}{R'}. \quad (5.31)$$

From these, the Ricci scalar can be calculated as the trace of the Ricci tensor:

$$\mathcal{R} \equiv g^{\mu\nu} R_{\mu\nu} = \frac{2k}{R^2} + \frac{2k'}{RR'} + \frac{2\ddot{R}'}{R'} + \frac{4\ddot{R}}{R} + \frac{4\dot{R}\dot{R}'}{RR'} + \frac{2\dot{R}^2}{R^2}. \quad (5.33)$$

The Einstein tensor $G_{\mu\nu} \equiv R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu}$ (eq. 2.19) can now be obtained. First, calculating the temporal component G_{00} and simplifying the expression:

$$G_{00} = -\frac{\ddot{R}'}{R'} - 2\frac{\ddot{R}}{R} + \frac{1}{2} \left(\frac{2k}{R^2} + \frac{2k'}{RR'} + \frac{2\ddot{R}'}{R'} + \frac{4\ddot{R}}{R} + \frac{4\dot{R}\dot{R}'}{RR'} + \frac{2\dot{R}^2}{R^2} \right) \quad (5.34)$$

$$= \frac{k}{R^2} + \frac{k'}{RR'} + \frac{2\dot{R}\dot{R}'}{RR'} + \frac{\dot{R}^2}{R^2}. \quad (5.35)$$

Similarly, the radial component G_{11} of the Einstein tensor is obtained and is simplified by expanding the parentheses:

$$G_{11} = \frac{R'\ddot{R}'}{1-k} + 2\frac{\dot{R}R'\dot{R}'}{R(1-k)} + \frac{R'k'}{R(1-k)} \quad (5.36)$$

$$- \frac{(R')^2}{2(1-k)} \left(\frac{2k}{R^2} + \frac{2k'}{RR'} + \frac{2\ddot{R}'}{R'} + \frac{4\ddot{R}}{R} + \frac{4\dot{R}\dot{R}'}{RR'} + \frac{2\dot{R}^2}{R^2} \right) \\ = -\frac{k(R')^2}{R^2(1-k)} - \frac{2\ddot{R}(R')^2}{R(1-k)} - \frac{\dot{R}^2(R')^2}{R^2(1-k)}. \quad (5.37)$$

The rest of the spatial components (G_{22} and G_{33}) will be the same as G_{11} and will therefore not be derived here.

5.2.3 Obtaining the dynamical equation for $R(t, r)$

To obtain the dynamical equation for the areal radius $R(t, r)$, Einstein's field equation is solved for G_{11} . Assuming dust cosmology, the only non-zero component of the energy-momentum tensor (eq. 2.28) is the temporal component, thus making the right-hand side of Einstein's field equation for G_{11} zero (see 2.18). Together with a non-zero cosmological constant, this leads to the following expression:

$$G_{11} + \Lambda g_{11} = 8\pi GT_{11} = -\frac{k(R')^2}{R^2(1-k)} - \frac{2\ddot{R}(R')^2}{R(1-k)} - \frac{\dot{R}^2(R')^2}{R^2(1-k)} + \Lambda \frac{(R')^2}{1-k} = 0. \quad (5.38)$$

Simplifying this by multiplying with $R^2(1-k)/(R')^2$, then multiplying with $\dot{R}$:

$$-k\dot{R} - 2R\dot{R}\ddot{R} - \dot{R}^3 + \Lambda R^2\dot{R} = 0. \quad (5.39)$$

Rewriting using $\frac{\partial}{\partial t}\dot{R}^2 = 2\dot{R}\ddot{R}$ and integrating with respect to time:

$$-kR - \int R \frac{\partial \dot{R}^2}{\partial t} dt - \int \dot{R}^3 dt + \int \Lambda R^2 \dot{R} dt = 0. \quad (5.40)$$

Calculating the first integral using integration by parts:

$$- \int R \frac{\partial \dot{R}^2}{\partial t} dt = -R\dot{R}^2 + \int \dot{R}^3 dt. \quad (5.41)$$

The integral on the right-hand side is recognised to cancel the second integral on the left-hand side of equation 5.40. The last integral in equation 5.40 is computed similarly, using integration by parts and $\frac{\partial}{\partial t}R^3 = 3R^2\dot{R}$:

$$\Lambda \int R^2 \dot{R} dt = \Lambda R^3 - \Lambda \int 2R^2 \dot{R} dt \Leftrightarrow \Lambda \int R^2 \dot{R} dt = \Lambda \frac{R^3}{3}. \quad (5.42)$$

Inserting the solution to the integrals back in, the expression simplifies to:

$$-kR - R\dot{R}^2 + \Lambda \frac{R^3}{3} + 2M(r) = 0, \quad (5.43)$$

where $2M(r)$ is introduced as a constant of integration with $M(r)$ being an arbitrary function depending on radius r . Dividing by R and solving for $\dot{R}^2$ then leads to the dynamical equation for $R(t, r)$:

$$\dot{R}^2 = -k + \Lambda \frac{R^2}{3} + \frac{2M}{R}. \quad (5.44)$$

This equation resembles the first Friedmann equation (eq. 2.26): $H^2 = \left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G}{3}\rho + \frac{\Lambda}{3} - \frac{\kappa}{a^2}$. The resemblance can be seen by rewriting equation 5.44

$$\frac{\dot{R}^2}{R^2} = \frac{2M}{R^3} + \frac{\Lambda}{3} - \frac{k}{R^2}. \quad (5.45)$$

To determine an expression for the density of the LTB model ρ_{LTB} , the spatial derivative of the dynamical equation for $R(t, r)$ (eq. 5.44) is considered:

$$\frac{\partial \dot{R}^2}{\partial r} = 2\dot{R}\dot{R}' = -k' + \Lambda \frac{2RR'}{3} + \frac{2M'}{R} - \frac{2MR'}{R^2}. \quad (5.46)$$

First, the temporal component of Einstein's field equations (eq. 2.18) is considered:

$$G_{00} + \Lambda g_{00} = 8\pi G T_{00}. \quad (5.47)$$

Inserting the temporal components G_{00} (eq. 5.35), T_{00} (eq. 2.28) and g_{00} (eq. 5.5) leads to:

$$\frac{k}{R^2} + \frac{k'}{RR'} + \frac{2\dot{R}\dot{R}'}{RR'} + \frac{\dot{R}^2}{R^2} - \Lambda = 8\pi G \rho. \quad (5.48)$$

Using equation 5.46 the term $2\dot{R}\dot{R}'$ can be substituted:

$$\frac{k}{R^2} + \frac{k'}{RR'} + \frac{1}{RR'} \left(-k' + \Lambda \frac{2RR'}{3} + \frac{2M'}{R} - \frac{2MR'}{R^2} \right) + \frac{\dot{R}^2}{R^2} - \Lambda = 8\pi G \rho. \quad (5.49)$$

Simplifying leads to the following equation:

$$\frac{k}{R^2} + \frac{2}{3}\Lambda + \frac{2M'}{R^2 R'} - \frac{2M}{R^3} + \frac{\dot{R}^2}{R^2} - \Lambda = 8\pi G \rho. \quad (5.50)$$

Substituting $\dot{R}^2/R^2$ with equation 5.45 results in:

$$\frac{k}{R^2} + \frac{2}{3}\Lambda + \frac{2M'}{R^2 R'} - \frac{2M}{R^3} + \frac{2M}{R^3} + \frac{\Lambda}{3} - \frac{k}{R^2} - \Lambda = 8\pi G \rho. \quad (5.51)$$

Simplifying leads to the following relation:

$$\frac{2M'}{R^2 R'} = 8\pi G \rho. \quad (5.52)$$

Denoting $\rho = \rho_{LTB}$, the density inside the void is given by:

$$\rho_{LTB} = \frac{1}{4\pi G} \frac{M'}{R^2 R'}. \quad (5.53)$$

Outside the void, the density is the one of an Einstein–de Sitter universe (eq. 2.39).

5.3 Dynamical equations used in the standard LTB model

The LTB model is now considered such that it describes a spherical void of radius r_b placed at the centre of the model universe (at $r = 0$). Outside the void (for $r > r_b$), the universe is described by an Einstein–de Sitter universe. Inside the void, the LTB model is specified by defining three functions, which are all free in time; the effective gravitational mass $M(r)$, the curvature $k(r)$, and the Big Bang time $t_B(r)$. [20]

The effective gravitational mass is defined such that it depends only on the radius:

$$M(r) \equiv \frac{1}{2} H_0^2 r^3 . \quad (5.54)$$

The spatial derivative of the expression for the effective gravitational mass is therefore:

$$\frac{dM(r)}{dr} = \frac{3}{2} H_0^2 r^2 . \quad (5.55)$$

The curvature is defined to be radially dependent and approach zero at the centre- and boundary of the void:

$$k(r) \equiv \begin{cases} -r^2 k_{max} \left(\left(\frac{r}{r_b} \right)^n - 1 \right)^m & \text{for } r \leq r_b \\ 0 & \text{for } r > r_b , \end{cases} \quad (5.56)$$

where k_{max} is the maximal value of the curvature. Further, m and n are constants, in this case, chosen to equal $m = n = 4$. The curvature as a function of radius can be seen in figure 4 (b). The spatial derivative of the curvature function is:

$$\frac{dk(r)}{dr} = \begin{cases} -2r k_{max} \left(\left(\frac{r}{r_b} \right)^n - 1 \right)^m - r k_{max} n m \left(\left(\frac{r}{r_b} \right)^n - 1 \right)^{m-1} \left(\frac{r}{r_b} \right)^n & \text{for } r \leq r_b \\ 0 & \text{for } r > r_b . \end{cases} \quad (5.57)$$

Further, the areal radius of the LTB model is defined. Outside the void, the areal radius depends linearly on the radius as equation 2.25: $a_{EdS} r$, where the scale factor of the Einstein–de Sitter universe is given by equation 2.32. Inside the void, the areal radius is defined by solving the dynamical equation for $R(t, r)$ (eq. 5.44). Choosing the cosmological constant $\Lambda = 0$, the solution to the dynamical equation is: [20]

$$R(t, r) = \begin{cases} \frac{M}{k} (1 - \cosh \eta) & \text{for } r \leq r_b \\ a_{EdS} r & \text{for } r > r_b , \end{cases} \quad (5.58)$$

where $\eta(t, r)$ is a parameter defined by the following expression:

$$\sinh \eta - \eta - (t - t_B) \frac{(-k)^{\frac{3}{2}}}{M} = 0 , \quad (5.59)$$

where $t_B(r)$ is the third free function specified in the standard LTB model. This function determines the time of the Big Bang and is dependent on r , thus the Big Bang time can change depending on spatial coordinates. In this case, the Big Bang time $t_B(r)$, is chosen to be zero.[20]

As found in equation 5.4, the areal radius $R(t, r)$ approximately describes the proper distance, that is the physical distance between two points in the model universe. Denoting $\frac{dM(r)}{dr} = M'$ and $\frac{dk(r)}{dr} = k'$, the spatial derivative of $R(t, r)$ is found by applying both the product- and chain rule:

$$\frac{\partial R(t, r)}{\partial r} = \begin{cases} \frac{M'}{k} (1 - \cosh \eta) - \frac{M}{k^2} k' (1 - \cosh \eta) - \frac{M}{k} \sinh(\eta) \eta' & \text{for } r \leq r_b \\ a_{EdS} & \text{for } r > r_b , \end{cases} \quad (5.60)$$

where $\eta' = \frac{\partial \eta(t, r)}{\partial r}$ denotes the spatial derivative of $\eta(t, r)$. This is found by taking the derivative with respect to r of both sides of equation 5.59. Isolating η' then leads to:

$$\eta'(t, r) = \frac{1}{\cosh(\eta) - 1} \frac{(-k)^{\frac{1}{2}}}{M^2} \left(k M' - \frac{3}{2} M k' \right) . \quad (5.61)$$

The areal radius It is further possible to determine the time derivative of the areal radius $\dot{R}$. This satisfies the dynamical equation for $R(t, r)$ (eq. 5.44):

$$\dot{R}^2 = -k + 2 \frac{M}{R} + \frac{\Lambda}{3} R^2 . \quad (5.62)$$

Considering $\Lambda = 0$, the time derivative of $R(t, r)$ can be isolated:

$$\dot{R}(t, r) = \sqrt{-k + 2\frac{M}{R}} . \quad (5.63)$$

The derivative of $R(t, r)$ with respect to both r and t can be computed as well. Therefore the derivative with respect to r is taken on both sides of equation 5.62 by applying the chain rule:

$$\frac{\partial \dot{R}^2}{\partial r} = \frac{\partial}{\partial r} \left(-k + 2\frac{M}{R} \right) \quad (5.64)$$

$$\Downarrow \quad 2\dot{R}'\dot{R} = -k' - 2\frac{M}{R^2}R' + 2\frac{M'}{R} , \quad (5.65)$$

where $\dot{R}' = \frac{\partial^2 R(t, r)}{\partial r \partial t}$ denotes the derivative of $R(t, r)$ with respect to both r and t . Isolating $\dot{R}'$ then leads to the expression:

$$\dot{R}'(t, r) = \frac{1}{2\dot{R}} \left(-k' - 2\frac{M}{R^2}R' + 2\frac{M'}{R} \right) . \quad (5.66)$$

The derived dynamical equations for the standard LTB model are implemented numerically in section 5.4.

5.3.1 Volume change in LTB models

To consider whether backreaction occurs in the LTB model, it is first of interest to determine the volume of an arbitrary domain $\mathcal{D}$ in the void compared to the volume of the domain $\mathcal{D}$ outside the void, thus in the Einstein–de Sitter universe.

First, the volume of the domain $\mathcal{D}$ in the Einstein–de Sitter universe outside the void is determined. When considering the Einstein–de Sitter universe, there is no curvature ($\kappa \rightarrow 0$) such that the square root of the determinant of the metric (eq. 5.73) can be approximated as: [19]

$$\sqrt{\det({}^{(3)}g_{\mu\nu})} \rightarrow \frac{|R'|}{\sqrt{1}} R^2 \sin \theta = |R'| R^2 \sin \theta . \quad (5.67)$$

Since the areal radius in the Einstein–de Sitter model is monotonically rising, the absolute value is neglected. The volume of the domain $\mathcal{D}$ can then be computed in spherical coordinates using equation 2.46:

$$\begin{aligned} V_{EdS}(t) &= \int_0^\pi \int_0^{2\pi} \int_0^{r_b} R' R^2 \sin \theta dr d\theta d\varphi \\ &= 4\pi \int_0^{r_b} R' R^2 dr . \end{aligned} \quad (5.68)$$

The term inside the integral is rewritten recalling the following derivative of a composite function:

$$R' R^2 = \frac{1}{3} (R^3)' . \quad (5.69)$$

Inserting the above expression leads to:

$$\begin{aligned} V_{EdS}(t) &= 4\pi \frac{1}{3} \int_0^{r_b} (R^3)' dr \\ &= \frac{4\pi}{3} [R^3(t, r)]_0^{r_b} \\ &= \frac{4\pi}{3} (R^3(t, r_b) - R^3(t, 0)) . \end{aligned} \quad (5.70)$$

By equation 2.25, the proper distance equals zero at the centre, that is at $r = 0$. Further, the proper distance at $r = r_b$ can be rewritten in terms of the scale factor, thus obtaining the volume of the domain $\mathcal{D}$ in the Einstein–de Sitter universe:

$$V_{EdS}(t) = \frac{4\pi}{3} a^3(t) r_b^3 . \quad (5.71)$$

It is further possible to determine the volume of the domain $\mathcal{D}$ inside the void described by the LTB metric. Therefore the determinant of the LTB metric (eq. 5.5) is first considered. Since the metric is diagonal, the determinant is the product of the diagonal terms. Considering only the spatial terms therefore leads to: [19]

$$\det({}^{(3)}g_{\mu\nu}) = \frac{(R')^2}{(1-k)} (R^2) (R^2 \sin^2 \theta) . \quad (5.72)$$

Taking the square root of both sides:

$$\sqrt{\det({}^{(3)}g_{\mu\nu})} = \frac{|R'|}{\sqrt{1-k}} R^2 \sin \theta . \quad (5.73)$$

The areal radius $R(t, r)$ (eq. 5.58) is monotonically rising and therefore the absolute value is neglected. It is then possible to determine the volume of the domain $\mathcal{D}$ in the LTB metric using equation 2.46. Considering the void in spherical coordinates:

$$\begin{aligned} V_{LTB}(t, r) &= \int_0^\pi \int_0^{2\pi} \int_0^{r_b} \sqrt{\det({}^{(3)}g_{\mu\nu})} d^3 X \\ &= \int_0^\pi \int_0^{2\pi} \int_0^{r_b} \frac{R'}{\sqrt{1-k}} R^2 \sin \theta dr d\theta d\varphi \\ &= 4\pi \int_0^{r_b} \frac{R'}{\sqrt{1-k}} R^2 dr . \end{aligned} \quad (5.74)$$

In the standard LTB model, where the curvature function is small, the volume integral can be approximated as the Einstein–de Sitter volume (eq. 5.71). This indicates that no overall backreaction will occur in this model, since the volumes are approximately equal.

5.3.2 Theoretical backreaction in LTB models

Besides looking at the volume fractions to determine whether backreaction occurs in the LTB model, it is also of interest to derive an expression for the backreaction Q (eq. 3.18) in terms of the expansion rate Θ and the shear squared σ^2 . Further, this also gives insights into numerical errors when implementing the model numerically, which can be used to evaluate the results of the alternative LTB model later examined, the Tardis model[2].

To implement the backreaction in the standard LTB model the expansion rate is first considered: [12]

$$\Theta = u^\alpha_{;\alpha} , \quad (5.75)$$

In comoving coordinates the four-velocity is $u^\alpha = (1, 0, 0, 0)$. Using the definition of the covariant derivative of an arbitrary vector field V : [12]

$$\nabla_\mu V^\nu = \partial_\mu V^\nu + \Gamma^\nu_{\mu\lambda} V^\lambda . \quad (5.76)$$

Contracting therefore leads to:

$$u^\alpha_{;\alpha} = u^\alpha_{,\alpha} + \Gamma^\alpha_{\alpha\lambda} u^\lambda , \quad (5.77)$$

where the first term $u^\alpha_{,\alpha} = 0$. The only contribution to the second term $\Gamma^\alpha_{\alpha\lambda} u^\lambda$ is for $\lambda = 0$ since only the temporal component of the four-velocity is non-zero. For the Christoffel symbols (see section 5.2.2) to be non-zero this requires $\alpha = 1, 2, 3$ respectively:

$$\Theta = \Gamma^\alpha_{\alpha\lambda} u^\lambda = \Gamma^1_{10} + \Gamma^2_{20} + \Gamma^3_{30} . \quad (5.78)$$

The Christoffel symbols are symmetric in their lower indices, and therefore the previously computed Christoffel symbols (eq. 5.13d, eq. 5.13h, and eq. 5.13k) are recalled. This leads to the following expression for the expansion rate:

$$\Theta(t, r) = \frac{\dot{R}'}{R'} + \frac{\dot{R}}{R} + \frac{\dot{R}}{R} = \frac{\dot{R}'}{R'} + 2\frac{\dot{R}}{R} . \quad (5.79)$$

Further, squaring 5.79 leads to:

$$\Theta^2(t, r) = \left(\frac{\dot{R}'}{R'} + 2\frac{\dot{R}}{R} \right)^2 = \left(\frac{\dot{R}'}{R'} \right)^2 + 4 \left(\frac{\dot{R}}{R} \right)^2 + 4 \frac{\dot{R}'}{R'} \frac{\dot{R}}{R} . \quad (5.80)$$

To get an expression for shear, the expansion tensor is first found: [8]

$$\Theta_{\mu\nu} = \Theta_{(\mu\nu)} = \sigma_{\mu\nu} + \frac{1}{3}\Theta h_{\mu\nu} , \quad (5.81)$$

where Θ is the square root of equation 5.80. First $\Theta_{\mu\nu}$ is found using the expression (eq. 2.17):

$$\Theta_{\mu\nu} = \bar{\nabla}_{(\mu} u_{\nu)} . \quad (5.82)$$

Using the definition of the covariant derivative (eq. 2.7) and that the expansion tensor is symmetric $\Theta_{\mu\nu}$ simplifies the equation:

$$\begin{aligned} \Theta_{\mu\nu} &= \frac{1}{2} (h^\gamma_\mu h^\sigma_\nu \nabla_\gamma u_\sigma + h^\gamma_\nu h^\sigma_\mu \nabla_\gamma u_\sigma) \\ &= h^\gamma_\mu h^\sigma_\nu \nabla_\gamma u_\sigma , \end{aligned} \quad (5.83)$$

where h^α_β is the projection tensor which projects into three dimensions.

Using the definition of the projection tensor (eq. 2.14) and the metric tensor (eq. 5.5), h^α_β can be found by raising one index of $h_{\mu\nu}$:

$$h_{\mu\nu} = g_{\mu\nu} + u_\mu u_\nu = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & \frac{(R')^2}{1-k} & 0 & 0 \\ 0 & 0 & R^2 & 0 \\ 0 & 0 & 0 & R^2 \sin^2 \theta \end{pmatrix} \quad (5.84)$$

$$h^\mu_\nu = g^{\mu\sigma} h_{\sigma\nu} = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} , \quad (5.85)$$

where $u_\alpha = (-1, 0, 0, 0)$. The contravariant tensor is simply the inverse of the covariant $g^{\mu\nu} g_{\mu\nu} = 4$ (eq. 2.10):

$$g^{\mu\nu} = \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & \frac{1-k}{(R')^2} & 0 & 0 \\ 0 & 0 & \frac{1}{R^2} & 0 \\ 0 & 0 & 0 & \frac{1}{R^2 \sin^2 \theta} \end{pmatrix} . \quad (5.86)$$

The derivative $\nabla_\gamma u_\sigma$ is the covariant derivative, using the definition from equation 2.7:

$$\Theta_{\mu\nu} = h^\gamma_\mu h^\sigma_\nu \left(\partial_\gamma u_\sigma - \Gamma^\lambda_{\sigma\gamma} u_\lambda \right). \quad (5.87)$$

Since $u_\mu = (-1, 0, 0, 0)$, the derivative, and thus the first term in the parenthesis, will always be zero. The index of the four-vector must be zero, that is $\lambda = 0$, since the only non-zero component of the four-velocity is the temporal component. This ensures that the second term in the parenthesis differs from zero. The Christoffel symbols computed in section 5.2.2 with $\lambda = 0$ are only non-zero for equal non-zero lower indices. Thus, the sum reduces to three non-zero terms:

$$\Theta_{\mu\nu} = -h^1_\mu h^1_\nu \Gamma^0_{11} u_0 - h^2_\mu h^2_\nu \Gamma^0_{22} u_0 - h^3_\mu h^3_\nu \Gamma^0_{33} u_0. \quad (5.88)$$

Inserting the computed Christoffel symbols, the projection tensor in three dimensions, and the trace of the expansion tensor into equation 5.81:

$$\begin{aligned} \sigma_{\mu\nu} &= \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & \frac{\dot{R}'R'}{1-k} & 0 & 0 \\ 0 & 0 & R\dot{R} & 0 \\ 0 & 0 & 0 & R\dot{R}\sin^2\theta \end{pmatrix} - \frac{1}{3} \left(\frac{\dot{R}'}{R'} + 2\frac{\dot{R}}{R} \right) \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & \frac{(\dot{R}')^2}{1-k} & 0 & 0 \\ 0 & 0 & R^2 & 0 \\ 0 & 0 & 0 & R^2 \sin^2\theta \end{pmatrix} \\ &= \text{diag} \left(0, \frac{R'}{1-k} \left(\dot{R}' - \frac{1}{3}\dot{R}' - \frac{2}{3}\frac{\dot{R}R'}{R} \right), R \left(\dot{R} - \frac{1}{3}\frac{\dot{R}'R}{R'} - \frac{2}{3}\dot{R} \right), R \sin^2\theta \left(\dot{R} - \frac{1}{3}\frac{\dot{R}'R}{R'} - \frac{2}{3}\dot{R} \right) \right), \end{aligned} \quad (5.89)$$

where the notation $\text{diag}(a, b, c, d)$ simply denotes a diagonal matrix where the first component along the diagonal is a , the second component is b , and so on. To find σ^μ_ν , the metric $g^{\mu\nu}$ is used to raise one index of $\sigma_{\lambda\nu}$:

$$\sigma^\mu_\nu = g^{\mu\lambda} \sigma_{\lambda\nu} = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & \frac{\dot{R}'}{R'} - \frac{1}{3}\frac{\dot{R}'}{R'} - \frac{2}{3}\frac{\dot{R}}{R} & 0 & 0 \\ 0 & 0 & \frac{\dot{R}}{R} - \frac{1}{3}\frac{\dot{R}'}{R'} - \frac{2}{3}\frac{\dot{R}}{R} & 0 \\ 0 & 0 & 0 & \frac{\dot{R}}{R} - \frac{1}{3}\frac{\dot{R}'}{R'} - \frac{2}{3}\frac{\dot{R}}{R} \end{pmatrix}. \quad (5.90)$$

This can be simplified:

$$\sigma^\mu_\nu = g^{\mu\lambda} \sigma_{\lambda\nu} = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & \frac{2}{3} \left(\frac{\dot{R}'}{R} - \frac{\dot{R}}{R} \right) & 0 & 0 \\ 0 & 0 & \frac{1}{3} \left(\frac{\dot{R}}{R} - \frac{\dot{R}'}{R} \right) & 0 \\ 0 & 0 & 0 & \frac{1}{3} \left(\frac{\dot{R}}{R} - \frac{\dot{R}'}{R} \right) \end{pmatrix} \quad (5.91)$$

$$= \text{diag} \left(0, -\frac{2}{3}, \frac{1}{3}, \frac{1}{3} \right) \left(\frac{\dot{R}}{R} - \frac{\dot{R}'}{R} \right). \quad (5.92)$$

Contracting σ^μ_ν with σ^ν_μ :

$$\sigma^\mu_\nu \sigma^\nu_\mu = (0 \cdot 0) + \frac{4}{9} \left(\frac{\dot{R}'}{R'} - \frac{\dot{R}}{R} \right)^2 + \frac{2}{9} \left(\frac{\dot{R}'}{R'} - \frac{\dot{R}}{R} \right)^2 = \frac{2}{3} \left(\frac{\dot{R}'}{R'} - \frac{\dot{R}}{R} \right)^2. \quad (5.93)$$

Using that $\sigma = \sqrt{\frac{1}{2} \sigma^\mu_\nu \sigma^\nu_\mu}$ to calculate σ^2

$$\sigma(t, r) = \sqrt{\frac{1}{2} \sigma^\mu_\nu \sigma^\nu_\mu} = \sqrt{\frac{1}{2} \frac{2}{3} \left(\frac{\dot{R}'}{R'} - \frac{\dot{R}}{R} \right)^2} = \frac{1}{\sqrt{3}} \left(\frac{\dot{R}'}{R'} - \frac{\dot{R}}{R} \right) \Rightarrow \quad (5.94)$$

$$\sigma^2(t, r) = \frac{1}{3} \left(\frac{\dot{R}'}{R'} - \frac{\dot{R}}{R} \right)^2. \quad (5.95)$$

Thus, the shear squared is obtained. This is necessary to examine the backreaction.

It is then possible to consider the backreaction term Q occurring in the Buchert equations (eq. 3.18). Rearranging leads to:

$$Q(t, r) = \frac{2}{3} (\langle \Theta^2 \rangle_{\mathcal{D}} - \langle \Theta \rangle_{\mathcal{D}}^2 - 2\langle \sigma^2 \rangle_{\mathcal{D}}) = \frac{2}{3} \langle \Theta^2 \rangle_{\mathcal{D}} - 2\langle \sigma^2 \rangle_{\mathcal{D}} - \frac{2}{3} \langle \Theta \rangle_{\mathcal{D}}^2. \quad (5.96)$$

To determine the backreaction term, it is therefore necessary to determine the spatial averages $\langle \Theta^2 \rangle_{\mathcal{D}}$, $\langle \Theta \rangle_{\mathcal{D}}^2$ and $\langle \sigma^2 \rangle_{\mathcal{D}}$ respectively. Using equation 2.47 to obtain the spatial average of Θ (eq. 5.79):

$$\begin{aligned} \langle \Theta \rangle_{\mathcal{D}} &= \frac{1}{V_{\mathcal{D}}} \int_{\mathcal{D}} \sqrt{\det({}^{(3)}g_{\mu\nu})} \Theta d^3X \\ &= \frac{1}{V_{\mathcal{D}}} \int_0^\pi \int_0^{2\pi} \int_0^{r_b} \frac{R^2 R'}{\sqrt{1-k}} \sin \theta \left(\frac{\dot{R}'}{R'} + 2 \frac{\dot{R}}{R} \right) dr d\theta d\varphi \\ &= \frac{4\pi}{V_{\mathcal{D}}} \int_0^{r_b} \frac{R^2 R'}{\sqrt{1-k}} \left(\frac{\dot{R}'}{R'} + 2 \frac{\dot{R}}{R} \right) dr. \end{aligned} \quad (5.97)$$

The term $\langle \Theta \rangle_{\mathcal{D}}^2$ is thus:

$$\langle \Theta \rangle_{\mathcal{D}}^2 = \left[\frac{4\pi}{V_{\mathcal{D}}} \int_0^{r_b} \frac{R^2 R'}{\sqrt{1-k}} \left(\frac{\dot{R}'}{R'} + 2 \frac{\dot{R}}{R} \right) dr \right]^2. \quad (5.98)$$

Equation 5.80 and 5.94 are then used to calculate $\frac{2}{3} \langle \Theta^2 \rangle_{\mathcal{D}} - 2\langle \sigma^2 \rangle_{\mathcal{D}}$. This expression simplifies the backreaction term since $\langle \Theta^2 \rangle_{\mathcal{D}}$ and $\langle \sigma^2 \rangle_{\mathcal{D}}$ have common terms which cancel: [13]

$$\begin{aligned} \frac{2}{3} \langle \Theta^2 \rangle_{\mathcal{D}} - 2\langle \sigma^2 \rangle_{\mathcal{D}} &= 2 \left(\frac{1}{V_{\mathcal{D}}} \int_0^{r_b} \int_0^\pi \int_0^{2\pi} \frac{R^2 R'}{\sqrt{1-k}} \sin(\theta) \left(\frac{1}{3} \Theta^2 - \sigma^2 \right) dr d\theta d\phi \right) \\ &= \frac{8\pi}{V_{\mathcal{D}}} \int_0^{r_b} \frac{R^2 R'}{\sqrt{1-k}} \left[\frac{1}{3} \left(\frac{\dot{R}'}{R'} \right)^2 + \frac{4}{3} \left(\frac{\dot{R}}{R} \right)^2 + \frac{4}{3} \frac{\dot{R}' \dot{R}}{R' R} - \frac{1}{3} \left(\frac{\dot{R}'}{R'} \right)^2 - \frac{1}{3} \left(\frac{\dot{R}}{R} \right)^2 + \frac{2}{3} \frac{\dot{R}' \dot{R}}{R' R} \right] dr \\ &= \frac{8\pi}{V_{\mathcal{D}}} \int_0^{r_b} \frac{R^2 R'}{\sqrt{1-k}} \left[\left(\frac{\dot{R}}{R} \right)^2 + \frac{6}{3} \frac{\dot{R}' \dot{R}}{R' R} \right] dr \\ &= \frac{8\pi}{V_{\mathcal{D}}} \int_0^{r_b} \frac{R^2 R'}{\sqrt{1-k}} \left[\left(\frac{\dot{R}}{R} \right)^2 + 2 \frac{\dot{R}' \dot{R}}{R' R} \right] dr. \end{aligned} \quad (5.99)$$

Thus, inserting the above expressions in the backreaction term (eq. 5.96) to obtain: [13]

$$Q(t, r) = \frac{8\pi}{V_{\mathcal{D}}} \int_0^{r_b} \frac{R^2 R'}{\sqrt{1-k}} \left[\left(\frac{\dot{R}}{R} \right)^2 + 2 \frac{\dot{R}' \dot{R}}{R' R} \right] dr - \frac{2}{3} \left[\frac{4\pi}{V_{\mathcal{D}}} \int_0^{r_b} \frac{R^2 R'}{\sqrt{1-k}} \left(\frac{\dot{R}'}{R'} + 2 \frac{\dot{R}}{R} \right) dr \right]^2, \quad (5.100)$$

where these integrals can be solved numerically.

5.4 Numerical implementation of the standard LTB model

The standard LTB model is implemented numerically considering a void with a boundary at $r_b = 40$ Mpc. This boundary ensures that the diameter of the void is smaller than large-scale structures at ~ 100 Mpc. The above theoretical expressions derived for the standard LTB metric (section 5.3) are applied inside the void ($r \leq r_b$), and outside the void ($r > r_b$), the expressions for an Einstein–de Sitter universe (section 2.2.3) are applied. The model universe is considered for different radii starting from the centre of the void. Using $r = 0$ would however result in division by zero and therefore the radii are approximated to start near the centre at $r_i = 10^{-3}$ Mpc. The model is examined up to $r_f = r_b + 10$ Mpc = 50 Mpc

(10 Mpc after the boundary of the void). Further, the model universe is considered at different times, where the initial time t_i is determined by equation 2.32:

$$t_i = t_0 a_i^{\frac{3}{2}}, \quad (5.101)$$

where present time is denoted t_0 and is given by: [6]

$$t_0 = \frac{2}{3} \frac{1}{H_0}, \quad (5.102)$$

where H_0 is the Hubble parameter at present time, which has the value $H_0 = 71.6 \cdot 10^{-3} \text{Gyr}^{-1}$. The initial scale factor is denoted a_i . This value is determined by equation 2.32 assuming the initial time t_i to be the time at which the cosmic microwave background (CMB) was formed, that is approximately at 380000 years after the Big Bang.[6] This leads to the value $a_i = 1/1100$. Further, the gravitational constant is approximated to have the value $G = 4.50 \cdot 10^{-15} \frac{\text{Mpc}}{\text{M}_\odot \text{Gyr}^2}$ (see appendix A.2).

The dynamical equations derived in section 5.3 are implemented numerically in appendix A.6. This is done by iterating over the considered radii. It is then possible to compute values for both the curvature $k(r)$ (eq. 5.56) and its spatial derivative $k'(r)$ (eq. 5.57). Choosing the maximal value of the curvature $k_{max} = 2 \cdot 10^{-7}$ and the constants $n = m = 4$, the curvature as a function of radius can be seen in figure 4 (b). The curvature equals zero both at the centre and outside the void ($r > r_b = 40 \text{ Mpc}$), thus describing the considered Einstein–de Sitter universe. The curvature is seen to be negative inside the void ($r < r_b$).

Further, values for the effective gravitational mass $M(r)$ (eq. 5.54) and its spatial derivative $M'(r)$ (eq. 5.55) are likewise generated by iterating over the considered radii. In figure 4 (a) the effective gravitational mass can be seen.

Using the computed values for $M(r)$ and $k(r)$, it is then of interest to compute numerical values for the areal radius $R(t, r)$ (eq. 5.58). Therefore the corresponding values of $\eta(t, r)$ are computed. This is done by root finding on equation 5.59, using SciPy's bisection function `scipy.optimize.bisect()`, where the interval considered is chosen to be from $a = 0 \text{ Mpc}$ to $b = 100 \text{ Mpc}$. Thus, iterating over both time and radii leads to values $\eta(t, r)$.

Effective gravitational mass function $M(r)$ and curvature $k(r)$

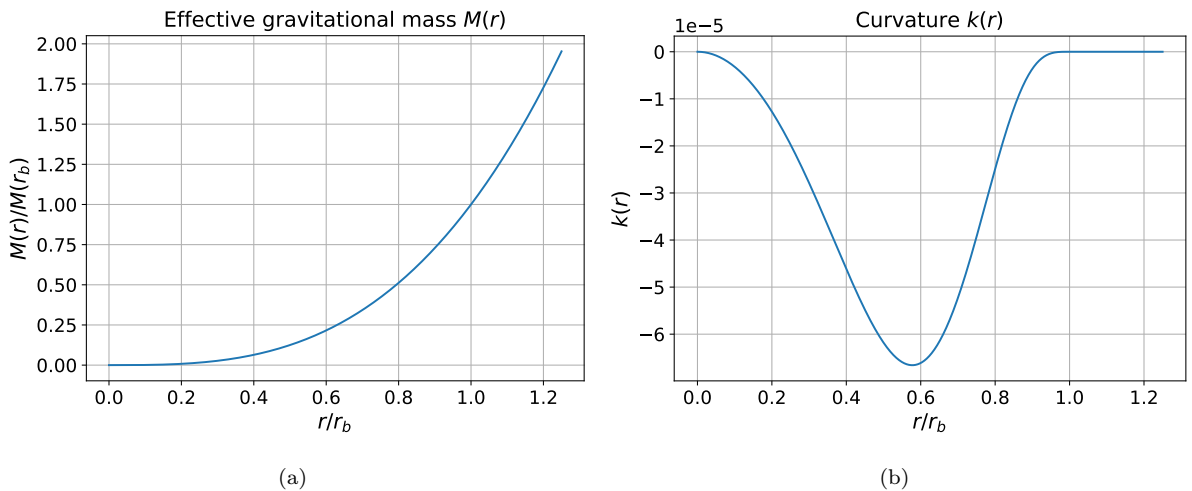


Figure 4: (a) The effective gravitational mass function $M(r)$ divided by the function's value at void radius $M(r_b)$ as a function of radius $\frac{r}{r_b}$ and (b) the curvature $k(r)$ as a function of radius for a void with $r_b = 40 \text{ Mpc}$.

5.5 Results of the standard LTB model

The numerical values for $\eta(t, r)$, generated using root finding, are used to compute the values for the areal radius $R(t, r)$. The areal radius at present time divided by the proper distance $a(t) \cdot r$ can be seen in figure 5 as a function of radius r . Likewise, the values for the spatial- and time derivative of the areal radius (eq. 5.60 and eq. 5.63 respectively) are computed as well as the derivative with respect to both time and radius (eq. 5.66). The computation of the values for $R'(t, r)$ includes generating the spatial derivative of $\eta(t, r)$ (eq. 5.61).

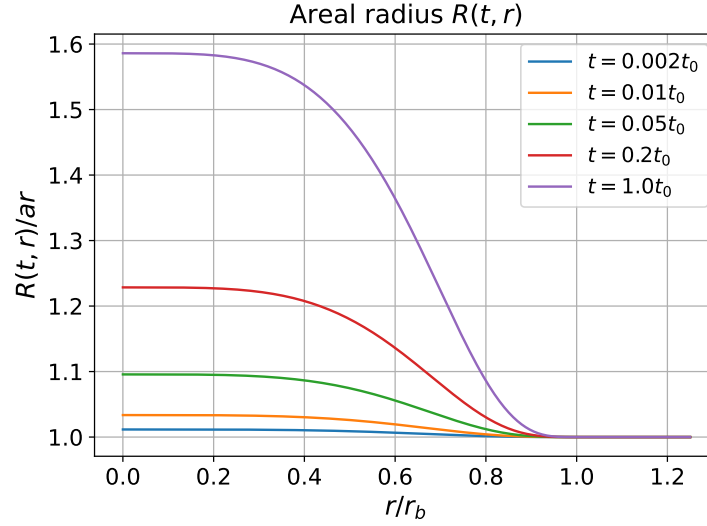


Figure 5: The areal radius $R(t, r)$ at five different times divided by the void radius r as a function of radius $\frac{r}{r_b}$

Considering figure 5 the areal radius $R(t, r)$ at different times is seen to equal the void radius r outside the void (that is $r/r_b > 1$) as the fraction $\frac{R(t, r)}{ar}$ is seen to equal one. Inside the void (for $r < r_b$), the fraction $\frac{R(t, r)}{ar}$ is seen to be approximately constant at small radii with $\frac{R(t, r)}{ar} > 1$. As the radii increase, the fraction $\frac{R(t, r)}{ar}$ decreases until it equals one at the boundary $r = r_b$. Initially, the areal radius $R(t, r)$ is thus larger than the proper distance $d_p = a(t) \cdot r$ (eq. 2.25) and decreases until it equals the proper distance of the Einstein–de Sitter universe outside the void. The maximum value of the areal radius is further, seen to increase as time increases.

5.5.1 Density in the standard LTB model

It is further of interest to consider the ratio of the density fractions, LTB and Einstein–de Sitter respectively, at different time points. Iterating over both time and radius, the values for both density fractions, ρ_{LTB} and ρ_{EdS} , are computed. The ratio of the density fractions $\frac{\rho_{LTB}}{\rho_{EdS}}$ are then depicted as a function of radius for different time steps. This can be seen on figure 6 (a), where $\frac{\rho_{LTB}}{\rho_{EdS}}$ is depicted at different times. Further, figure 6 (b) correspondingly depicts the ratio of the density fractions $\frac{\rho_{LTB}}{\rho_{EdS}}$ as a function of the areal radius $R(t, r)$ instead of the radius r .

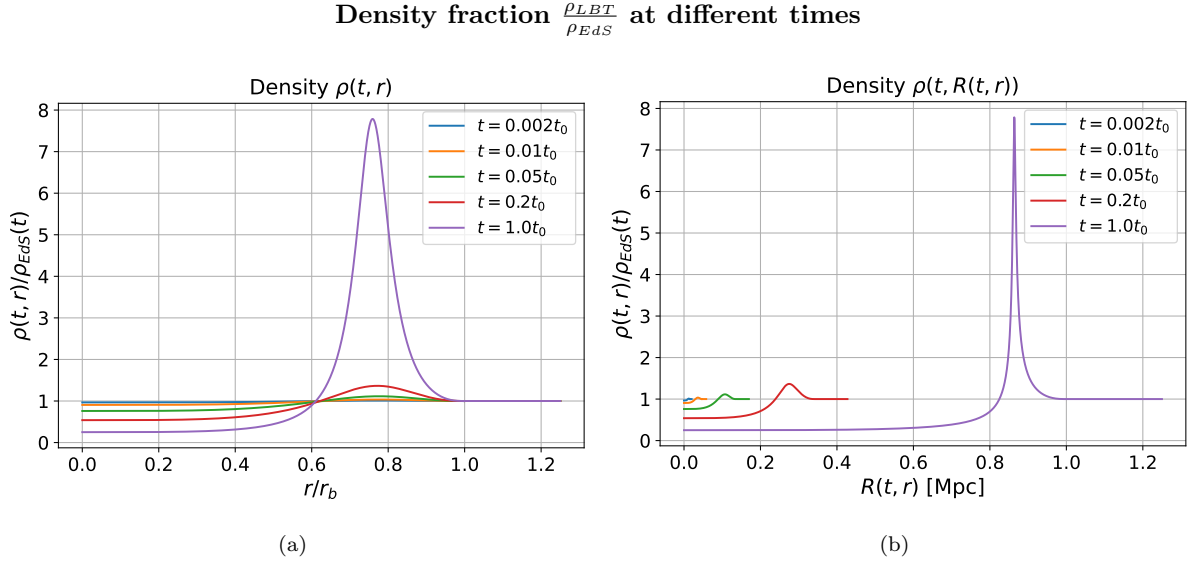


Figure 6: Density fraction $\frac{\rho_{LTB}}{\rho_{EdS}}$ at five different times for a void with $r_b = 40\text{Mpc}$ as a function of (a) radius r and (b) areal radius $R(t, r)$.

Considering the radial dependence of the density fraction $\frac{\rho_{LTB}}{\rho_{EdS}}$ in figure 6 (a), the density fraction is one outside the void. Thus as expected, the density outside the void is described by an Einstein–de Sitter universe ($\rho_{LTB} = \rho_{EdS}$) for all considered times (initial-, intermediate- and end time). Considering the inside of the void, the density fraction is seen to be less than one, that is $\rho_{LTB} < \rho_{EdS}$, initially for all considered times. After this it increases such that it is larger than one, that is $\rho_{LTB} > \rho_{EdS}$, until it peaks at $r \approx 0.75r_b$ and decreases. The maximal value of the peak is further seen to increase as time increases.

The density fraction $\frac{\rho_{LTB}}{\rho_{EdS}}$ is further considered as a function of the areal radius $R(t, r)$ in figure 6 (b). Outside the void, the density ρ_{LTB} is still seen to equal the density of an Einstein–de Sitter universe ρ_{EdS} . The boundary of the void is however seen to increase as time increases.

5.5.2 Backreaction in the standard LTB model

Two methods are implemented numerically to determine whether backreaction occurs in the LTB model; the volumes of both the void V_{LTB} and the surrounding Einstein–de Sitter universe V_{EdS} are considered as well as the theoretical backreaction term Q occurring in the Buchert equations.

The backreaction is first considered by examining the volume. To implement numerically whether backreaction occurs in the considered LTB model, values for the volume fraction V_{LTB} and V_{EdS} are generated.

The volume of the standard LTB model (eq. 5.74) is approximated as a Riemann sum: [21]

$$V_{LTB}(t, r) \approx 4\pi \sum_{i=1}^{Nr} \frac{(R')_i}{\sqrt{1-k_i}} R_i^2 \Delta r, \quad (5.103)$$

where Nr is implemented numerically as the number steps radial steps Δr considered from the centre ($r = 0$) to 10 Mpc after the void ($r = r_f$) and i denotes the i 'th radius point.

Iterating over both time and radius, the values for V_{LTB} are computed using equation 5.103 approximating the volume at time t as a sum over all considered radial steps. The values for V_{EdS} are computed using equation 5.71. The volume fractions are depicted as a function of time in figure 7, for an initial

volume considered from the centre (r_i) to (a) the boundary of the void ($r_f = r_b$) and (b) halfway to the boundary of the void ($r_f = 0.5r_b$), thus within the void.

Volumes at different end-radius

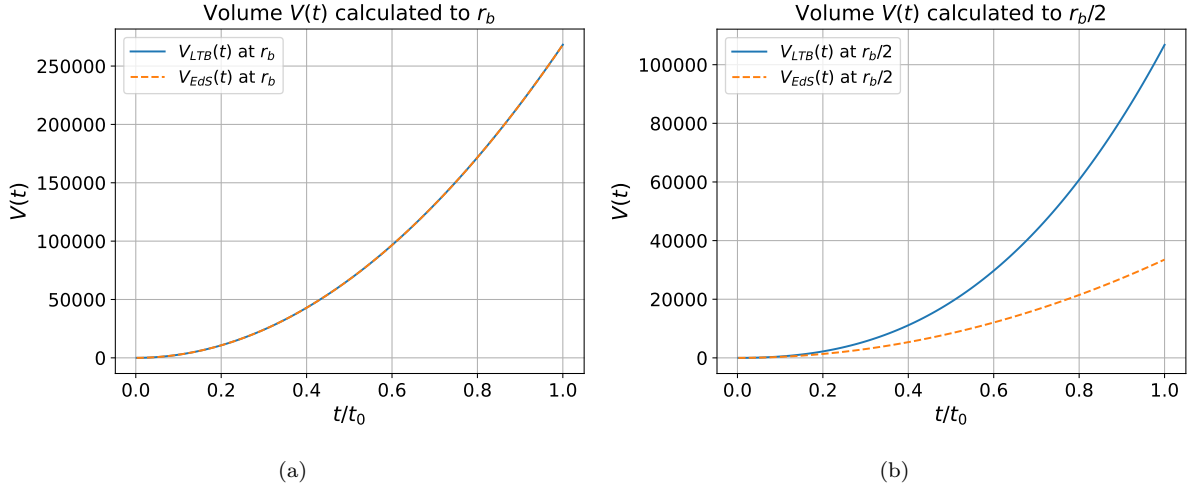


Figure 7: Volumes V_{LTB} (blue) and V_{EdS} (orange), for a void with $r_b = 40$ Mpc as a function of time t integrating over r from initial radius r_i to end radius at (a) $r_f = r_b$ and (b) $r_f = 0.5r_b$.

Considering figure 7 (a), the difference between the volume of both the void and the Einstein–de Sitter universe, V_{LTB} and V_{EdS} respectively, is seen to be negligible when integrating over the entire void. This is in agreement with the volume element of the LTB metric reducing to the one of the Einstein–de Sitter universe when $k(r) \ll 1$.

In contrast, in figure 7 (b), the volumes are seen to differ when only integrating halfway to the boundary r_b , thus ending within the void.

This is further depicted when considering the volume fraction $\frac{V_{LTB}}{V_{EdS}}$ as a function of time t . This can be seen in figure 8 when integrating over (a) the entire void and (b) halfway to the boundary r_b .

Volume fraction

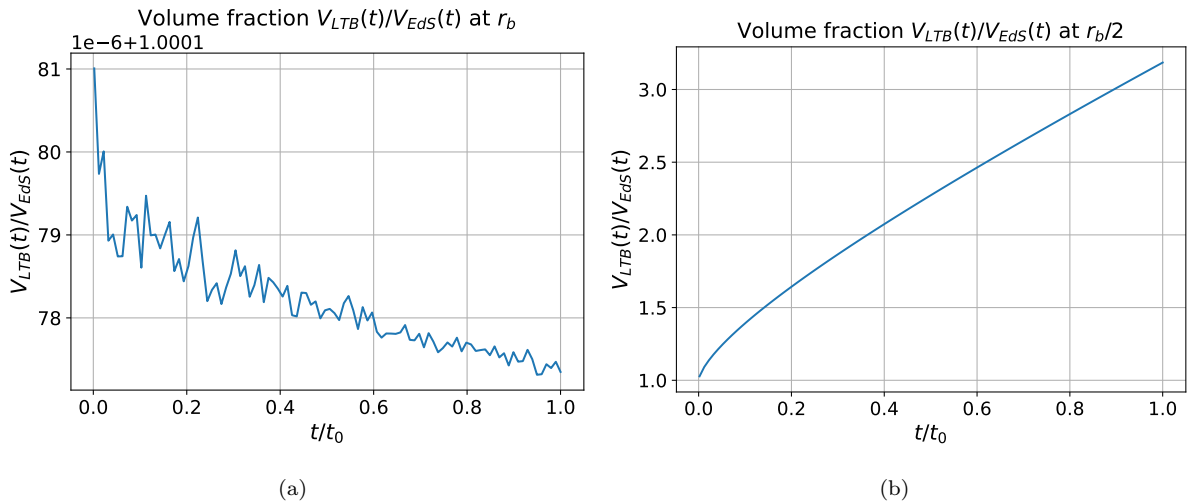


Figure 8: Volume fraction $\frac{V_{LTB}}{V_{EdS}}$ as a function of time t for a void with $r_b = 40$ Mpc when integrating (a) over the entire void and (b) halfway to the boundary.

Considering figure 8 (a), the volume $\frac{V_{LTB}}{V_{EdS}}$ is seen to be $1.0001 + 10^{-6} \approx 1$, that is $V_{LTB} \approx V_{EdS}$, when integrating over the entire void. As expected, figure 8 (b) on the other hand shows the volume fraction to be larger, that is $V_{LTB} > V_{EdS}$, when integrating only halfway to the boundary, thus ending within the void. This corresponds to intermediate scales.

The backreaction in the standard LTB model can be directly examined by evaluating the dynamical equation for the backreaction term. The values for the backreaction term Q are computed numerically. Iterating over time, these are generated by equation 5.96. For each time t , the two definite integrals in equation 5.96 are solved numerically by approximating them as Riemann sums. Figure 9 (a) depicts the backreaction term Q as a function of radius r . Further, the backreaction of the entire volume $Q(t, r_b)$ is depicted as a function of time t when integrating up to the boundary of the void, thus $Q(r_b)$, and further than the boundary, that is $Q(r_b + 10 \text{ Mpc})$ respectively. This can be seen in figure 9 (b).

Backreaction

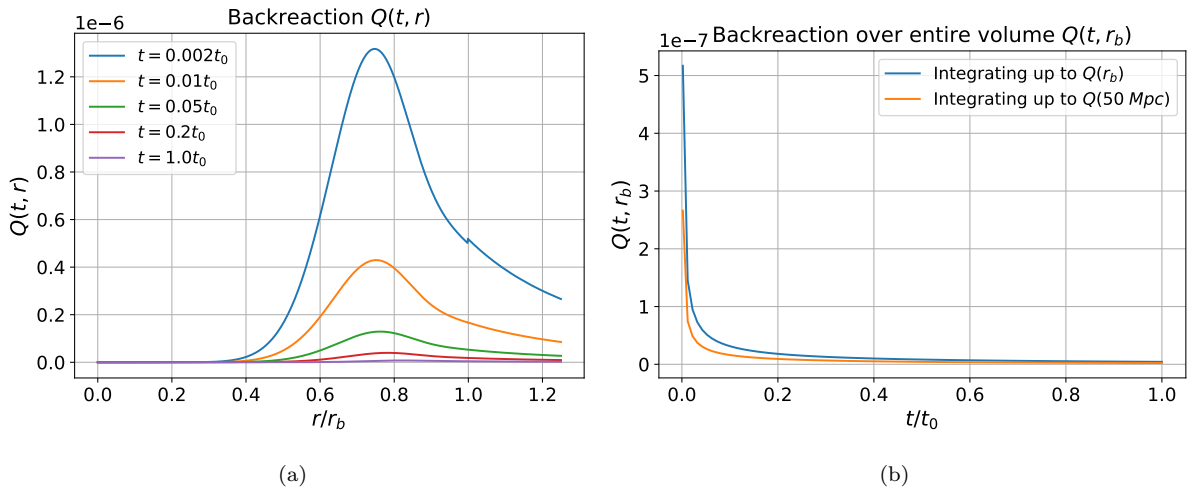


Figure 9: Backreaction term Q as function of (a) radius r and for five different time steps, and (b) time t for a void with $r_b = 40 \text{ Mpc}$ when integrating to the boundary $Q(r_b)$ and beyond the boundary $Q(r_b + 10 \text{ Mpc})$ respectively.

Considering figure 9 (a), the backreaction term Q of the standard LTB model is seen as a function of radii for different times. The maximal value of the backreaction Q is seen to decrease with time from $Q \approx 1.3 \cdot 10^{-6}$ close to the initial time ($t \approx 0.02t_0$) to $Q \approx 0.05 \cdot 10^{-6}$ closer to present time ($t = 0.2t_0$). All values of the backreaction term Q are however seen to be of the order of 10^{-6} or smaller, which is approximately zero.

Considering 9 (b), the backreaction term $Q(t, r_b)$ of the entire void is seen as a function of time t . When integrating up to the boundary of the void $Q(r_b)$, the backreaction is seen to be larger, than when integrating further than the boundary of the void $Q(r_b + 10 \text{ Mpc})$. The backreaction is thus seen to decrease when integrating over a larger volume (beyond the boundary of the void r_b). The values for the backreaction term are however small $Q \propto 10^{-7}$, congruent with the volumes (fig. 7), that no backreaction occurs in the standard LTB model when considering the entire void.

6 The Tardis model

6.1 Introduction to the Tardis model

Another LTB model is examined closer, the so-called *Tardis model*. This model was developed by Mikko Lavinto, Syksy Räsänen, and Sebastian J. Szybka in 2014[2]. The model is an LTB model, thus it considers an inhomogeneous universe. Furthermore, it considers the universe using the *Swiss Cheese* description. This means that the dust-filled universe will have multiple spherical voids which in this model are called *Tardis regions*. These regions have a larger volume than expected from the outside area of the voids. This property will be examined when implementing the model numerically. The background is a flat matter-dominated universe, the Einstein–de Sitter universe (see section 2.2.3).[2]

This thesis only considers one void and thus the Swiss Cheese model is not examined here. It should be noted that the Tardis article starts by using the *Szekeres model*¹⁵ as a solution to Einstein’s field equations, however before they examine their model they take the LTB limit of the Szekeres model, thus the previously derived general LTB model still applies.

The Tardis model has a slightly different line element than the standard LTB model:

$$ds^2 = -dt^2 + \frac{R'(t, r)^2}{1 + E(r)} dr^2 + R(t, r)^2 d\theta + R(t, r)^2 \sin^2(\theta) d\varphi, \quad (6.1)$$

where the definition of the curvature function $E(r)$ has changed from the standard LTB model’s $k(r)$. The relation between the two is $-k(r) = E(r)$. The areal radius of the Tardis model is further defined as a piecewise function. When the curvature is larger than zero, the areal radius is the same as in the standard LTB model for the case $E \geq 0$. [20] When the curvature is zero, the areal radius is given by the following expression: [20]

$$R(t, r) = \begin{cases} \left(\frac{9}{2}M(t - t_B)^2\right)^{1/3} & \text{for } E(r) = 0 \\ \frac{M}{E}(\cosh(\eta) - 1) & \text{for } E(r) > 0. \end{cases} \quad (6.2)$$

In the Tardis model, there are five assumptions. These restrict the origin, the monotonicity of R , and R at $t = t_0$ and assume the boundary between the Tardis region and the background to be continuous. Further, only one time-singularity, the Big Bang singularity, until present time is assumed. These assumptions prove the restriction of the curvature function: $|E(r)| \ll 1$. [2] This results in the average expansion rate being close to the background value of the Einstein–de Sitter universe. Therefore, to ensure that there will be backreaction in the Tardis model one of these assumptions should be broken to ensure a different outcome than the standard LTB model. [2] This is chosen to be the assumption regarding the monotonicity of R . This means that the numerical value of R' is important to take into account. The areal radius function will start by increasing and then changing sign two times. Thus, $R(t, r)$ will have one maximum and one minimum.¹⁶ These will be called r_1 and r_2 respectively. This creates three intervals of R' , which become useful when expressing averages.

$$\begin{aligned} \text{For } 0 \leq r \leq r_1 & \quad R'(t, r) > 0 \\ \text{For } r_1 < r \leq r_2 & \quad R'(t, r) < 0 \\ \text{For } r_2 < r & \quad R'(t, r) > 0. \end{aligned} \quad (6.3)$$

It should be noted that this sign change is equivalent to a step function which, when differentiated, is a delta function. The delta function is dropped, like in the Tardis article. [2] It will be shown later when

¹⁵This model is not used in this thesis, so we will not go into depth regarding what it assumes. The major difference is that the Szekeres model does not assume spherical symmetry in the regions. [22]

¹⁶This can be seen when implementing the model numerically (see figure 12).

calculating the average of the density over the domain $\mathcal{D}$, that the effective gravitational mass function will have a maximum and minimum at the same values of r as $R(t, r)$ (eq. 6.22).

The Tardis model has two regions, one where the curvature function is zero and one where it is not. These regions differ in the solution of Einstein's field equations, which can be seen for the areal radius (eq. 6.2). Looking at the case $E(r) > 0$ the parameter η appears, which in this case is defined by the equation: [20]

$$\sinh(\eta) - \eta = \frac{E^{3/2}}{M}(t - t_B), \quad (6.4)$$

where the functions $E(r)$, $M(r)$ and $t_b(r)$ appears. All of these can be chosen freely as in the standard LTB model. First, the $t_B(r)$ function is chosen to be 0. The effective gravitational mass $M(r)$ is defined as: [2]

$$M(r) = \frac{1}{2}H_0^2 r^3 \left[1 + e^{-\left(\frac{r-r_{peak}}{A_{val}r_b}\right)^{10}} \right], \quad (6.5)$$

where r_{peak} affects the corresponding radius of the peak of $M(r)$ and A_{val} is an arbitrary constant. The values for r_{peak} and A_{val} used in this thesis are taken from the Tardis article[2]: $r_{peak} = 0.85r_b$ and $A_{val} = 10^{-1.2}$. The function is related to the curvature function by a condition on $M(r)$'s extremum points. The derivative of $M(r)$ is calculated:

$$M'(r) = \frac{dM(r)}{dr} = \frac{1}{2}H_0^2 \left(3r^2 \left[1 + e^{-\left(\frac{r-r_{peak}}{A_{val}r_b}\right)^{10}} \right] + \frac{10}{(A_{val}r_b)^{10}} r^3 (r - r_{peak})^9 e^{-\left(\frac{r-r_{peak}}{A_{val}r_b}\right)^{10}} \right). \quad (6.6)$$

The visualisation of this can be seen in figure 10.

The effective gravitational mass $M(r)$ and its derivative $M'(r)$

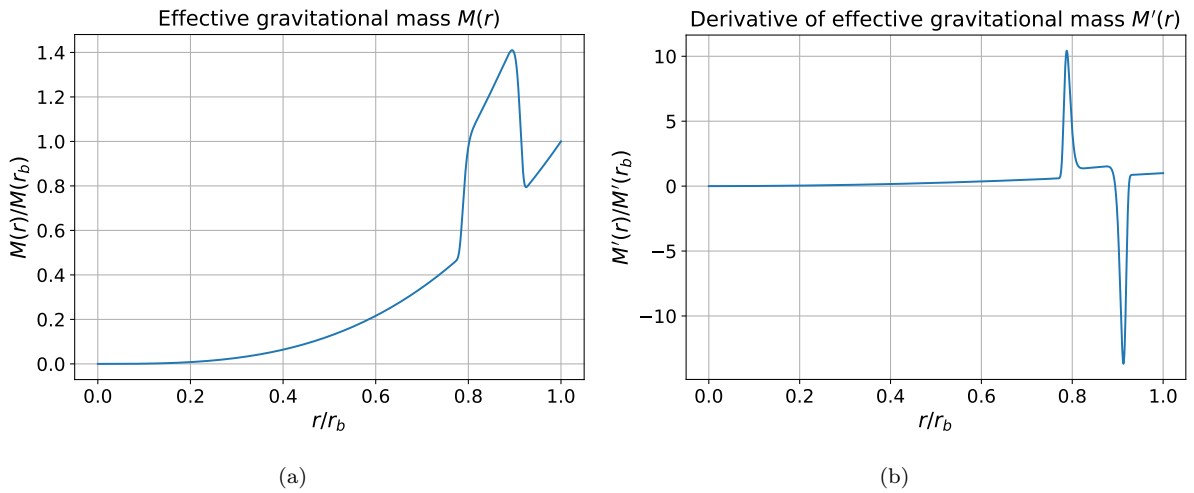


Figure 10: The effective gravitational mass $M(r)$ (a) and the radial derivative of the effective gravitational mass $M'(r)$ (b) are plotted as a function of r and are normalised with respect to the value of the function at the boundary r_b .

6.1.1 Cubic spline

At $M(r)$'s first extremum point r_1 (the maximum of $M(r)$, which can be seen in figure 10(a)) the differential of the curvature function should equal zero. This point becomes the maximum in the curvature function with a value of $E(r_1) = 0.001$, fitting with the condition of $|E(r)| \ll 1$ and the curvature function is zero at the second extremum r_2 of $M(r)$. The $E(r)$ function is chosen as in the Tardis article[2] to be

a cubic spline[23], with these restrictions, and is symmetric around the point r_1 . First, these points are found by using the bisection algorithm `scipy.optimize.bisect()` on the $M'(r)$ function. For r_1 the roots are found between $a = 0.875r_b$ and $b = 0.91r_b$, where a is the lower boundary, b is the upper boundary and r_b is the boundary of the Tardis region. This is chosen to have the value $r_b = 20$ Mpc.[2] For the second extremum r_2 , the roots are found between $a = 0.91r_b$ and $b = 0.95r_b$. The result from this is $r_1 = 0.894r_b$ and $r_2 = 0.925r_b$. These points can then be used to make the symmetric cubic spline.

The theoretical background of the cubic spline is further explained in the appendix A.3. The expressions which are explained there can then be used to build a cubic spline algorithm in the Tardis model between the points r_1 and r_2 . Another point, $r_0 = 0.864r_b$, is introduced, such that it is the reflection of r_2 in r_1 . This can be seen in figure 11.

The curvature function $E(r)$ and its derivative $E'(r)$

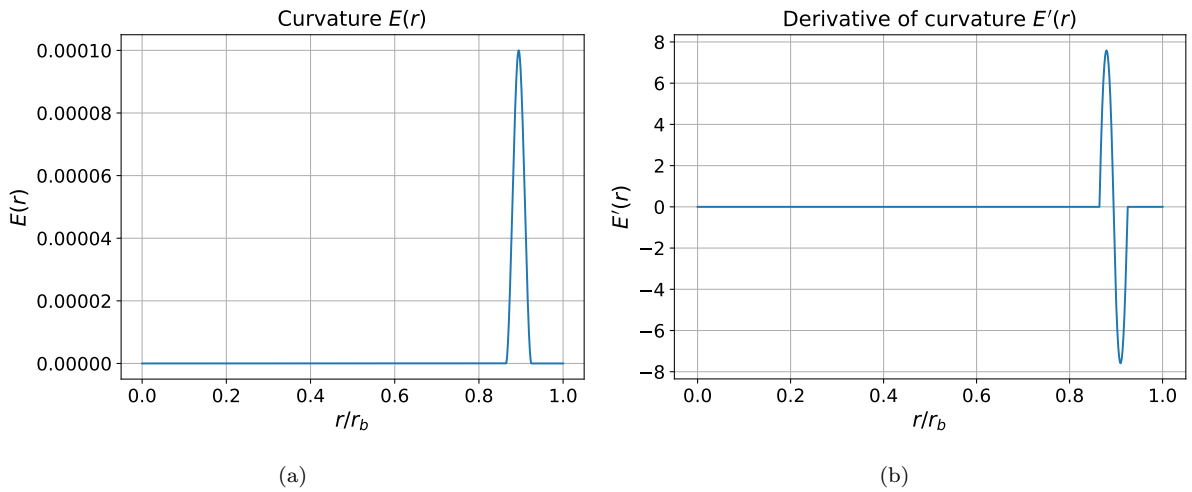


Figure 11: (a) The curvature function $E(r)$ created as a cubic spline between $r_0 \approx 0.86r_b$, $r_1 \approx 0.89r_b$, and $r_2 \approx 0.92r_b$, and (b) the derivative $E'(r)$.

6.2 Dynamical equations used in the Tardis model

Before implementing the Tardis model numerically, the dynamical equations specifying it are derived analogously to the standard LTB model, with some differences. (see 5.3)

First, an important thing to note is that the areal function is non-monotonic and that the curvature function in this model has a sign difference compared to the standard LTB model. This will change some of the dynamical equations.

First, the volume is found using equation 2.46. This element will change from the standard LTB model because of the broken assumption regarding the monotonicity of the areal function, which means that the areal radius function will change sign. This only affects functions where the square root of the three-dimensional metric is included, because the absolute value of the position-differentiated areal radius will appear:

$$V_{\mathcal{D}}(t, r) = \int_0^{r_b} \sqrt{\det({}^{(3)}g)} d^3X. \quad (6.7)$$

By computing the determinant, taking the square root and resolving the angular integrals one obtains:

$$V_{\mathcal{D}}(t, r) = 4\pi \int_0^{r_b} \frac{1}{\sqrt{1+E}} |R'| R^2 dr. \quad (6.8)$$

Using that $|E(r)| \ll 1$, the integral is similar to the volume found for the other LTB model (eq. 5.74). However, due to the broken assumption, the derivative of R is not always positive. Therefore, the integral is split up into three intervals (see equation 6.3), where the sign is changed for intervals in which $R(t, r)$ is decreasing. The first from 0 to r_1 , the next from r_1 to r_2 and the third from r_2 to r_b , where $R(r_1) > R(0)$, $R(r_1) > R(r_2)$ and $R(r_b) > R(r_2)$:

$$\begin{aligned} V_{\mathcal{D}}(t, r) &\approx 4\pi \left[\int_0^{r_1} R' R^2 dr - \int_{r_1}^{r_2} R' R^2 dr + \int_{r_2}^{r_b} R' R^2 dr \right] \\ &= \frac{4\pi}{3} \left(2R^3(r_1, t) + R^3(r_b, t) - 2R^3(r_2, t) \right). \end{aligned} \quad (6.9)$$

These intervals are chosen due to the absolute value of $R'(t, r)$ (see figure 13(b)). The change of monotonicity of the areal function can be seen in figure 12. It can be seen from the graph that the areal radius will increase until around $0.9r_b$ where it will start to decrease and then again increase. The points where R' changes sign are the same as in equation 5.69. Further, it is used that $R(t, 0) = 0$.

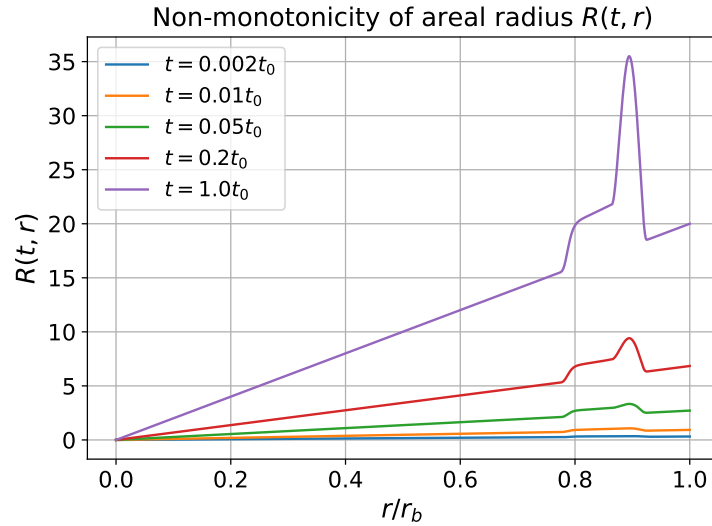


Figure 12: The areal radius as a function of r in Mpc at different times, where the three intervals can be seen as described in equation 6.3.

The expansion rate (eq. 5.79) and the shear (eq. 5.94) are the same as the ones derived in the standard LTB model since they are neither averaged nor depend on the curvature function directly:

$$\Theta(t, r) = \frac{\dot{R}'}{R'} + 2 \frac{\dot{R}}{R}, \quad (6.10)$$

$$\sigma(t, r) = \frac{1}{\sqrt{3}} \left(\frac{\dot{R}'}{R'} - \frac{\dot{R}}{R} \right). \quad (6.11)$$

The equations for $\dot{R}$, the density ρ , η' and the backreaction term Q are the same as for the standard LTB model. However, due to the definition of the metric and thus the line element, there are a few differences:

$$\dot{R}(t, r) = \sqrt{E + 2 \frac{M}{R}} \quad (6.12)$$

$$\rho(t, r) = \frac{1}{4\pi G} \frac{M'}{R^2 R'} \quad (6.13)$$

$$\eta'(t, r) = \frac{1}{\cosh(\eta) - 1} \frac{(E)^{\frac{1}{2}}}{M^2} \left(\frac{3}{2} M E' - E M' \right) \quad (6.14)$$

$$Q(t, r) = \frac{8\pi}{V_{\mathcal{D}}} \int_0^{r_b} \frac{R^2 |R'|}{\sqrt{1+E}} \left[\left(\frac{\dot{R}}{R} \right)^2 + 2 \frac{\dot{R}' R}{R' R} \right] dr - \frac{2}{3} \left[\frac{4\pi}{V_{\mathcal{D}}} \int_0^{r_b} \frac{R^2 |R'|}{\sqrt{1+E}} \left(\frac{\dot{R}'}{R'} + 2 \frac{\dot{R}}{R} \right) dr \right]^2. \quad (6.15)$$

The piecewise functions will also differ slightly since $E(r) = 0$ in most of the interval. As seen in equation 6.2, the areal radius function changes, and thus its derivative changes as well:

$$R'(t, r) = \begin{cases} \frac{3}{2} M' t^2 \left(\frac{9}{2} M t^2 \right)^{-2/3} & \text{for } E(r) = 0 \\ \frac{M'}{E} (\cosh(\eta) - 1) - \frac{M}{E^2} E' (\cosh(\eta) - 1) + \frac{M}{E} \sinh(\eta) \eta' & \text{for } E(r) > 0. \end{cases} \quad (6.16)$$

Then the density averaged over the domain $\mathcal{D}$ is computed using the definition 2.47, first calculating the angular integral:

$$\langle \rho \rangle_{\mathcal{D}} = \frac{1}{V_{\mathcal{D}}} \int_0^{r_b} \int_0^{\pi} \int_0^{2\pi} \frac{R^2 |R'|}{\sqrt{1+E}} \sin(\theta) \rho dr d\theta d\phi \quad (6.17)$$

$$= \frac{4\pi}{V_{\mathcal{D}}} \int_0^{r_b} \frac{R^2 |R'|}{\sqrt{1+E}} \left(\frac{1}{4\pi G} \frac{M'}{R^2 R'} \right) dr. \quad (6.18)$$

Again as for the volume the absolute value of R' is simplified using the intervals from equation 6.3:

$$\langle \rho \rangle_{\mathcal{D}} = \frac{1}{V_{\mathcal{D}} G} \left(\int_0^{r_l} \frac{R^2 R'}{\sqrt{1+E}} \frac{M'}{R^2 R'} dr - \int_{r_l}^{r_u} \frac{R^2 R'}{\sqrt{1+E}} \frac{M'}{R^2 R'} dr + \int_{r_u}^{r_b} \frac{R^2 R'}{\sqrt{1+E}} \frac{M'}{R^2 R'} dr \right) \quad (6.19)$$

$$= \frac{1}{V_{\mathcal{D}} G} \left(\int_0^{r_l} \frac{M'}{\sqrt{1+E}} dr - \int_{r_l}^{r_u} \frac{M'}{\sqrt{1+E}} dr + \int_{r_u}^{r_b} \frac{M'}{\sqrt{1+E}} dr \right). \quad (6.20)$$

This can be further simplified using that $|E(r)| \ll 1$. Evaluating the integrals, and using the volume $V_{\mathcal{D}}$ calculated in equation 6.9:

$$\langle \rho \rangle_{\mathcal{D}} = \frac{1}{V_{\mathcal{D}} G} \left(\int_0^{r_1} M' dr - \int_{r_1}^{r_2} M' dr - \int_{r_2}^{r_b} M' dr \right) \quad (6.21)$$

$$= \frac{M(r_1) - M(0) - (M(r_2) - M(r_1)) + M(r_b) - M(r_2)}{\frac{4}{3} \pi G (2R^3(r_1, t) + R^3(r_b, t) - 2R^3(r_2, t))} \\ = \frac{3}{4\pi G} \frac{2M(r_1) + M(r_b) - 2(M(r_2))}{2R^3(r_1, t) + R^3(r_b, t) - 2R^3(r_2, t)}, \quad (6.22)$$

where it has been used that $M(0) = 0$.

The average of the spatial curvature $\langle {}^{(3)}R \rangle_{\mathcal{D}}$ can be calculated similarly. This is done in the Tardis article[2], and gives the following results:

$$\langle {}^{(3)}R \rangle_{\mathcal{D}} = -6 \frac{E(r_1)R(r_1)}{R(r_b)^3 + 2R(r_1)^3 - 2R(r_2)^3}. \quad (6.23)$$

6.3 Numerical implementation of the Tardis model

To further scrutinize the Tardis model it is implemented numerically (see appendix A.7) similarly to the standard LTB model (see section 5.4). The constants used are the same except for the radial boundary which is only 20 Mpc similar to the Tardis article[2]. Additionally, a few different approaches are used in some cases. As mentioned above the curvature function is not given but found using the roots of the effective gravitational mass function and then implementing a cubic spline. Similar to the standard LTB model, the functions are piecewise depending on whether the curvature function is zero. In the Tardis model, the curvature function is zero in large parts of the region which results in a different areal radius function. Moreover, as seen in the previous section, the integrals, where the numerical value of R' is included, are split up into three regions.

Implementing the dynamical equations of the Tardis model numerically is expected to give rise to a non-zero backreaction Q , in contrast to the standard LTB model. This is examined by implementing

both the volume and the theoretical backreaction term numerically. The backreaction is further examined using symbolic regression to obtain a model fitting to the values for Q . This is done using the as-yet unpublished SR-Benchmark tool made by Mattias Ermakov Thing[24] and Sofie Marie Koksang[22] to do the symbolic regression. Time t/t_0 is used as the input parameter, thus resulting in an expression for the backreaction Q related to time t/t_0 .

6.4 Results in the Tardis model

To examine the Tardis model's dynamical equations, the curvature function $E(r)$ and the effective gravitational mass $M(r)$ are used to compute the areal radius $R(t, r)$ (eq. 6.2). Here root finding of η in equation 6.4 is performed similarly to the standard LTB model (section 5.4). The result is plotted at different times in figure 13 (a) and is similar to the result from the Tardis article[2]. Figure 13 (a) shows that at later times the maximum of the areal radius is larger. For $r < 0.86r_b$, thus when the curvature function is zero, the areal radius is seen to be $\sim 25\%$ larger than the background value. When the curvature function is non-zero, the areal radius increases and then again equals the background value when the curvature equals zero. This shows that the proper distance will increase inside the Tardis region. Further, at later times the peak of the areal radius is larger.

The areal radius $R(t, r)$ and its spatial derivative $R'(t, r)$

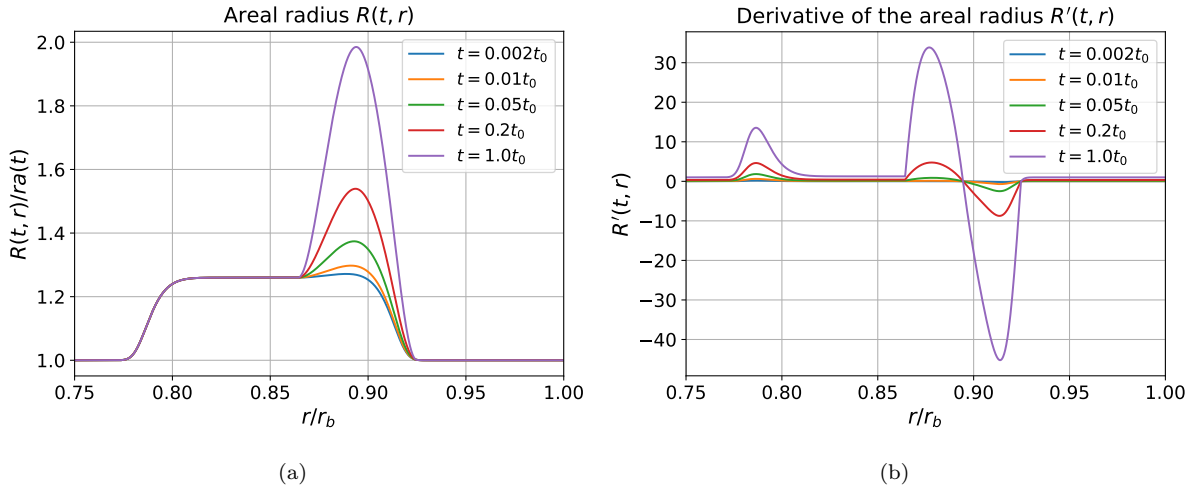


Figure 13: (a) The areal radius at different times divided by $ra(t)$, where $a(t)$ is the Einstein–de Sitter scale factor. (b) The derivative $R'(t, r)$ both figures is plotted versus $\frac{r}{r_b}$.

6.4.1 Density in the Tardis model

Using equation 6.13 the density is plotted at different times. Here the result differs from the Tardis article[2], when comparing the results from $r/r_b \approx 0.9$ and above. In the Tardis article[2] the density function smoothly and monotonically rises to one after the extremum, whereas in figure 14 the density is first increasing, then decreasing and then increasing abruptly to the background Einstein–de Sitter density. To further examine this difference from the article the number of points in r is changed. This can be seen in figure 15.

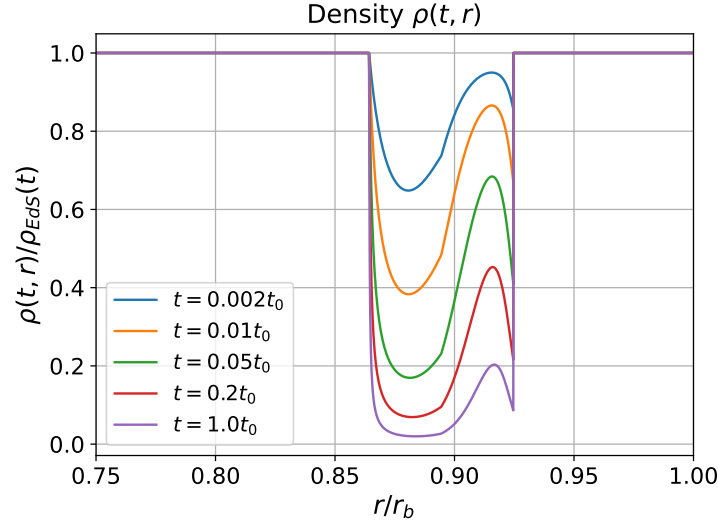


Figure 14: The density ρ normalised to an Einstein–de Sitter universe at different times.

As seen, the graphs for 1000 and 10000 points in r are both very similar to figure 14. However, using only 100 points in r gives a different graph, that more closely resembles the one in the article[2]. This does not bear any physical meaning but is simply because the points differing from figure 14 are not plotted due to the low number of points. The number of points in r is not specified in the Tardis article[2], but a look at the (unpublished) code indicates that 1000 points have been used.

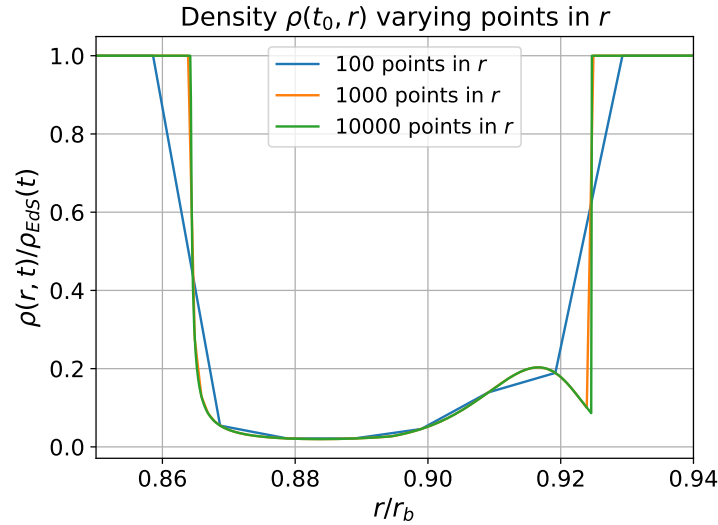
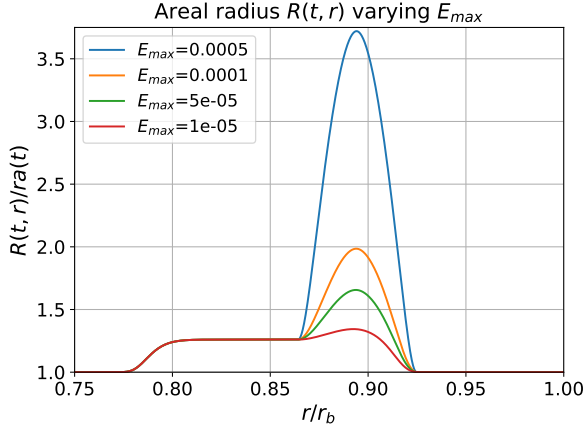


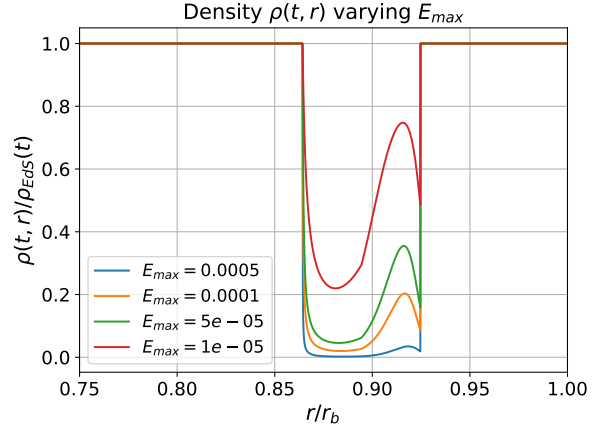
Figure 15: The density ρ plotted with a varying number of points in r .

To further evaluate the model, the constants used are varied to get a better understanding of the model's parameters. This can be seen in figure 16. These constants include A_{val} , E_{max} and r_b . As seen on the graphs the value of the constant in the effective gravitational mass function A_{val} (figure 16 (c)+(d)) moves the graphs of the areal and curvature function to the left for smaller values and the opposite for larger values. Moreover, the extremum of the areal radius function is greater for smaller values. The maximum value of the curvature (figure 16 (a)+(b)) also controls the values of the extrema of the graphs. For smaller values, the areal function has a lower extremum and a smaller dip can be seen in the density graph.

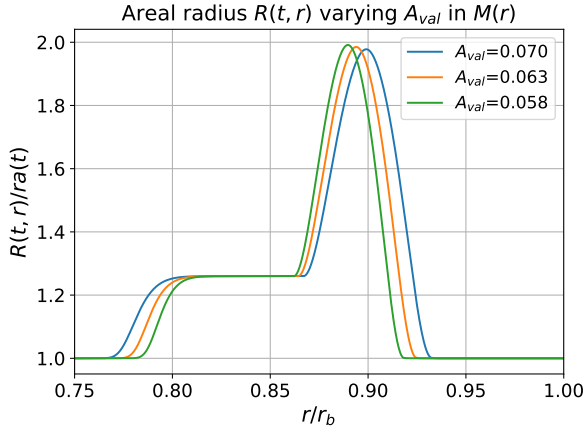
Variation of parameters



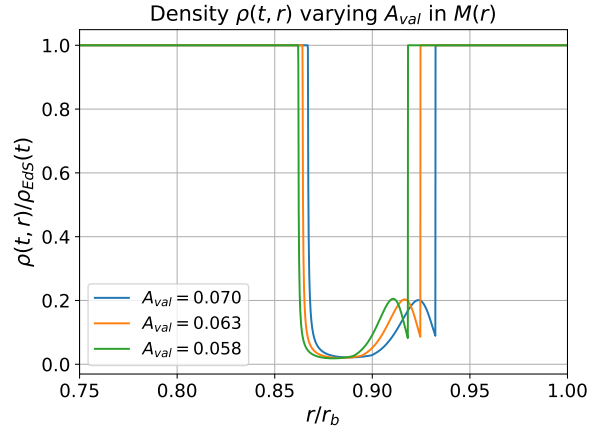
(a) Areal radius and variation of E_{max}



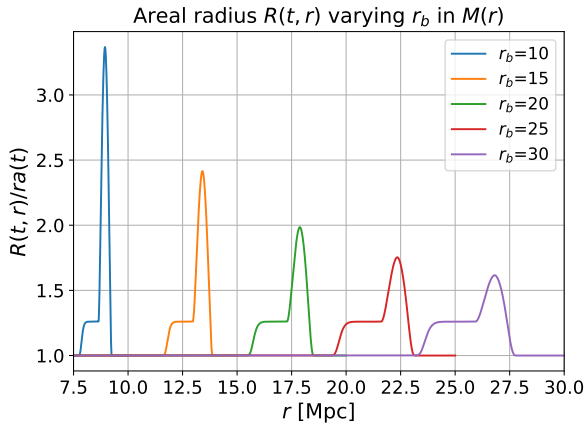
(b) Density and variation of E_{max}



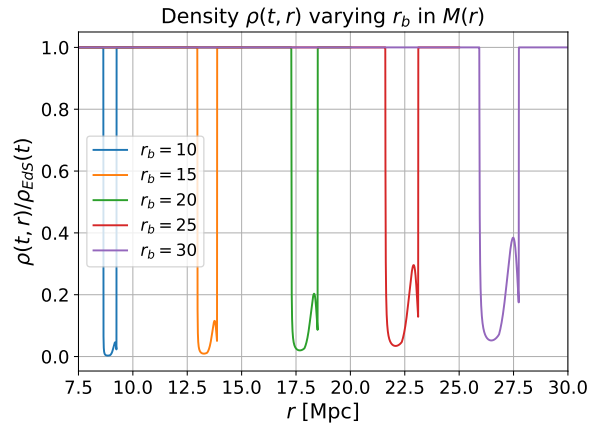
(c) Areal radius and variation of A_{val}



(d) Density and variation of A_{val}



(e) Areal radius and variation of r_b



(f) Density and variation of r_b

Figure 16: Variations of the maximum value of the curvature function E_{max} , the value of the constant in the gravitational mass function A_{val} , and the boundary of the void r_b . The colour difference indicates different values of the parameters, where the yellow lines in (a), (b), (c) and (d) are the original values. In (e) and (f) it is the green graph which has the original value.

Additionally, for a larger value of E_{max} , the error in the density graph becomes smaller. Lastly, varying the size of the Tardis region by varying r_b also shifts the graph along the x-axis. This can be seen, since figure 16 (e) and (f) are plotted only as a function of r and are not normalised. Further, changing r_b squeezes the peak of the areal radius function for smaller r_b . This results in a smaller error in the density graph when compared to the continuous density graph in the Tardis article[2]. This could again be an alternative explanation of the difference of the density graph 14 in this thesis from the Tardis article. [2]

6.4.2 Expansion rate Θ and shear σ in the Tardis model

The expansion rate Θ is obtained using equation 6.10. Similarly, the shear σ is obtained from equation 6.11. These are plotted for varying time steps in figure 17.

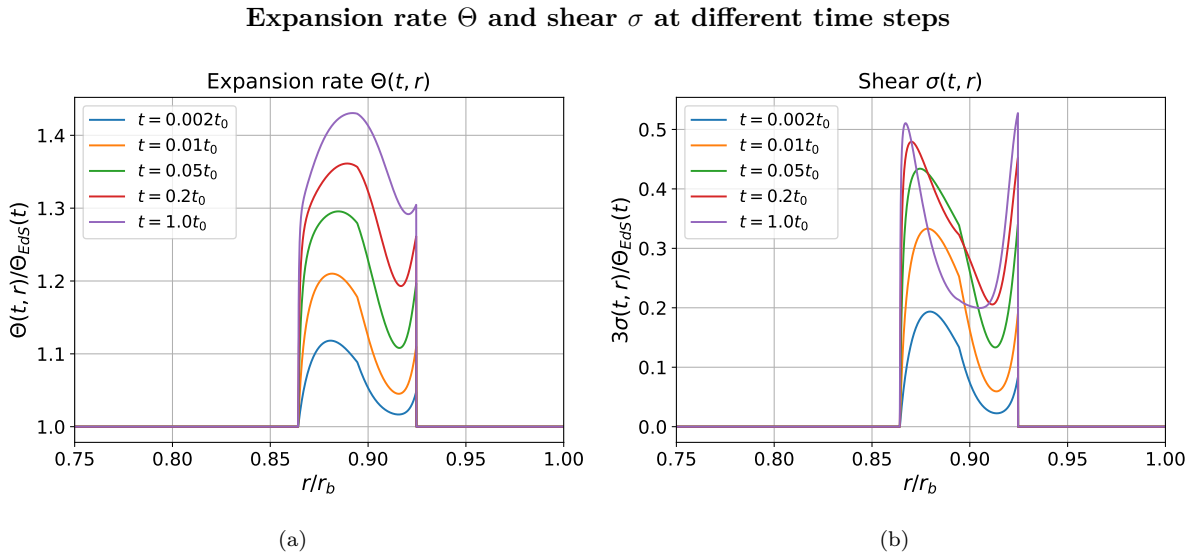


Figure 17: (a) The expansion rate Θ and (b) The shear σ .

The expansion rate Θ is of similar size to the expansion rate from the Tardis article[2] but has the same difference as rho, where it behaves differently above $0.9r_b$. The shear σ is also impacted by this and is furthermore around twice as big as in the Tardis article[2].

The averages of these two components can be used to calculate the backreaction term Q (eq. 3.18), which is further elaborated on in the following section.

6.4.3 Backreaction in the Tardis model

The volume from equation 6.9 is plotted against the volume of one at the same radius for the background. This can be seen in figure 18. The graph shows how the fraction of the underdense Tardis region increases in time, thus making backreaction possible in the Tardis model. The volume of the underdense region will be more than ten times as large as an Einstein–de Sitter volume of the same radius at present time. This also shows how the Tardis model is defined by introducing these underdense regions which have a larger volume, than what appears on the boundary of the region. It should also be noted that the background Einstein–de Sitter volume is not equal to the Tardis volume at early times. As done in the standard LTB model the backreaction term Q is also examined. In contrast to the standard LTB model where the volume graph (figure 7) showed that no backreaction occurs, the volume graph (see figure 18) indicates that backreaction does occur. However, the backreaction term Q is plotted in the standard LTB

model (see figure 9). This is done to get the precision of the numerical implementation of the standard LTb model to get a bound on the precision of the backreaction in the Tardis model.

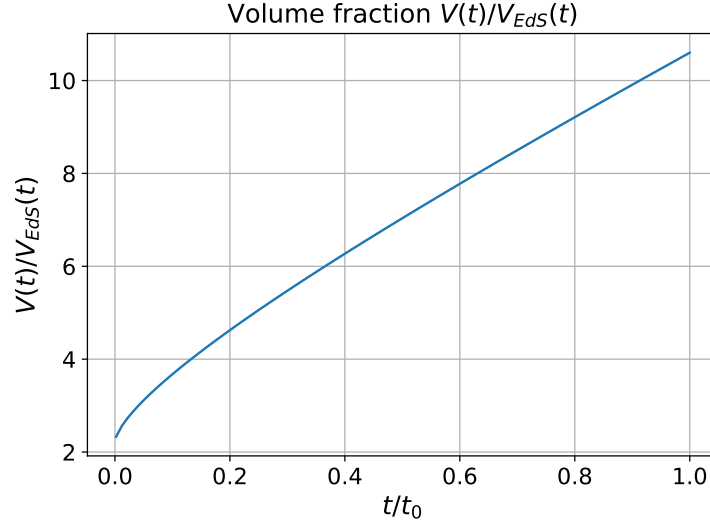


Figure 18: The volume $V(t)$ of the Tardis region divided with the Einstein–de Sitter background at the same radius r_b is plotted with respect to the time.

The backreaction $Q(t, r)$ as a function of time and radii

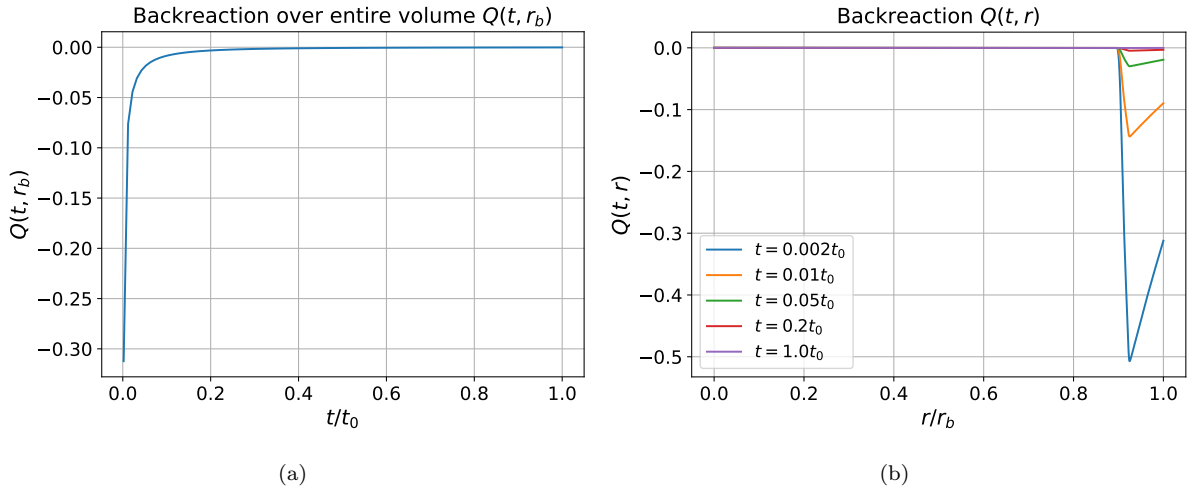


Figure 19: (a) Backreaction Q as a function of time, and (b) the backreaction Q as a function of radius for different time steps.

The backreaction term for the Tardis model can be seen in figure 19. The backreaction is plotted with respect to radii (figure 19 (b)), is zero until it reaches a radii ~ 17.5 Mpc. Here it drops and then rises again. The drop happens around the maximum of the curvature function $E(r)$ (see figure 11). The backreaction is also plotted versus time (figure 19 (a)). Here the backreaction is also negative but goes to zero at later times. As seen in equation 3.26 the backreaction term is positive, thus a negative backreaction leads to a decelerating effect on the Universe.

Next, the backreaction is further analysed using symbolic regression. As seen in figure 20 the symbolic regression perfectly fits the backreaction even though only the first 180 time-steps were used out of 200, thus emitting 10% of the point to test if the prediction of the symbolic regression matched the graph.

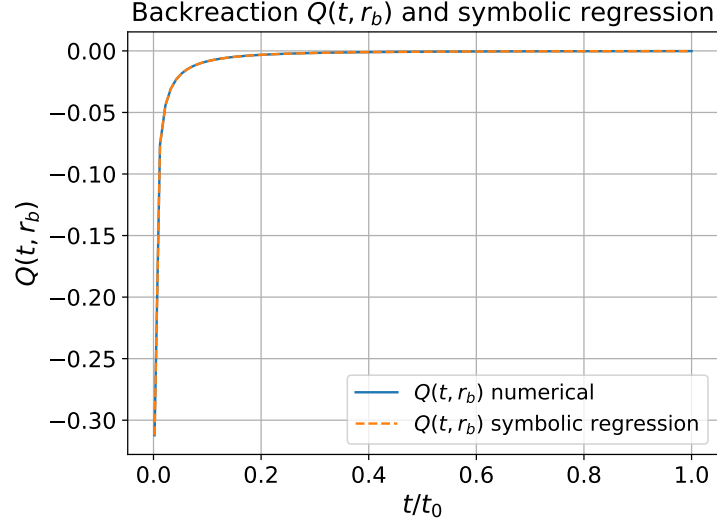


Figure 20: The figure shows the backreaction of the Tardis model, where the dashed, orange line shows the symbolic regression and the blue line shows the same as in figure 19(a)

Using the SR-Benchmark tool mentioned in 6.3, choosing ITEA[25] as the symbolic regression tool. The full expression obtained by the symbolic regression is:

$$Q(t) = -2.5575 \cdot 10^{-2} + 5.27566 \cdot 10^{-2} \sqrt{|t/t_0|} - 1.02771 \cdot 10^{-2} \sqrt{|(t/t_0)^2|} \\ + 4.89687 \cdot 10^{-3} \ln \left((t/t_0)^{-5} \right) - 1.23558 \cdot 10^{-4} \sqrt{|(t/t_0)^{-2}|} - 1.69657 \cdot 10^{-2} \sqrt{|(t/t_0)^{-1}|}, \quad (6.24)$$

with a mean squared error of $3.06 \cdot 10^{-12}$.

To properly evaluate the backreaction term, the density parameter Ω is calculated for all parameters: mass (eq 3.30), curvature (eq. 3.31) and backreaction (eq. 3.32). This can be seen in figure 21 (a). As seen on the graph, the different density parameters do not add up to one as they should (see eq. 3.29). In figure 21 (b) the density parameter for the backreaction term can be seen. The figure shows, that the density parameter is positive and increases until around $\sim t = 0.4t_0$.

The density parameter for all components $\Omega(t)$ and of the backreaction $\Omega_Q(t)$

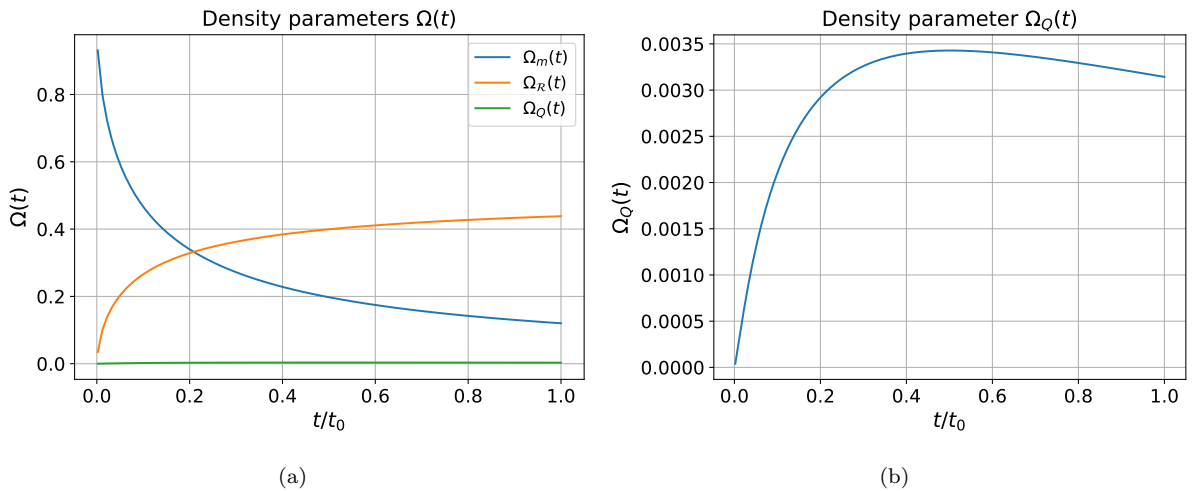


Figure 21: (a) The density parameter for all components Ω_R , Ω_m and Ω_Q and (b) the density parameter of the backreaction both plotted versus time t/t_0

7 Discussion

7.1 Inhomogeneous universes

This thesis has examined inhomogeneous, cosmological models which differ from the standard cosmological model. The standard model assumes isotropy and homogeneity by the approximation that inhomogeneity at small- and intermediate scales does not affect the large-scale dynamics of the Universe. This model explains the accelerated expansion of the Universe using the cosmological constant Λ , which is the only term leading to an accelerated expansion.

An alternative model is the LTB model, which discards the assumption of homogeneity and isotropy as this only holds for large scales around 100 Mpc and above. It can also be discussed if it is a good approximation on large scales with voids having an average size of 10 – 100 Mpc.[26] One possibility to examine this assumption is by using an inhomogeneous metric, such as the LTB metric. This metric takes the inhomogeneity of the density distribution of the energy in the Universe into account. To describe an inhomogeneous universe, the Buchert equations are used. These equations are analogous to the Friedmann equations in the homogeneous model. The Buchert equations are first averaged after the local equations are found, oppositely to the FLRW model. This leads to an additional term in the Buchert equations, which is the cosmic backreaction:

$$Q \equiv \frac{2}{3} \text{Var}(\Theta) - 2\langle \sigma^2 \rangle . \quad (7.1)$$

This depends both on the variance of the expansion rate and the shear. The variance of the expansion rate will be positive and non-zero for a non-homogeneous expansion. The average of the shear will however contribute to deceleration of the expansion when it is non-zero. Thus, the cosmic backreaction in the LTB model is, together with the cosmological constant, a possible explanation of the acceleration of the expansion of the Universe, as long as the variance of the expansion is larger than the average of the shear. To further examine cosmic backreaction, a simple toy model, a two-region model, is first considered.

7.2 The two-region model

The two-region model is a simple model which takes two regions; an overdense and an underdense region, and develops them over time. These regions locally decelerate, however the model shows an overall accelerated expansion.

Figure 2 shows the deceleration parameter as a function of time. The deceleration parameter of the two-region model follows a similar development in time as the one of the standard cosmological model with the optimal value of the volume at the turnover point $f_2 = 0.3$. Even though the exact values of the model cannot be trusted, because it is a simple model, the behaviour is still worth discussing since the resulting graph of q is close to Λ CDM. The fact that an overall acceleration happens can be deduced from the volume fraction (figure 1). The plot of the volume fraction in the two-region model shows that eventually, the volume fraction of the underdense region will dominate at the turnover point. Thus, contributing to overall acceleration instead of deceleration, since the overdense region expands slower, than the underdense region, which contains less mass, and thus a smaller gravitational decelerating factor. Additionally, as seen in figure 2 the acceleration can become larger in the two-region model than in the Λ CDM model. However, as the underdense region becomes larger and larger, the overdense region will become less and less significant, leading to an accelerated expansion. When the overdense becomes insignificant, the constant expansion of the underdense region will dominate.

7.3 The standard LTB model

To further examine the backreaction a standard LTB model is considered. This model makes it possible to include shear, which does not appear in the two-region model where the mass distribution is constant. This is also important since shear is the factor which decelerates the cosmic backreaction and should thus be taken into account.

The standard LTB model is a more complex model. However, the model is simplified by setting the cosmological constant to zero. The model represents a spherically symmetric inhomogeneous universe with an LTB void at its centre and an Einstein–de Sitter background.

Three functions are defined to specify the LTB model; the effective gravitational mass $M(r)$ (eq. 5.54), the curvature $k(r)$ (eq. 5.56) which are both continuous, and the Big Bang time $t_B(r) = 0$.

7.3.1 Density in the standard LTB model

Numerical implementation of the above expressions then allows the density fractions of both the void ρ_{LTB} and the Einstein–de Sitter universe ρ_{EdS} to be considered. In figure 6 (a), the ratio of the density fractions $\frac{\rho_{LTB}}{\rho_{EdS}}$ is seen at different times. For all considered time steps, the void of the standard LTB model is seen to consist of an underdense region ($\rho_{LTB} < \rho_{EdS}$) and an overdense region ($\rho_{LTB} > \rho_{EdS}$). The density of the overdense region peaks at $r \approx 0.75r_b$. The maximum value of the peak increases with time, as the overdense region must attract additional matter gravitationally and thus increase the density in this part of the region.

The areal radial dependence of the ratio of density fractions $\frac{\rho_{LTB}}{\rho_{EdS}}$ is further considered in figure 6 (b). The boundary of the void r_b is seen to increase as time increases. Thus, as time progresses, the areal radius at which the boundary is placed $R(t, r_b)$ is seen to increase. The areal radius can be interpreted as the proper distance and thus shows that the physical distance to the boundary of the void increases. This can be interpreted as the expansion of the void itself, as the proper distance depends on the scale factor of the Universe.

7.3.2 Backreaction in the standard LTB model

The dynamical equations specifying the considered standard LTB model are all continuous. In contrast to the two-region model, this gives rise to shear throughout the LTB model. In the two-region model, shear only occurs at the boundary between the two regions and not within each region. Due to continuity, the LTB model on the other hand gives rise to shear continuously throughout the constructed inhomogeneous universe which affects the backreaction of the LTB models.

To determine whether backreaction occurs in the standard LTB model, the volume of both regions (V_{LTB} and V_{EdS} respectively) are considered. In figure 7 (b), the initial volumes are integrated radially from the centre of the void to the boundary ($r_f = 0.5r_b$), thus ending within the void described by the LTB metric. The values of the volume V_{LTB} are seen to be larger than the ones of the Einstein–de Sitter volume V_{EdS} . Thus locally, backreaction occurs when iterating over time since the LTB volume expands faster than the Einstein–de Sitter volume.

Considering figure 7 (a), the initial volumes are instead integrated radially to the boundary of the void p ($r_f = r_b$). The difference between the volume V_{LTB} and V_{EdS} are seen to be approximately zero when integrated over the considered times. This can further be seen in figure 8 (a), where the fraction $\frac{V_{LTB}}{V_{EdS}}$ is seen to be approximately one ($1.0001 + 10^{-6} \approx 1$) when considering the entire void. This is in contrast to figure 8 (b), where the volume is considered halfway to the boundary of the void. In this case, $\frac{V_{LTB}}{V_{EdS}}$ is seen to be larger than one, that is $V_{LTB} > V_{EdS}$.

Thus backreaction can only occur in the standard LTB model, when considering volumes ending within the void, in congruence with the theoretical derivation of the volumes. This corresponds to the small-scale structures of the universe and thus shows how small-scale inhomogeneity can lead to backreaction. In contrast, no backreaction occurs in the standard LTB model when considering the entire void, as this describes the expansion of the overall universe, and thus the large-scale structures.

To further determine the backreaction and to examine the numerical implementation and methods used, the theoretical backreaction term Q arising in the Buchert equations (eq. 3.18) is implemented. Considering the dynamical equation for Q in figure 9 (a), the backreaction is seen to be of the order of 10^{-6} , which is approximately zero. This is thus consistent with what is seen when considering the expansion of volumes over the entire void (figure 7 (a)). The backreaction term Q is however not exactly zero as expected. This could be due to numerical errors when implementing the model numerically, including approximating the volume V_{LTB} as a Riemann sum (eq. 5.103) and solving the integrals in Q (eq. 3.18). By trial and error, these numerical errors are seen to decrease when the number of radial steps Nr is increased up to a threshold value of $Nr = 10000$ steps. Therefore, when implementing the standard LTB model, $Nr = 10000$ is used, as this reduces the numerical error on the dynamical equation for the backreaction term Q depicted as a function of time.

Thus when considering the entire void, both methods agree that no cosmic backreaction occurs in the constructed standard LTB model describing an inhomogeneous universe.

7.4 The Tardis model

Since the standard LTB model does not result in backreaction, the Tardis model is examined. It contains voids which are bigger on the inside, thus allowing them to expand faster than the background universe. These voids are implemented using the LTB metric with a different curvature $E(r)$ and effective gravitational mass $M(r)$. Further, the areal radius function is non-monotonic, this property should, in contrast to the monotonic areal function of the standard LTB model, introduce backreaction.

The curvature function $E(r)$ and effective gravitational mass $M(r)$ are made as defined in the Tardis article[2], and the obtained results are thus identical to the results in the article (figure 11 and 10). From these, the areal radius $R(t, r)$ and its derivative $R'(t, r)$ are calculated (figure 13). $R(t, r)$ is similar, but not identical to the plot in the Tardis article[2], as the top points in the article are lower than in figure 13. This might be due to slightly different values for c or H_0 , which can change the overall scaling of the function. Further, the constants are varied in figure 16. Here it can be seen that varying both E_{max} , A_{val} and r_b can change the maximum value of the areal function. A slight change of these could thus result in a different areal function which changes the size of both the shear and expansion rate of the model.

7.4.1 Density in the Tardis model

The density of the Tardis model shown in figure 14 differs from the results from the article[2]. The difference persists in the plots in section 6.4.3. It has been rigorously checked that all the former plots as well as the density function are correct. Even when numerically copying the plots from the Tardis article[2], and using these to compute the density, a similar plot to figure 14 is obtained. Hence the difference must lie in how the numerical work is implemented. In the unpublished code from Tardis article[2], a Taylor expansion to high orders is made for the functions in the Tardis model when doing the numerical work, presumably to avoid division by zero. This might have led to different results. In this thesis, the Taylor expansion has not been used, instead, the functions are implemented directly. As the functions are defined as piecewise functions, dividing by zero never occurs. When using a large number

of positional points ($\sim 10^6$) division by small numbers occurs at the boundary of the piece-wise function. This problem is simply avoided by using a lower number of positional points ($\sim 10^4$).

The density is plotted for different numbers of points in r in figure 15. The graph for 100 r points looks similar to the results from the Tardis article[2], but this is because the points that differ from this are not included in the 100 points in r . The code on which the Tardis model[2] is based indicates that the number of points used for r is 1000, which, according to figure 15, is enough to obtain the density function seen in figure 14. Further, the constant varied in 16 could also contribute to a different density curve and could together with other changes yield a similar graph as the density graph in the Tardis article. The obtained results seem non-physical because the density rises and falls and then rises rapidly along the boundary of the underdense region. It is however the result obtained using several numerical and analytical methods for calculating $\rho(t, r)$, and must therefore be the mathematically correct result.

7.4.2 Backreaction in the Tardis model

The volume fraction shown in figure 18 shows an expanding volume inside the void compared to the background, similar, but not identical to Tardis[2], as the volume fraction only reaches 7 in the article, and 11 in figure 18. This is assumed to be related to the fact that the areal radius R in this thesis is also larger than the one in the article, as discussed above. This difference in volume results in the Tardis region being not only larger but both larger and growing on the inside, without ever crossing over to the outside. This behaviour is expected to lead to backreaction.

Backreaction is then considered by plotting it against both time and radii (figure 19 (a) and (b)). These plots show a small backreaction (in order of 0.1) which is largest at radii's around $\sim 0.9r_b$ and at early times. To further analyse the backreaction, symbolic regression is applied. When analysing the backreaction term it is possible to obtain a good estimate, however, one could argue that the curve is easy to fit and it could be interesting to use fewer points in the symbolic regression. This could test how great it is at predicting backreaction in the future from the model used. The method thus introduces a way to find the scale factor dependency of the backreaction which can be converted to redshift that can be observed. It should be noted that in this thesis symbolic regression is done with respect to time.

The density parameters of the Tardis model are also examined (figure 21 (a) and (b)). The density parameters plotted show that the backreaction is negligible at all times with respect to the density parameters of mass and curvature. However, another aspect of backreaction in this model is that backreaction might affect the curvature density parameter $\Omega_{\mathcal{R}}$. Backreaction could thus possibly appear in the curvature density parameter.[2]

Comparing the backreaction in this thesis to the one in the Tardis article, both are small ($|\Omega_Q| < 0.01$ at the boundary r_b) but there is a sign difference, as Ω_Q is negative in the Tardis article, thus making Q contribute positively to the acceleration due to the sign in the averaged Raychaudhuri equation (eq. 3.26), which describes the acceleration.[2] The backreaction term Q obtained in this thesis results in a negative contribution to the acceleration. The main difference between the two is that the shear σ calculated in this thesis (figure 17 (b)) is around twice as big as the shear calculated in the Tardis article[2], giving a larger negative contribution to the backreaction term. The expansion rate Θ calculated in this thesis (figure 17 (a)) is of similar size to the one calculated in the Tardis article[2], but suffers from the same problems as the density ρ . Halving the shear σ but otherwise using the results from this thesis leads to a backreaction similar to the one calculated in the Tardis article[2] (see appendix A.4).

The results obtained in our version of the Tardis model correspond with the results of our supervisor.[22] Thus, the Tardis article's results cannot be reproduced. Further, the density parameter of the backreaction is negligible.

8 Conclusion

This thesis has investigated an alternative or additional explanation for the accelerated expansion of the Universe to the Λ CDM model assumed in standard cosmology, where the accelerated expansion is explained by dark energy. This standard cosmological model further assumes that small- and intermediate-scale inhomogeneity does not affect the large-scale dynamics of the Universe, by deriving the local equations after taking the spatial average of the universe. An alternative model can therefore be derived by first deriving the local equations and then taking the spatial average. This approach introduces a set of dynamical equations known as *the Buchert equations*[\[1\]](#), which gives rise to an additional term called the *cosmic backreaction*. The backreaction term can contribute to positive acceleration when the variance of the expansion rate is significantly larger than the decelerating terms; the energy density and the shear.

To examine backreaction, a two-region toy model of a simple inhomogeneous universe is constructed introducing an overdense and an underdense region evaluated over time. Numerical implementation of this model shows that despite locally each of the regions can only decelerate, accelerating expansion can occur in the overall toy model universe. This accelerated expansion is similar to the one of the Λ CDM model making the theory of backreaction, occurring due to inhomogeneity, a possible explanation.

This is further studied by taking shear into account introducing a standard Lemaître–Tolman–Bondi (LTB) model with two regions. The model is specified by defining three continuous functions; the effective gravitational mass, the curvature and the Big Bang time. This leads to dynamical equations for multiple functions, including the areal radius, the density, and the volume. Both numerical implementation and the theoretical expressions of the standard LTB model show that no backreaction occurs when considering the entire toy model universe.

To investigate an LTB model where the backreaction does not vanish, the Tardis model is considered. The areal radius describing the model is non-monotonic, which can lead to backreaction. The Tardis model is further defined by having Tardis regions, which have a larger volume on the inside than expected from the outside. The density graph of the Tardis model considered in this thesis differs from the one in the original article[\[2\]](#). This deviation of the density function stems from the derivatives of the areal radius, which affects the components of the backreaction; the expansion rate and the shear.

The backreaction in the Tardis model is first examined through the volume fraction. This shows both that non-zero backreaction occurs and that the Tardis regions are larger on the inside and expanding faster than the background. Further examination of the backreaction through the density parameters however results in a negligible backreaction. The results of the Tardis model are thus both indecisive and contradictory.

Despite the results on backreaction, the use of symbolic regression could be an important tool for inhomogeneous cosmological models in general, making it possible to relate backreaction to observations through the scale factor and thus redshift.

Even though cosmic backreaction does not explain the accelerating expansion of the Universe from the models considered in this thesis, cosmic backreaction of significant size could be a necessary addition to the standard model of cosmology and aid in explaining the accelerated expansion of the Universe.

9 Acknowledgements

The completion of this thesis was supported by several people, with whom we want to share our gratitude. First, to our supervisor, assistant professor Sofie Marie Koksang, who always gave us a helping hand and shared her great knowledge of cosmology and general relativity. Further, she made the framework for our thesis. Next, we want to thank PhD student Alexander Oestreicher for helping us with a further understanding of cosmology and for helpful comments on how we could improve this thesis. Lastly, we are grateful that we were given Adam Skovbjerg Knudsen's and Lea Krarup's bachelor thesis from 2022, which derived the foundation of the local equations used as a starting point in this thesis.

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A Appendix

A.1 Derivations for the two-region model

A.1.1 Deriving H for the two-region model

$$H = \frac{\dot{a}}{a} = \frac{\frac{d}{dt} \left(\sqrt[3]{a_1^3 + a_2^3} \right)}{\sqrt[3]{a_1^3 + a_2^3}} = \frac{\frac{1}{3}(a_1^3 + a_2^3)^{-2/3}(3a_1^2\dot{a}_1 + 3a_2^2\dot{a}_2)}{\sqrt[3]{a_1^3 + a_2^3}} \quad (\text{A.1})$$

$$= \frac{a_1^2\dot{a}_1 + a_2^2\dot{a}_2}{a^3} = \frac{a_1^3\frac{\dot{a}_1}{a_1} + a_2^3\frac{\dot{a}_2}{a_2}}{a^3} = \frac{a_1^3H_1 + a_2^3H_2}{a^3} = \frac{a_1^3}{a^3}H_1 + vH_2 \quad (\text{A.2})$$

Using the following definitions:

$$H_1 = \frac{\dot{a}_1}{a_1}, H_2 = \frac{\dot{a}_2}{a_2} \quad (\text{A.3})$$

$$a^3 = a_1^3 + a_2^3 \quad (\text{A.4})$$

$$v = \frac{a_2^3}{a_1^3 + a_2^3} = \frac{a_2^3}{a^3} \quad (\text{A.5})$$

$$h = \frac{H_2}{H_1} \quad (\text{A.6})$$

H can be rewritten into the following.

$$H = \frac{a_1^3}{a^3}H_1 + vH_2 = \frac{a_1^3}{a^3}H_1 + \frac{a_2^3}{a^3}H_1 - \frac{a_2^3}{a^3}H_1 + vH_2\frac{H_2}{H_1} = \frac{a_1^3 + a_2^3}{a^3}H_1 - vH_1 + vhH_1 = H_1(1 - v + vh) \quad (\text{A.7})$$

A.1.2 Deriving q_2 for the two-region model

Using the definition of the deceleration parameter q , it can be found for the overdense region:

$$q_2 = -\frac{\ddot{a}_2 a_2}{\dot{a}_2^2}, \quad (\text{A.8})$$

where the derivatives denote the time derivative. Thus differentiating the composite function $a_2 \propto 1 - \cos \varphi$:

$$\dot{a}_2 = \frac{da_2}{d\varphi} \frac{d\varphi}{dt} = \sin \varphi \frac{d\varphi}{dt}. \quad (\text{A.9})$$

Using that the inverse of the derivative is equal to the derivative of t with respect to φ :

$$\dot{a}_2 = \sin \varphi \frac{1}{\frac{dt}{d\varphi}} \quad (\text{A.10})$$

$$= \frac{\sin \varphi}{1 - \cos \varphi}. \quad (\text{A.11})$$

The same approach can be used to find the double differentiation of the scale factor of the overdense region:

$$\ddot{a}_2 = \frac{d\dot{a}_2}{d\varphi} \frac{d\varphi}{dt} \quad (\text{A.12})$$

$$= \left(\frac{\cos \varphi}{1 - \cos \varphi} + \sin \varphi \left(-\frac{1}{(1 - \cos \varphi)^2} \sin \varphi \right) \right) \frac{1}{\frac{dt}{d\varphi}} \quad (\text{A.13})$$

$$= \left(\frac{\cos \varphi}{1 - \cos \varphi} - \frac{\sin^2 \varphi}{(1 - \cos \varphi)^2} \right) \frac{1}{(1 - \cos \varphi)} \quad (\text{A.14})$$

$$= \frac{\cos \varphi}{1 - \cos \varphi} - \frac{\sin^2 \varphi}{(1 - \cos \varphi)^3}. \quad (\text{A.15})$$

Inserting this in the equation for q_2 :

$$q_2 = -\frac{\cos \varphi}{1 - \cos \varphi} - \frac{\sin^2 \varphi}{(1 - \cos \varphi)^3} \frac{\ddot{a}_2 a_2}{\dot{a}^2} \frac{(1 - \cos \varphi)^2}{\sin^2 \varphi} (1 - \cos \varphi), \quad (\text{A.16})$$

which can be reduced to:

$$q_2 = -\frac{\cos \varphi (1 - \cos \varphi)}{\sin^2 \varphi} + 1 \quad (\text{A.17})$$

$$= \frac{-\cos \varphi + \cos^2 \varphi + \sin^2 \varphi}{\sin^2 \varphi} \quad (\text{A.18})$$

$$= \frac{1 - \cos \varphi}{\sin^2 \varphi}. \quad (\text{A.19})$$

Thus, an expression for the deceleration parameter of the overdense region is obtained.

A.1.3 Deriving q for the two-region model

The deceleration parameter q is defined as following and calculated using the product rule:

$$q = \frac{-1}{H^2} \frac{\ddot{a}}{a} = \frac{-1}{H^2 a} \frac{d}{dt} \left(\frac{\dot{a}_1 a_1^2 + \dot{a}_2 a_2^2}{a^2} \right) \quad (\text{A.20})$$

$$= \frac{-1}{H^2 a} \left(\frac{da^{-2}}{dt} + \frac{d\dot{a}_1}{dt} \frac{a_1^2}{a^2} + \frac{da_1^2}{dt} \frac{\dot{a}_1}{a^2} + \frac{d\dot{a}_2}{dt} \frac{a_2^2}{a^2} + \frac{da_2^2}{dt} \frac{\dot{a}_2}{a^2} \right) \quad (\text{A.21})$$

Calculating the time derivatives using $\frac{da^{-2}}{dt} = -2a^{-3}\dot{a}$ and $\frac{da_i^2}{dt} = 2a_i\dot{a}_i$.

$$q = \frac{-1}{H^2 a} \left(\frac{-2\dot{a}}{a^3} (\dot{a}_1 a_1^2 + \dot{a}_2 a_2^2) + \ddot{a}_1 \frac{a_1^2}{a^2} + 2a_1 \dot{a}_1 \frac{\dot{a}_1}{a^2} + \ddot{a}_2 \frac{a_2^2}{a^2} + 2a_2 \dot{a}_2 \frac{\dot{a}_2}{a^2} \right) \quad (\text{A.22})$$

Factoring out a common factor of a^{-2} and collecting terms.

$$q = \frac{-1}{H^2 a^3} \left(\frac{-2\dot{a}}{a} (\dot{a}_1 a_1^2 + \dot{a}_2 a_2^2) + \ddot{a}_1 a_1^2 + 2a_1 \dot{a}_1^2 + \ddot{a}_2 a_2^2 + 2a_2 \dot{a}_2^2 \right) \quad (\text{A.23})$$

Calculating $\dot{a}$ explicitly:

$$\dot{a} = \frac{d\sqrt{a_1^3 + a_2^3}}{dt} = \frac{3a_1^2 \dot{a}_1 + 3a_2^2 \dot{a}_2}{3(a_1^3 + a_2^3)^{2/3}} = \frac{a_1^2 \dot{a}_1 + a_2^2 \dot{a}_2}{a^2} \quad (\text{A.24})$$

Inserting this into the expression for q .

$$q = \frac{-1}{H^2 a^3} \left(-2 \frac{a_1^2 \dot{a}_1 + a_2^2 \dot{a}_2}{a^3} (\dot{a}_1 a_1^2 + \dot{a}_2 a_2^2) + \ddot{a}_1 a_1^2 + 2a_1 \dot{a}_1^2 + \ddot{a}_2 a_2^2 + 2a_2 \dot{a}_2^2 \right) \quad (\text{A.25})$$

Rewriting this:

$$q = \frac{2}{H^2 a^6} ((a_1^2 \dot{a}_1 + a_2^2 \dot{a}_2)^2 - a^3 (a_1 \dot{a}_1^2 + a_2 \dot{a}_2^2)) - \frac{\ddot{a}_1 a_1^2}{H^2 a^3} - \frac{\ddot{a}_2 a_2^2}{H^2 a^3} \quad (\text{A.26})$$

Looking at the parenthesis:

$$(a_1^2 \dot{a}_1 + a_2^2 \dot{a}_2)^2 - a^3 (a_1 \dot{a}_1^2 + a_2 \dot{a}_2^2) = a_1^4 \dot{a}_1^2 + a_2^4 \dot{a}_2^2 + 2a_1^2 \dot{a}_1 a_2^2 \dot{a}_2 - a_1^4 \dot{a}_1^2 - a_2^4 \dot{a}_2^2 - a_2^3 a_1 \dot{a}_1^2 - a_1^3 a_2 \dot{a}_2^2 \quad (\text{A.27})$$

$$= 2a_1^2 \dot{a}_1 a_2^2 \dot{a}_2 - a_2^3 a_1 \dot{a}_1^2 - a_1^3 a_2 \dot{a}_2^2 \quad (\text{A.28})$$

The remaining terms are inserted back into the expression for q .

$$q = \frac{2}{H^2 a^6} (2a_1^2 \dot{a}_1 a_2^2 \dot{a}_2 - a_2^3 a_1 \dot{a}_1^2 - a_1^3 a_2 \dot{a}_2^2) - \frac{\ddot{a}_1 a_1^2}{H^2 a^3} - \frac{\ddot{a}_2 a_2^2}{H^2 a^3} \quad (\text{A.29})$$

Factoring out a $a_1^3 a_2^3$ and simplifying by recognising a square of binomials.

$$q = \frac{2a_1^3 a_2^3}{H^2 a^6} \left(2 \frac{\dot{a}_1 \dot{a}_2}{a_1 a_2} - \frac{\dot{a}_1^2}{a_1^2} - \frac{\dot{a}_2^2}{a_2^2} \right) - \frac{\ddot{a}_1 a_1^2}{H^2 a^3} - \frac{\ddot{a}_2 a_2^2}{H^2 a^3} = -\frac{2a_1^3 a_2^3}{H^2 a^6} \left(\frac{\dot{a}_1}{a_1} - \frac{\dot{a}_2}{a_2} \right)^2 - \frac{\ddot{a}_1 a_1^2}{H^2 a^3} - \frac{\ddot{a}_2 a_2^2}{H^2 a^3} \quad (\text{A.30})$$

Replacing $\frac{a_i}{a_i}$ with H_i :

$$q = -\frac{2a_1^3 a_2^3}{H^2 a^6} (H_1 - H_2)^2 - \frac{\ddot{a}_1 a_1^2}{H^2 a^3} - \frac{\ddot{a}_2 a_2^2}{H^2 a^3} \quad (\text{A.31})$$

Using the following definitions:

$$q_1 = -\frac{1}{H_1^2} \frac{\ddot{a}_1}{a_1}, \quad q_2 = -\frac{1}{H_2^2} \frac{\ddot{a}_2}{a_2} \quad (\text{A.32})$$

Rewriting the first term:

$$-\frac{2a_1^3 a_2^3}{H^2 a^6} (H_1 - H_2)^2 = -\frac{2}{H_1^2 (1-v+vh)^2} \frac{a_1^3 a_2^3}{a^3 a^3} (H_1^2 + H_2^2 - 2H_1 H_2) \quad (\text{A.33})$$

$$\begin{aligned} &= -\frac{2}{(1-v+vh)^2} (1-v)v \frac{(H_1^2 + H_2^2 - 2H_1 H_2)}{H_1^2} \\ &= -\frac{2}{(1-v+vh)^2} (1-v)v(1+h^2-2h) = -\frac{2v(1-v)(1-h)^2}{(1-v+vh)^2} \end{aligned} \quad (\text{A.34})$$

And the second term:

$$-\frac{\ddot{a}_1 a_1^2}{H^2 a^3} = -\frac{1}{H_1^2} \frac{\ddot{a}_1 a_1^2}{(1-v+vh)^2 (a_1^3 + a_2^3)} \frac{a_1}{a_1} \quad (\text{A.35})$$

$$= -\frac{1}{H_1^2} \frac{\ddot{a}_1}{a_1} \frac{a^3 - a_2^3}{(1-v+vh)^2 (a_1^3 + a_2^3)} = q_1 \frac{1-v}{(1-v+vh)^2} \quad (\text{A.36})$$

And, at last, the third term:

$$-\frac{\ddot{a}_2 a_2^2}{H^2 a^3} = -\frac{1}{H_1^2} \frac{\ddot{a}_2 a_2^2}{(1-v+vh)^2 (a_1^3 + a_2^3)} \frac{H_2^2 a_2}{H_2^2 a_2} \quad (\text{A.37})$$

$$= -\frac{H_2^2}{H_1^2 H_2^2} \frac{\ddot{a}_2 a_2^3}{(1-v+vh)^2 (a_1^3 + a_2^3) a_2} = q_2 \frac{vh^2}{(1-v+vh)^2} \quad (\text{A.38})$$

Thus q becomes:

$$q = -\frac{2v(1-v)(1-h)^2}{(1-v+vh)^2} + q_1 \frac{1-v}{(1-v+vh)^2} + q_2 \frac{vh^2}{(1-v+vh)^2} \quad (\text{A.39})$$

a_1 is proportional to time. For the void region, which is empty, $q_1 = 0$. Source: [6] and [7]. q is from [12]

A.2 The used constants c , H_0 , and G with alternative units

When implementing the dynamical equations of both the standard LTB model and the Tardis model numerically, values for the speed of light c , the Hubble parameter at present time H_0 and the gravitational constant G are needed. In SI-units, these are: [6]

$$c = 299792458 \frac{\text{m}}{\text{s}}, \quad H_0 = 70 \frac{\text{km/s}}{\text{Mpc}}, \quad G = 6.673 \cdot 10^{-11} \frac{\text{m}^3}{\text{kg} \cdot \text{s}^2}. \quad (\text{A.40})$$

Since both the standard LTB- and Tardis model are cosmological models, it is of interest to convert the units of the constants c , H_0 and G to proper cosmological units, namely $\text{s} \rightarrow \text{Gyr}$, $\text{m} \rightarrow \text{Mpc}$ and $\text{kg} \rightarrow \text{M}_\odot$:

$$1\text{Gyr} = 3.156 \cdot 10^{16} \text{s}, \quad 1\text{Mpc} = 3.086 \cdot 10^{22} \text{m}, \quad 1\text{M}_\odot = 1.989 \cdot 10^{30} \text{kg}. \quad (\text{A.41})$$

This leads to the following values for c , H_0 and G respectively:

$$\begin{aligned} c &= 299792458 \frac{\text{m}}{\text{s}} \cdot \frac{\left(3.156 \cdot 10^{16} \frac{\text{s}}{\text{Gyr}}\right)}{\left(3.086 \cdot 10^{22} \frac{\text{m}}{\text{Mpc}}\right)} \\ &= 306.6 \left(\frac{\text{m}}{\text{s}}\right) \left(\frac{\text{s}}{\text{Gyr}}\right) \left(\frac{\text{Mpc}}{\text{m}}\right) \\ &= 306.6 \frac{\text{Mpc}}{\text{Gyr}} \end{aligned} \tag{A.42}$$

$$\begin{aligned} H_0 &= 70 \frac{\text{km/s}}{\text{Mpc}} \cdot \frac{\left(3.156 \cdot 10^{16} \frac{\text{s}}{\text{Gyr}}\right)}{\left(3.086 \cdot 10^{22} \frac{\text{m}}{\text{Mpc}}\right)} \\ &= 71.6 \cdot 10^{-6} \left(\frac{\text{m}}{\text{s}}\right) \left(\frac{\text{s}}{\text{Gyr}}\right) \left(\frac{\text{Mpc}}{\text{m}}\right) \\ &= 71.6 \cdot 10^{-3} \frac{1}{\text{Gyr}} \end{aligned} \tag{A.43}$$

$$\begin{aligned} G &= 6.673 \cdot 10^{-11} \frac{\text{m}^3}{\text{kg} \cdot \text{s}^2} \cdot \frac{\left((1.989 \cdot 10^{30} \frac{\text{kg}}{\text{M}_\odot}) \left(3.156 \cdot 10^{16} \frac{\text{s}}{\text{Gyr}}\right)^2\right)}{\left(3.086 \cdot 10^{22} \frac{\text{m}}{\text{Mpc}}\right)^3} \\ &= 4.50 \cdot 10^{-15} \left(\frac{\text{m}^3}{\text{kg} \cdot \text{s}^2}\right) \left(\frac{\text{kg}}{\text{M}_\odot}\right) \left(\frac{\text{s}^2}{\text{Gyr}^2}\right) \left(\frac{\text{Mpc}^3}{\text{m}^3}\right) \\ &= 4.50 \cdot 10^{-15} \frac{\text{Mpc}^3}{\text{M}_\odot \cdot \text{Gyr}^2} . \end{aligned} \tag{A.44}$$

A.3 The cubic spline

Before generating a cubic spline the theoretical background is explained. A spline is a function which is made up of multiple polynomials connecting different points. These points are called *knots*. This concept is used in interpolation sometimes preferred over polynomial interpolation, which finds the lowest possible degree polynomial which connects these points. [27] Thus, the method of spline interpolation often succeeds with a lower degree of polynomial since it fits these polynomials within small subsets. This results in a similar fit as the polynomial interpolation but with a smaller degree making the method useful. In the evaluation of when to use which method, *Runge's phenomena* is important. This describes that for a higher degree polynomial with equidistant¹⁷ points the error can diverge, in this case, a spline interpolation is one of the methods which solves the problem. [28]

The algorithm for a cubic spline interpolation thus creates n cubic polynomials between a sequence of knots with the length $n + 1$. Creating a polynomial $q_i(x) = y_i$ between the two knots (x_{i-1}, y_{i-1}) and (x_i, y_i) . Where the index i is defined as $i = 1, \dots, n$ which makes the final cubic spline function a function between the points (x_0, y_0) and (x_n, y_n) . [23] The algorithm aims to have the lowest possible bending of each polynomial this can be described by the curvature: [29]

$$\kappa = \frac{q''}{(1 + q'^2)^{3/2}} , \tag{A.45}$$

where the primes denote derivative and the given formula for the curvature applies for any graph of a function. This also requires the q' and q'' to exist for the polynomials at the knots. Thus, the cubic spline interpolation creates a differentiable function which is a piece-wise polynomial function with degree 3 that takes an interval $[a, b]$ and maps it to the set of real numbers $\mathbb{R}$: [27]

$$Q : [a, b] \mapsto \mathbb{R} . \tag{A.46}$$

¹⁷Where equidistant describes points with the same distance separating them.

A.3.1 Algorithm of a cubic spline

The algorithm requires six inputs. First, it requires two given knots thus both x and y coordinates. These are called (x_1, y_1) and (x_2, y_2) . Further, it requires the slopes k_1 and k_2 at these knots. Where the polynomial function has to satisfy: [23]

$$q(x_1) = y_1, \quad q(x_2) = y_2 \quad (\text{A.47})$$

$$q'(x_1) = k_1, \quad q'(x_2) = k_2. \quad (\text{A.48})$$

The polynomial can be expressed using three definitions. The knot vector t and the coefficients a, b :

$$t(x) = \frac{x - x_1}{x_2 - x_1} \quad (\text{A.49})$$

$$a = k_1(x_2 - x_1) - (y_2 - y_1) \quad (\text{A.50})$$

$$b = -k_2(x_2 - x_1) + (y_2 - y_1). \quad (\text{A.51})$$

This gives the cubic polynomial: [23]

$$q(x) = (1 - t(x))y_1 + t(x)y_2 + t(x)(1 - t(x))((1 - t(x))a + t(x)b). \quad (\text{A.52})$$

To ensure the k values of the polynomial also match the given values the derivative of q needs to be computed. This can be done using the chain rule:

$$q'(x) = \frac{dq}{dx} = \frac{dq}{dt} \frac{dt}{dx} = \frac{y_2 - y_1}{x_2 - x_1} (1 - t(x))y_1 + t(x)y_2 + t(x)(1 - t(x))((1 - t(x))a + t(x)b). \quad (\text{A.53})$$

A.4 Backreaction in the Tardis model using a smaller σ

Halving the values of the shear σ to make it more similar to the values in the Tardis article[2] leads to a positive backreaction term Q . This resembles the backreaction obtained in the article and is shown in figure 22.

The backreaction $Q(t, r)$ using $\sigma/2$

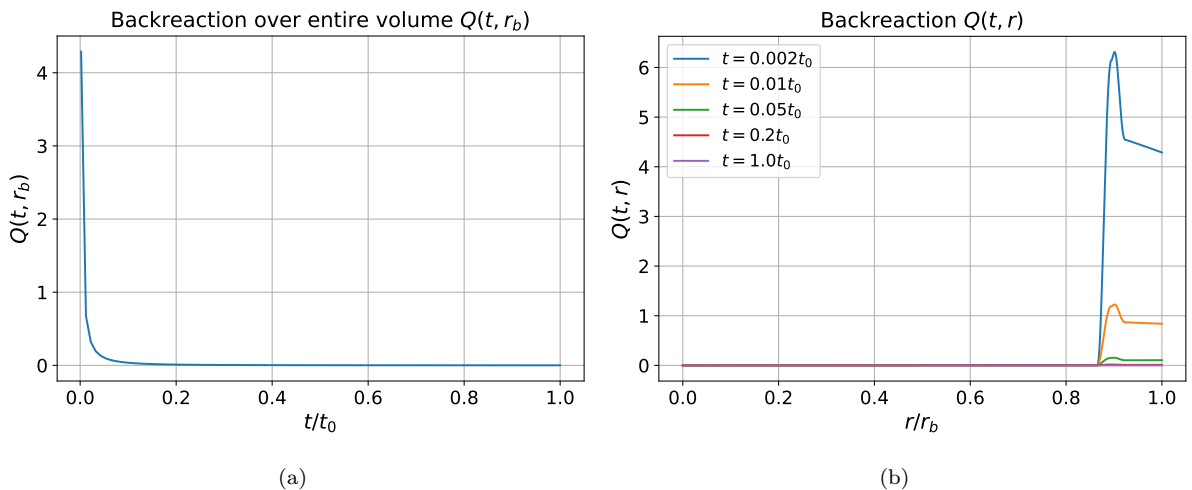


Figure 22: (a) Backreaction Q as a function of time, and (b) The backreaction Q as a function of radius for different time steps, both with the shear σ divided by two.

Furthermore, this has been tested by taking the values from the plots of Θ and σ in the Tardis article and integrating these over the volume. This results in the following, similar-looking plots shown in figure 23:

The backreaction $Q(t, r)$ using the plots from the Tardis article

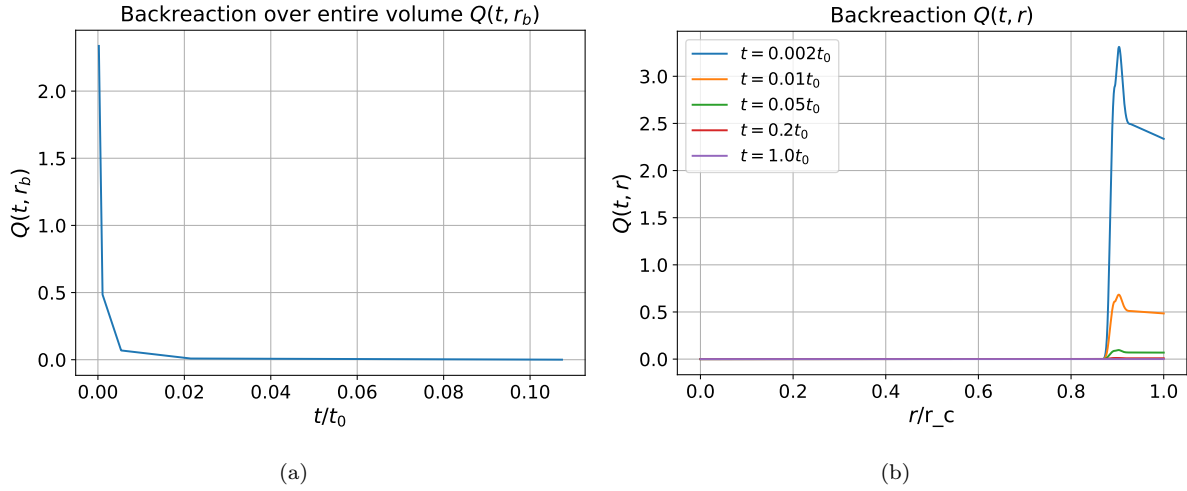


Figure 23: (a) Backreaction Q as a function of time, and (b) The backreaction Q as a function of radius for different time steps, both using the plots from the Tardis article[2] to obtain Θ and σ .

This shows the same overall result for the backreaction, which, however, is half of what is obtained when dividing σ by two.

A.5 Code for the Two-region model

```
1 import numpy as np
2 import matplotlib.pyplot as plt
3 from scipy.integrate import solve_ivp
4
5 plt.rc('axes', titlesize=16)      # fontsize of the axes title
6 plt.rc('axes', labelsz=16)       # fontsize of the x and y labels
7 plt.rc('xtick', labelsz=14)      # fontsize of the tick labels
8 plt.rc('ytick', labelsz=14)      # fontsize of the tick labels
9 plt.rc('legend', fontsize=13)    # legend fontsize
10 plt.rc('figure', titlesize=16)   # fontsize of the figure title
11
12 # Defining constants used throughout
13
14 H0 = 0.07
15 Omegar0 = 0
16 Omegam0 = 0.3
17 Omegal0 = 0.7
18 Omega0 = 1
19 a0 = (1,)
20 wr=1/3
21 wm=0
22 wl=-1
23 t0 = 1/H0*0.964 # From Ryden
24
25 # theta defined as linspace
26 N = 100000
27 thetastart = 0# 1e-7
28 theta0 = 3/2*np.pi
29 thetaend = 2*np.pi
30 theta = np.linspace(thetastart, thetaend, N)
```

```

31
32 f2 = 0.3 # as Rasanen - overdense region
33 f1 = 1 - f2
34
35 # Equations found to plot our q
36 t = (theta-np.sin(theta))
37 q1 = 0
38 q2 = (1-np.cos(theta))/((np.sin(theta))**2)
39 h = np.sin(theta)*(theta-np.sin(theta))/((1-np.cos(theta))**2)
40 H1 = 1/(t)
41 v = (np.pi)**3*f2*(1-np.cos(theta))**3/(8*f1*(theta-np.sin(theta))**3+(np.pi)**3*f2*(1-np
    .cos(theta))**3)
42 v1 = 8*f1*(theta-np.sin(theta))**3/(8*f1*(theta-np.sin(theta))**3+(np.pi)**3*f2*(1-np.cos
    (theta))**3)
43 p = (1-v+h*v)
44 p2 = p**2 #parentese squared
45 H = H1*p
46 q = q1*(1-v)/(p2)+q2*v*h**2/(p2)-2*v*(1-v)*(1-h)**2/(p2)
47
48 # Volume fraction over time
49 plt.figure()
50 plt.plot(theta,v)
51 plt.plot(theta,v1)
52 # Plot custimization
53 plt.grid(),plt.xlabel("$\\varphi$"), plt.ylabel("$v\\varphi$"),plt.legend(["Overdense
    region","Void"])
54 plt.xlim(left=0)
55 plt.tight_layout()
56 plt.title("Volume fraction of the two regions as a function of $\\varphi$")
57 plt.savefig("Volume fraction.pdf", format="pdf", bbox_inches="tight")
58 plt.show()
59
60 # Toy model with different values of f
61 plt.figure()
62 f = [0.1,0.2,0.3,0.4,0.5,0.6,0.7,0.75] # different values of f2
63 for i in range(0,len(f)):
64     vf = (np.pi)**3*f[i]*(1-np.cos(theta))**3/(8*(1-f[i])*(theta-np.sin(theta))**3+(np.pi)
        **3*f[i]*(1-np.cos(theta))**3)
65     pf = (1-vf+h*vf)
66     pf2 = pf**2 #parentese squared
67     qf = q1*(1-vf)/(pf2)+q2*vf*h**2/(pf2)-2*vf*(1-vf)*((1-h)**2)/(pf2)
68     plt.plot(theta,qf, label=f'$f_2={f[i]}$')
69
70
71 #plt.style.use("seaborn-v0_8-whitegrid")
72 plt.title("Varying volume percentage at turnover point $f_2$")
73 plt.xlabel("$\\varphi$")
74 plt.ylabel("$q\\varphi$")
75 plt.legend(loc="lower left")
76 plt.xlim([0,2*np.pi])
77 plt.tight_layout()
78 plt.grid()
79 plt.savefig("Varying f2.pdf", format="pdf", bbox_inches="tight")
80 plt.show()
81
82 # LCDM model
83

```

```

84 def a(t):
85     return (np.sinh((t)*3*H0/2*np.sqrt(Omega10))*np.sqrt(Omega0/Omega10))**(2/3)
86 def da(t):
87     return H0*(Omega0/(a(t))+Omega10*(a(t)**2))**(1/2)
88 def dda(t):
89     denominator = Omega10+(1-Omega10-Omega0)*a(t)**(-2)+Omega0*a(t)**(-3)
90     Omega0= Omega0*a(t)**(-3)/denominator
91     Omega1= Omega10/denominator
92     return -da(t)**2/a(t)*1/2*(Omega0*(1+3*wm)+Omega1*(1+3*wl))
93 def qLCDM(t):
94     return -dda(t)*a(t)/(da(t)**2)
95
96 timespace = np.linspace(0,2*t0)
97 plt.plot(timespace,a(timespace))
98
99 ## With  $\theta$  instead of t
100
101 theta_array = np.linspace(thetastart,thetaend,N)
102 theta0 = 3/2*np.pi
103 theta0 = 3/2*np.pi
104 times = t0*(theta_array-np.sin(theta_array))/(theta0-np.sin(theta0))
105
106 plt.plot(theta_array,a(times))
107
108 plt.plot(theta_array,qLCDM(times))
109
110 import matplotlib.ticker as tck
111 fig, ax1 = plt.subplots(constrained_layout=True)
112 ax1.plot(theta/np.pi,q, label='Two-region model')
113 ax1.plot(theta_array/np.pi,qLCDM(times), label='$\\Lambda$CDM')
114 ax1.xaxis.set_major_formatter(tck.FormatStrFormatter('%g  $\pi$ '))
115 ax1.set_xlabel("$\\varphi$")
116 ax1.set_ylabel("$q(\\varphi)$")
117 plt.title('Deceleration parameter  $q$  of time  $t$  and  $\varphi$ ')
118
119 # Define the new tick locations for the second x-axis
120 new_tick_locations = np.array([0.0, 0.25, 0.5, 0.75, 1, 1.25, 1.5, 1.75, 2.0])
121
122 # Define a function to map old x-values to new x-values ( $x - \sin(x)$ )
123 def tick_function(X):
124     return t0*(X*np.pi - np.sin(X*np.pi))/(theta0-np.sin(theta0))
125
126 # Create the second x-axis (twin)
127 ax2 = ax1.twinx()
128 ax2.set_xlim(ax1.get_xlim()) # Set the same limits as the first x-axis
129 ax2.set_xticks(new_tick_locations)
130 ax2.set_xticklabels([f'{tick:.2f}' for tick in tick_function(new_tick_locations)])
131 ax2.set_xlabel('time [Gyr]')
132
133 ax1.legend()
134 ax1.grid()
135 plt.tight_layout()
136 plt.savefig("Deceleration vs theta and t.pdf", format="pdf", bbox_inches="tight")
137 plt.show()

```


A.6 Code for the standard LTB model

```
1 import numpy as np
2 import matplotlib.pyplot as plt
3
4 from scipy import optimize
5 from tqdm import tqdm
6
7 """# Definition of constants"""
8
9 k_max = 2e-7          # Maximal value of curvature
10 r_b = 40              # Boundary of void [Mpc]
11
12 m = 4                 # Constant for curvature function
13 n = 4                 # Constant for curvature function
14
15 small = 1e-3          # Initial radius [Mpc]
16
17 H_0 = 71.58781594e-3  # Hubble constant [1/Gyr]
18 c = 306.5926758       # Speed of light [Mpc/Gyr]
19 G = 4.498234911e-15   # G [Mpc^3/(M_sun*Gyr^2)]
20
21 # Initial time values
22 a_i = 1/1000          # Initial scale factor (CMB)
23 t_0 = (2/3)*(1/H_0)    # Present time [Gyr]
24 t_i = t_0*a_i**(3/2)   # Initial time [Gyr]
25
26 Nt = 100              # Number of time steps
27 Nr = 1000             # Number of radial steps
28
29 r_i = small           # Initial radius
30 r_f = r_b + 10        # Final radius
31 #r_f = r_b/2
32
33 a = 0                 # Boundary for root-finding
34 b = 100               # Boundary for root-finding
35
36 t = np.linspace(t_i, t_0, Nt + 1)  # Time values
37 r = np.linspace(r_i, r_f, Nr + 1)  # Radial values
38
39
40 """# Definition of functions"""
41
42 # Curvature function
43 def k(r, k_max, r_b, m, n):
44     return np.where(r <= r_b, -1 * k_max * r ** 2 * ((r / r_b) ** n - 1) ** m, 0)
45
46 # Mass function
47 def M(r, H0):
48     return 1/2*H0**2*(r**3)/(c**2)
49
50 # Development angle
51 def f_eta(eta, c, t, k, M):
52     return np.sinh(eta)-eta-c*t*(-k)**(3/2)/M
53
54 # Areal radius
55 def A(r, M, k, eta, r_b, t_0, t):
```

```

56     return np. where(r <= r_b, M/k*(1-np.cosh(eta)), a_func(t)*r)
57
58 # Scale factor
59 def a_func(t):
60     return (t/t_0)**(2/3)
61
62 """Spatial derivatives: """
63
64 # Spatial derivative of mass function
65 def drM(r, H_0, c):
66     return (3/2)*((H_0**2)/(c**2))*r**2
67
68 # Spatial derivative of curvature
69 def drk(r, k_max, r_b, m, n):
70     return np.where(r <= r_b, -2*r*k_max*((r/r_b)**n-1)**m - r*k_max*n*m*((r/r_b)**n-1)
71                     *(m-1)*(r/r_b)**n, 0)
72
73 # Spatial derivative of areal radius
74 def drA(M, drM, k, drk, eta, dreta, r, r_b, t, t_0):
75     return np.where(r <= r_b, drM/k * (1-np.cosh(eta)) - M/(k**2)*drk*(1-np.cosh(eta)) - M/
76                     k*np.sinh(eta)*dreta, a_func(t))
77
78 # Spatial derivative of eta
79 def dreta(M, drM, k, drk, eta, c, t, r, r_b):
80     return np.where(r <= r_b, 1/(np.cosh(eta)-1) * c*t*(-k)**(1/2)/(M**2) * ( k*drM - 3/2*M
81                     *drk ), 0)
82
83 """Density: """
84
85 # Rho LTB
86 def rho(c, G, drM, A, drA):
87     return (c**4)/(4*np.pi*G)*drM/(A**2*drA)
88
89 # Rho Eds
90 def rho_Eds(t):
91     return 1 / (6*np.pi*G*t**2)
92
93 """Volume: """
94
95 # Volume LTB
96 def V_LTB(r, k, A, drA):
97     return 4*np.pi* np.cumsum(drA / (np.sqrt( 1+k)) * A**2* np.diff(r)[0])
98
99 # Volume Eds
100 def V_Eds(t, r_f):
101     return 4*np.pi/3* (a_func(t)**3) * (r**3)
102
103 """# Pre-loop definitions
104
105 Inital vectors / matrices:
106 """
107
108 eta = np.zeros([Nr+1, Nt+1])
109 A_values = np.zeros([Nr+1, Nt+1])
110
111 # Spatial derivatives
112 drM_values = np.zeros(Nr+1)

```

```

110 drk_values = np.zeros(Nr+1)
111 dreta_values = np.zeros([Nr+1, Nt+1])
112 drA_values = np.zeros([Nr+1, Nt+1])
113
114 # Density
115 rho_values = np.zeros([Nr+1, Nt+1])
116 rho_Eds_values = np.zeros([Nr+1, Nt+1])
117
118 # Volume
119 V_LTB_values = np.zeros([Nr+1, Nt+1])
120 V_Eds_values = np.zeros([Nr+1, Nt+1])
121
122 """Calling functions: """
123
124 k_values = k(r, k_max, r_b, m, n)
125 M_values = M(r, H_0)
126
127 drM_values = drM(r, H_0, c)
128 drk_values = drk(r, k_max, r_b, m, n)
129
130 """# Master loop"""
131
132 for i in tqdm(range(0, Nt+1)):
133
134     for j in range(0, Nr+1):
135
136         def test_func(eta):
137             return f_eta(eta, c, t[i], k_values[j], M_values[j])
138
139         # Values for eta and A
140         eta[j,i] = optimize.bisect(test_func, a, b)
141
142         A_values[j,i] = A(r[j], M_values[j], k_values[j], eta[j,i], r_b, t_0, t[i])
143
144         # Values for spatial derivatives
145         dreta_values[j,i] = dreta(M_values[j], drM_values[j], k_values[j], drk_values[j], eta[j,i], c, t[i], r[j], r_b)
146
147         drA_values[j,i] = drA(M_values[j], drM_values[j], k_values[j], drk_values[j], eta[j,i], dreta_values[j,i], r[j], r_b, t[i], t_0)
148
149         # Values for density
150         rho_values[j,i] = rho(c, G, drM_values[j], A_values[j,i], drA_values[j,i])
151
152         rho_Eds_values[j,i] = rho_Eds(t[i])
153
154         # Values for volume
155         V_LTB_values[:,i] = V_LTB(r, k_values, A_values[:,i], drA_values[:,i])
156
157         V_Eds_values[:,i] = V_Eds(t[i], r)
158
159 """# Finding theta and sigma"""
160
161 #Funtions
162 def dtA(r, k, M, A):
163     return np.where(r <= r_b, np.sqrt(-k+2*M/A)*c, 0)
164

```

```

165 def dtrA(r, A, drA, dtA, drk, M, drM):
166     return np.where(r <= r_b, c**2/(2*dtA)*(-drk - 2*M*drA/(A**2) + 2*drM/A), 0)
167
168 def theta_sq(V_Eds, A, drA, dtA, dtrA):
169     return 4*np.pi/V_Eds * ( A**2*dtrA**2/drA + 4*drA*dtrA + 2* A*dtrA*dtA )
170
171 def theta(r, A, drA, dtA, dtrA, theta_Eds):
172     return np.where(r <= r_b, 2*dtA/A+dtrA/drA, theta_Eds)
173
174 def sigma(A, drA, dtA, dtrA):
175     return 1/np.sqrt(3)*(-dtA/A + dtrA/drA )
176
177 def dta_func(t):
178     return 2/3*(t**(-1/3)*t_0**(-2/3))
179
180 def theta_Eds(a_func, dta_func):
181     return 3*dta_func/a_func
182
183 #Initializing matrices
184 dtA_values = np.zeros([Nr+1, Nt+1])
185 dtrA_values = np.zeros([Nr+1, Nt+1])
186 theta_values = np.zeros([Nr+1, Nt+1])
187 theta_Eds_values = np.zeros(Nt+1)
188 sigma_values = np.zeros([Nr+1, Nt+1])
189
190 #Computing values
191 for i in tqdm(range(0, Nt+1)):
192
193     theta_Eds_values[i] = theta_Eds(a_func(t[i]), dta_func(t[i]))
194
195     for j in range(0, Nr+1):
196
197         dtA_values[j,i] = dtA(r[j], k_values[j], M_values[j], A_values[j,i])
198
199         dtrA_values[j,i] = dtrA(r[j], A_values[j,i], drA_values[j,i], dtA_values[j,i],
200                                drk_values[j], M_values[j], drM_values[j])
201
202         theta_values[j,i] = theta(r[j], A_values[j,i], drA_values[j,i], dtA_values[j,i],
203                                dtrA_values[j,i], theta_Eds_values[i])
204
205         sigma_values[j,i] = sigma(A_values[j,i], drA_values[j,i], dtA_values[j,i],
206                                dtrA_values[j,i])
207
208 """"# Q""""
209
210 #Defining determinant
211 def det(k, A, drA):
212     return (drA*A**2/np.sqrt(1-k))
213
214 #Initializing matrices
215 theta_int = np.zeros((Nr+1, Nt+1))
216 theta_norm_V = np.zeros((Nr+1, Nt+1))
217
218 theta2_int = np.zeros((Nr+1, Nt+1))
219 theta2_norm_V = np.zeros((Nr+1, Nt+1))
220
221 sigma2_int = np.zeros((Nr+1, Nt+1))

```

```

219 sigma2_norm_V = np.zeros((Nr+1, Nt+1))
220
221 V_vals = np.zeros((Nr+1, Nt+1))
222
223 # Vectorized implementation
224 for i in tqdm(range(Nt+1)):
225     dr = np.diff(r)[0] # Difference between consecutive r values
226
227     det_vals = det(k_values, A_values[:,i], drA_values[:,i])
228
229     V_vals[:, i] = 4*np.pi * np.cumsum(dr * det_vals)
230
231     # Compute integrals using cumulative sum
232     theta_int[:, i] = 4*np.pi * np.cumsum(dr * theta_values[:,i] * det_vals)
233     theta_norm_V[:, i] = theta_int[:, i] / V_vals[:, i]
234
235     theta2_int[:, i] = 4*np.pi * np.cumsum(dr * theta_values[:,i]**2 * det_vals)
236     theta2_norm_V[:, i] = theta2_int[:, i] / V_vals[:, i]
237
238     sigma2_int[:, i] = 4*np.pi * np.cumsum(dr * sigma_values[:,i]**2 * det_vals)
239     sigma2_norm_V[:, i] = sigma2_int[:, i] / V_vals[:, i]
240
241 Q_alt = 2/3*(theta2_norm_V-theta_norm_V**2)-2*sigma2_norm_V
242
243 """# Plots"""
244
245 plt.plot(r, k_values, label="$k(r)$")
246 plt.title("Curvature function $k(r)$")
247 plt.grid()
248 plt.xlabel("$r$ [Mpc]")
249 plt.ylabel("$k(r)$")
250 plt.legend(loc='upper left')
251 plt.tight_layout()
252 plt.savefig("k(r)_LTB.pdf", format="pdf", bbox_inches="tight")
253 plt.show()
254
255 plt.plot(r, M_values/ M_values[r_b], label="$M(r)/M(r_b)$")
256 plt.title("Effective gravitational mass $M(r)$")
257 plt.grid()
258 plt.xlabel("$r$ [Mpc]")
259 plt.ylabel("$M(r)/M(r_b)$")
260 plt.legend(loc='upper left')
261 plt.tight_layout()
262 plt.savefig("M(r)_LTB.pdf", format="pdf", bbox_inches="tight")
263 plt.show()
264
265 for i in tqdm(range(Nt+1)):
266     if i%(int(Nt/4)) == 0:
267         plt.plot(r, A_values[:, i]/(r*a_func(t[i])), label=f"$t={t[i]/t_0}$ $t_0$")
268 plt.legend()
269
270 plt.title("Areal radius $R(r,t)$ of LTB compared to EdS")
271 plt.grid()
272 plt.xlabel("$r$ [Mpc]")
273 plt.ylabel("$R(r,t)/(a(t)r)$")
274 plt.tight_layout()
275 plt.savefig("R(r,t)_EdS_LTB.pdf", format="pdf", bbox_inches="tight")

```

```

276 plt.show()
277
278 for i in tqdm(range(Nt+1)):
279     if i%(int(Nt/4)) == 0:
280         plt.plot(r, A_values[:, i], label=f"t={t[i]/t_0}$t_0$")
281
282 plt.legend()
283 plt.title("Areal radius $R(r,t)$")
284 plt.grid()
285 plt.xlabel("$r$ [Mpc]")
286 plt.ylabel("$R(r,t)$")
287 plt.tight_layout()
288 plt.savefig("R(r,t)_LTB.pdf", format="pdf", bbox_inches="tight")
289 plt.show()
290
291 for i in tqdm(range(0,Nt+1)):
292     if i%(int(Nt/4)) == 0:
293         plt.plot(r, rho_values[:,i]/rho_Eds_values[:,i], label=f'$t={t[i]/t_0}$t_0$')
294
295 plt.legend()
296 plt.title("Density function $\rho(r,t)$")
297 plt.grid()
298 plt.xlabel("$r$ [Mpc]")
299 plt.ylabel("$\rho_{LTB}(r,t)/\rho_{EdS}(t)$")
300 plt.tight_layout()
301 plt.savefig("rho_r-LTB.pdf", format="pdf", bbox_inches="tight")
302 plt.show()
303
304 for i in tqdm(range(0,Nt+1)):
305     if i%(int(Nt/4)) == 0:
306         plt.plot(A_values[:,i], rho_values[:,i]/rho_Eds_values[:,i], label=f'$t={t[i]/t_0}$t_0$')
307
308 plt.legend()
309 plt.title("Density function $\rho(R,t)$")
310 plt.grid()
311 plt.xlabel("$R(r,t)$ [Mpc]")
312 plt.ylabel("$\rho_{LTB}(r,t)/\rho_{EdS}(t)$")
313 plt.tight_layout()
314 plt.savefig("rho_R-LTB.pdf", format="pdf", bbox_inches="tight")
315 plt.show()
316
317 plt.plot(t/t_0, V_LTB_values[Nr,:], label = "$V_{LTB}$")
318 plt.plot(t/t_0, V_Eds_values[Nr,:], '--', label = "$V_{EdS}$")
319
320 plt.legend()
321 plt.title("Volume $V_{LTB}$ of the entire void")
322 plt.grid()
323 plt.xlabel("$t/t_0$")
324 plt.ylabel("$V_{LTB}$")
325 plt.tight_layout()
326 plt.savefig("V-LTB.pdf", format="pdf", bbox_inches="tight")
327 plt.show()
328
329 plt.plot(t/t_0, V_LTB_values[int(Nr/2),:], label = "$V_{LTB}$")
330 plt.plot(t/t_0, V_Eds_values[int(Nr/2),:], '--', label = "$V_{EdS}$")
331

```

```

332 plt.legend()
333 plt.title("Volume  $V_{\text{LTB}}$  for half of the void")
334 plt.grid()
335 plt.xlabel(" $t/t_0$ ")
336 plt.ylabel(" $V_{\text{LTB}}/V_{\text{EdS}}$ ")
337 plt.tight_layout()
338 plt.savefig("V_0.5_LTB.pdf", format="pdf", bbox_inches="tight")
339 plt.show()
340
341 plt.plot(t[1:]/t_0, V_LTB_values[Nr,1:]/ V_Eds_values[Nr,1:], label = " $V_{\text{LTB}}/V_{\text{EdS}}$ "
342 )
343
344 plt.legend()
345 plt.title("Fraction of volumes for the entire void")
346 plt.grid()
347 plt.xlabel(" $t/t_0$ ")
348 plt.ylabel(" $V_{\text{LTB}}/V_{\text{EdS}}$ ")
349 plt.tight_layout()
350 plt.savefig("V_fraction_LTB.pdf", format="pdf", bbox_inches="tight")
351 plt.show()
352
353 plt.plot(t[1:]/t_0, V_LTB_values[int(Nr/2),1:]/ V_Eds_values[int(Nr/2),1:], label = " $V_{\text{LTB}}/V_{\text{EdS}}$ "
354 )
355
356 plt.legend()
357 plt.title("Fraction of volumes for half of the void")
358 plt.grid()
359 plt.xlabel(" $t/t_0$ ")
360 plt.ylabel(" $V_{\text{LTB}}/V_{\text{EdS}}$ ")
361 plt.tight_layout()
362 plt.savefig("V_fraction_0.5_LTB.pdf", format="pdf", bbox_inches="tight")
363 plt.show()
364
365 for i in range(1,Nt+1):
366     if i%(int(Nt/5)) == 0:
367         plt.plot(r[:-2], Q_alt[:-2,i], label=f' $t={t[i]}/t_0$ ')
368
369 plt.legend()
370 plt.title("Backreaction  $Q$ ")
371 plt.grid()
372 plt.xlabel(" $r$  [Mpc]")
373 plt.ylabel(" $Q$ ")
374 plt.tight_layout()
375 plt.savefig("Q_LTB.pdf", format="pdf", bbox_inches="tight")
376 plt.show()

```

A.7 Code for the Tardis model

```

1 import numpy as np
2 from scipy.interpolate import CubicSpline
3 import matplotlib.pyplot as plt
4 from scipy import optimize
5 from tqdm import tqdm
6
7 plt.rc('axes', titlesize=16)      # fontsize of the axes title
8 plt.rc('axes', labelsiz=16)      # fontsize of the x and y labels
9 plt.rc('xtick', labelsiz=14)      # fontsize of the tick labels
10 plt.rc('ytick', labelsiz=14)     # fontsize of the tick labels
11 plt.rc('legend', fontsize=13)    # legend fontsize
12 plt.rc('figure', titlesize=16)   # fontsize of the figure title
13
14 """Constants"""
15
16 # Defining constants (from https://arxiv.org/pdf/1308.6731.pdf)
17 small = 1e-3 #do not make smaller
18
19 r_b = 20 #Mpc - largest is 280 Mpc
20 r_peak = 0.85 * r_b
21 A_val = 10**(-1.2)
22
23 H_0 = 71.58781594e-3      # Hubble constant [1/Gyr]
24 c = 306.5926758           # Speed of light [Mpc/Gyr]
25 G = 4.498234911e-15       # G in Mpc^3/(M_sun*Gyr^2)
26
27 # Initial time values
28 a_i = 1/1100              # Initial scale factor (CMB)
29 t_0 = (2/3)*(1/H_0)        # Present time [Gyr]
30 t_i = t_0*a_i**(3/2)       # Initial time [Gyr]
31
32 cs_top = 0.0001           # Top point of cubic spline
33
34 # Space
35 N = 10_000                # Number of steps
36 r = np.linspace(small, r_b, N)
37 dr = np.append(np.diff(r)[0], np.diff(r))
38
39 # Time
40 t_i = 0.002*t_0
41 Nt = 20
42 t = np.linspace(t_i, t_0, Nt)
43
44 # Time TARDIS
45 Nt = 5
46 t = np.array([0.002, 0.01, 0.05, 0.2, 1])*t_0
47 #Nt = 10
48 #t = np.array([0.002, 0.004, 0.008, 0.016, 0.032, 0.064, 0.128, 0.256, 0.512, 1.0])*t_0
49
50 """Functions"""
51
52 def alwaysone(i):
53     return 1
54 def alwaysr(r, i):
55     return r

```



```

56
57 def plot_of_r_t(function, divisor = 1, divisorfunc = alwaysone, title="Lorem ipsum",
    xlabel="$r/r_b$", ylabel="y", legend=True, x=alwaysr, x_lim=False, name="saved plot")
    :
58     for i in tqdm(range(Nt)):
59         plt.plot(x(r,i)/r_b,function(r,i)/divisor/divisorfunc(i), label=f'$t={t[i]/t_0}t_0$')
60     if legend==True:
61         plt.legend(loc='best')
62     if x_lim==True:
63         plt.xlim([0.75, 1.00])
64     plt.title(title)
65     plt.grid()
66     plt.xlabel(xlabel)
67     plt.ylabel(ylabel)
68     plt.tight_layout()
69     plt.savefig(f"{title}.pdf", format="pdf", bbox_inches="tight")
70     plt.show()
71
72 def plot_of_r(function, divisor = 1, title="Lorem ipsum", xlabel="$r/r_b$", ylabel="y",
    legend=False, x=r, x_lim=False, name="saved plot"):
73     plt.plot(x/r_b,function(x)/divisor)
74     if legend==True:
75         plt.legend(loc='best')
76     if x_lim==True:
77         plt.xlim([0.75, 1.00])
78     plt.title(title)
79     plt.grid()
80     plt.xlabel(xlabel)
81     plt.ylabel(ylabel)
82     plt.tight_layout()
83     plt.savefig(f"{title}.pdf", format="pdf", bbox_inches="tight")
84     plt.show()
85
86 def M(r):
87     return 1/2 * H_0**2 * r**3 * (1+np.exp(-((r-r_peak)/(A_val*r_b))**10)) * c**(-2)
88
89 def M_dr(r):
90     return ( 3/2 * H_0**2 * r**2 * (1+np.exp(-((r-r_peak)/(A_val*r_b))**10)) - 5/((A_val*
        r_b)**10) * H_0**2 * r**3 *(r-r_peak)**9 * (np.exp(-((r-r_peak)/(A_val*r_b))**10)) )
        * c**(-2)
91
92 def E(r):
93     """
94     E is a cubic spline.
95     """
96     x0 = (r_1-(r_2-r_1))
97     y1 = cs_top
98     y_point = [0,y1,0]
99     x_point = [x0,r_1,r_2]
100     cs = CubicSpline(x_point, y_point, bc_type = (( 1,0.0 ), (1,0.0))) #bc_type =(( 1,0 ),
        (1,0)) 'not-a-knot')
101     return np.where(r < x0, 0, (np.where(r > r_2, 0, cs(r) )))
102
103 def E_dr(r):
104     """
105     The derivative of E(r).
106     """

```

```

107     x0 = (r_1-(r_2-r_1))
108     y1 = cs_top
109     y_point = [0,y1,0]
110     x_point = [x0,r_1,r_2]
111     cs = CubicSpline(x_point, y_point, bc_type = (( 1,0.0 ), (1,0.0))) #bc_type =(( 1,0 ),
112                               (1,0)) 'not-a-knot')
113
114     return np.where(r < x0, 0, (np.where(r > r_2, 0, cs(r,1) )))
115
116 def M_dr_root(r):
117     return M_dr(r)/M_dr(r_b)
118
119 r_1 = optimize.bisect(M_dr_root, a=0.88*r_b, b=0.90*r_b, xtol=1e-12)
120 r_2 = optimize.bisect(M_dr_root, a=0.90*r_b, b=0.94*r_b, xtol=1e-12)
121
122 # Defining a for einstein de sitter
123 def a_eds(i):
124     return (t[i]/t_0)**(2/3)
125
126 def f_eta(r, i):
127     eta = np.zeros(np.size(r))
128     x = np.ones(np.size(r))*(1+small)
129     fder = lambda x, r: np.cosh(x)-1
130     f = lambda x, r: np.sinh(x)-x- (E(r)**(3/2) / M(r) * c*t[i])
131     eta = optimize.newton(f, x, fprime=fder, args=(r, ), maxiter=200, tol=1e-12)
132     return eta
133
134 def R(r, i):
135     with np.errstate(divide='ignore', invalid='ignore'):
136         return np.where(E(r) == 0, (9/2*M(r)*(c*t[i])**2)**(1/3),
137                                (M(r) / E(r))*(np.cosh(f_eta(r,i))-1))
138
139 def eta_dr(r, i):
140     return 1/(np.cosh(f_eta(r,i))-1)*c*t[i]*(E(r))**(1/2)/((M(r))**2)*(-E(r)*M_dr(r)+3/2*M(
141         r)*E_dr(r))
142
143 def R_dr(r, i):
144     with np.errstate(divide='ignore', invalid='ignore'):
145         return np.where(E(r) == 0, 1/3*(9/2*M(r)*(c*t[i])**2)**(-2/3)*(9/2*M_dr(r)*(c*t[i])
146             **2),
147                                M_dr(r)/E(r)*(np.cosh(f_eta(r,i))-1) - M(r)/((E(r))**2)*E_dr(r)*(np.
148                cosh(f_eta(r,i))-1) + M(r)/E(r)*np.sinh(f_eta(r,i))*eta_dr(r, i))
149
150 def rho(r, i):
151     return (c**4)/(4*np.pi*G)*M_dr(r)/((R(r, i))**2*R_dr(r, i))
152
153 def rho_eds(i):
154     return 3 * c**2 * ((2/(3*t[i]))**2 / (8* np.pi * G))
155
156 def R_dt(r, i):
157     return np.sqrt(2*M(r)/R(r, i)+E(r))*c
158
159 def R_dr_dt(r,i):
160     return c**2/(2*R_dt(r, i))*(2*M_dr(r)/R(r, i)-2*M(r)*R_dr(r, i)/((R(r, i))**2)+E_dr(r))
161
162 def a_eds_dt(i):
163     return 2/3*(t[i])**(-1/3)*(t_0)**(-2/3)

```

```

160 def theta_eds(i):
161     return 3*a_eds_dt(i)/a_eds(i)
162
163 def theta(r, i):
164     return 2*R_dt(r, i)/R(r, i)+R_dr_dt(r, i)/R_dr(r, i)
165
166 def sigma(r, i):
167     return 1/np.sqrt(3)*(-R_dt(r, i)/R(r, i)+R_dr_dt(r, i)/R_dr(r, i)) #this is sigma**2
168     with units 1/s**2
169
170 def R3_avg(i):
171     return -6*(E(r_1)*R(r_1,i))/(R(r_b,i)**3+2*R(r_1,i)**3-2*R(r_2,i)**3)
172
173 def rho_avg(i): # units m^-2 G^-1 without any c's
174     return 3/(4*np.pi*G)*(M(r_b)+2*M(r_1)-2*M(r_2))/(R(r_b,i)**3+2*R(r_1,i)**3-2*R(r_2,i)
175     **3)
176
177 def deter(r, i):
178     return np.abs(R_dr(r,i))*R(r,i)**2/np.sqrt(1+E(r))
179
180 def deter_eds(r, i):
181     return a_eds(i)**3 * r**2
182
183 # Volume LTB
184 def V_Tardis(r, i):
185     return 4*np.pi* np.cumsum(deter(r, i) * dr)
186
187 # Volume Eds
188 def V_Eds(r, i):
189     return 4*np.pi/3* (a_eds(i)**3) * (r**3)
190
191 def integrating_Q(i):
192     # Vectorized implementation
193     theta_vals = theta(r, i)
194     deter_vals = deter(r, i)
195     sigma_vals = sigma(r, i)
196     V_vals = V_Tardis(r,i)
197
198     # Compute integrals using cumulative sum
199     theta_int = 4*np.pi * np.cumsum(dr * theta_vals * deter_vals) / V_vals
200
201     theta2_int = 4*np.pi * np.cumsum(dr * theta_vals**2 * deter_vals) / V_vals
202
203     sigma2_int = 4*np.pi * np.cumsum(dr * sigma_vals**2 * deter_vals) / V_vals
204     return theta_int, theta2_int, sigma2_int, V_vals
205
206 def Q(r, i):
207     theta_int, theta2_int, sigma2_int, V_vals = integrating_Q(i)
208     return 2/3*(theta2_int-theta_int**2)-2*sigma2_int
209
210 def H_avg(i):
211     theta_averaged = np.zeros(Nt)
212     theta_int, theta2_int, sigma2_int, V_vals = integrating_Q(i)
213     theta_averaged = theta_int[-1]
214     return theta_averaged/3
215
216 def theta_avg(i):

```

```

215     return theta_eds(i)*(1+2*R_dt(r_1,i)/R(r_1,i)*(R(r_1,i)/R(r_b,i))**3-2*R_dt(r_2,i)/R(
        r_2,i)*(R(r_2,i)/R(r_b,i))**3)/(1+2*(R(r_1,i)/R(r_b,i))**3-2*(R(r_2,i)/R(r_b,i))**3)
216
217
218 plot_of_r(M, M(r_b), title="Effective gravitational mass  $M(r)$ ", ylabel=" $M(r)/M(r_b)$ ")
219 plot_of_r(M_dr, M_dr(r_b), title="Derivative of effective gravitational mass  $M'(r)$ ",
        ylabel=" $M'(r)/M'(r_b)$ ")
220 plot_of_r(E, title="Curvature  $E(r)$ ", ylabel=" $E(r)$ ")
221 plot_of_r(E_dr, M_dr(r_b), title="Derivative of curvature  $E'(r)$ ", ylabel=" $E'(r)$ ")
222
223 plot_of_r_t(R, r, a_eds, title="Areal radius  $R(t,r)$ ", ylabel=" $R(t,r)/r_a(t)$ ", x_lim=
        True)
224 plot_of_r_t(R, title="Non-monotonicity of areal radius  $R(t,r)$ ", ylabel=" $R(t,r)$ ",
        x_lim=False)
225 plot_of_r_t(R_dr, title="Derivative of the areal radius  $R'(t,r)$ ", ylabel=" $R'(t,r)$ ",
        x_lim=True)
226 plot_of_r_t(rho, divisorfunc=rho_eds, title="Density  $\rho(t,r)$ ", ylabel=" $\rho(t,r) / \rho_{EdS}(t)$ ", x_lim=True)
227 plot_of_r_t(theta, divisorfunc=theta_eds, title="Expansion rate  $\Theta(t,r)$ ", ylabel=" $\Theta(t,r) / \Theta_{EdS}(t)$ ", x_lim=True)
228 plot_of_r_t(sigma, divisor= 1/3, divisorfunc=theta_eds, title="Shear  $\sigma(t,r)$ ",
        ylabel=" $3\sigma(t,r) / \Theta_{EdS}(t)$ ", x_lim=True)
229 plot_of_r_t(Q, title="Backreaction  $Q(t,r)$ ", ylabel=" $Q(t,r)$ ")
230
231 Nt = 100
232 t = np.linspace(t_i, t_0, Nt)
233
234 Q_array = np.zeros([N, Nt])
235 V_Tardis_array = np.zeros([N, Nt])
236 V_Eds_array = np.zeros([N, Nt])
237 H_avg_array = np.zeros([Nt])
238 for i in tqdm(range(Nt)):
239     Q_array[:,i] = Q(r,i)
240     V_Tardis_array[:,i] = V_Tardis(r, i)
241     V_Eds_array[:,i] = V_Eds(r, i)
242     H_avg_array[i] = H_avg(i)
243
244 # Q at r_b
245 plt.plot(t[:]/t_0, Q_array[-1, :])
246 plt.title("Backreaction over entire volume  $Q(t,r_b)$ ")
247 plt.grid()
248 plt.xlabel(" $t/t_0$ ")
249 plt.ylabel(" $Q(t,r_b)$ ")
250 plt.tight_layout()
251 plt.savefig("Backreaction_at_rb.pdf", format="pdf", bbox_inches="tight")
252 plt.show()
253
254 plt.plot(t/t_0, V_Tardis_array[-1, :]/V_Eds_array[-1, :])
255 plt.title("Volume fraction  $V(t)/V_{EdS}(t)$ ")
256 plt.grid()
257 plt.xlabel(" $t/t_0$ ")
258 plt.ylabel(" $V(t)/V_{EdS}(t)$ ")
259 #plt.xlim(0.75*r_b, 1.0*r_b)
260 plt.tight_layout()
261 plt.savefig("Volume plot.pdf", format="pdf", bbox_inches="tight")
262 plt.show()
263

```

```

264 import pandas as pd
265
266 df = pd.DataFrame({'target': Q_array[-1, :], 'features': t/t_0})
267 df.to_csv('Qdata.csv', index=False)
268
269 H_avg_array
270
271 plt.plot(t/t_0, -Q_array[-1, :]/(6*H_avg_array**2))
272 #plt.ylim(0, 1e-3)
273 plt.title("Density parameter  $\Omega_Q(t)$ ")
274 plt.grid()
275 plt.xlabel("$t/t_0$")
276 plt.ylabel(" $\Omega_Q(t)$ ")
277 #plt.xlim(0.75*r_b, 1.0*r_b)
278 plt.tight_layout()
279 plt.savefig("Density parameter  $\Omega_Q$ .pdf", format="pdf", bbox_inches="tight")
280 plt.show()
281
282 R3_array = np.zeros(Nt)
283 rho_array = np.zeros(Nt)
284 theta_avger = np.zeros(Nt)
285 for i in tqdm(range(Nt)):
286     R3_array[i] = R3_avg(i)
287     rho_array[i] = rho_avg(i)
288     theta_avger[i] = theta_avg(i)
289
290 plt.plot(t/t_0, rho_array*8*np.pi*G*c**2/(3*H_avg_array**2))
291 plt.plot(t/t_0, -R3_array*c**2/(6*H_avg_array**2))
292 plt.plot(t/t_0, -Q_array[-1, :]/(6*H_avg_array**2))
293 plt.title("Density parameters  $\Omega(t)$ ")
294 plt.legend([" $\Omega_m(t)$ ", " $\Omega_{\mathcal{R}}(t)$ ", " $\Omega_Q(t)$ "])
295 plt.grid()
296 plt.xlabel("$t/t_0$")
297 plt.ylabel(" $\Omega(t)$ ")
298 #plt.xlim(0.75*r_b, 1.0*r_b)
299 plt.tight_layout()
300 plt.savefig("Density parameters  $\Omega$ .pdf", format="pdf", bbox_inches="tight")
301 plt.show()
302
303 -2.5575e-2 + 5.27566e-2*np.sqrt(np.abs(t/t_0)) + -1.02771e-2*np.sqrt(np.abs((t/t_0)**(2)))
304 ) + 4.89687e-3*np.log((t/t_0)**(-5)) + -1.23558e-4*np.sqrt(np.abs((t/t_0)**(-2))) +
305 -1.69657e-2*np.sqrt(np.abs((t/t_0)**(-1)))
306
307 plt.plot(t/t_0, Q_array[-1, :])
308 plt.plot(t/t_0, -2.5575e-2 + 5.27566e-2*np.sqrt(np.abs(t/t_0)) + -1.02771e-2*np.sqrt(np.
309 abs((t/t_0)**(2))) + 4.89687e-3*np.log((t/t_0)**(-5)) + -1.23558e-4*np.sqrt(np.abs((t
310 /t_0)**(-2))) + -1.69657e-2*np.sqrt(np.abs((t/t_0)**(-1))), '--')
311
312 plt.title("Backreaction  $Q(t, r_b)$  and symbolic regression")
313 plt.grid()
314 plt.legend([" $Q(t, r_b)$  numerical", " $Q(t, r_b)$  symbolic regression"])
315 plt.xlabel("$t/t_0$")
316 plt.ylabel(" $Q(t, r_b)$ ")
317 plt.tight_layout()
318 plt.savefig("Backreaction_at_rb_SR.pdf", format="pdf", bbox_inches="tight")
319 plt.show()

```